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— a market-process perspective



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ESTEBAN F. THOMSEN

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Chapter 1

Introduction

In recent decades, as problems caused by imperfect information have received increased attention from economists, there has arisen an interest in the informational role prices may perform in an economy.¹ This book is an attempt to examine in some detail this role, and the ways in which it has been analysed. In this introduction, the arguments that will be presented in later chapters will first be briefly described.

Before the relatively recent interest in their informational role, prices were generally explained by economists as caused by scarcity and described as useful incentives for making (omniscient) economic agents adopt decisions consistent with such scarcity. Fully informed agents were assumed to optimize with respect to 'given' prices. Starting perhaps with Hayek's 1945 article 'The use of knowledge in society', prices have become increasingly perceived as more or less effective devices for the communication of information. Although the impact of this article was not fully felt in economics for about fifteen years, with the development in the 1960s of what is known as the 'economics of information', a relatively large literature has examined this informational aspect of prices in detail. Working from within a framework in which individuals are only partially informed, prices are now often seen either as 'sufficient statistics' for economic agents to make appropriate economic decisions, as sources from which they can infer other information, or as means by which they can attempt to signal information to others.

The economics of information approach is probably the most widespread and influential in economics today. But another approach to informational problems which has also had significant impact is the 'bounded rationality' theory of Herbert A.Simon. From both of these perspectives there have been critical analyses of the informational role of prices. This book will

analyse some examples believed to be representative of these two different theoretical approaches to the study of the informational role of prices in situations of imperfect information. The main examples chosen are the work on the informational efficiency of prices by Sanford J. Grossman and Joseph E. Stiglitz, representing a modern economics of information perspective, and writings of Simon and Richard R. Nelson, representing bounded rationality theory.

Besides presenting a survey of the field, the argument will be that although there has been much progress in appreciating some significant informational functions of prices, an important and perhaps central informational role of prices when they are in disequilibrium has not been recognized fully. This is not a surprising discovery, given that economic theory consists almost exclusively of the analysis of equilibrium situations. However, it is a conclusion with important implications, because this complete neglect of disequilibrium, and of the disequilibrium role of prices in particular, is likely to become a serious obstacle to a full understanding of markets in reality.

Some grounds for optimism are perhaps provided by the fact that a growing number of economists—including some of the founders of modern general equilibrium analysis—are starting to believe that an exclusive concern with equilibrium is misguided, and that disequilibrium situations also deserve serious attention from economists. In fact, a case can be made that situations of disequilibrium should now be the central concern of economic theory both because ample effort has been devoted already to the analysis of equilibrium and because situations of disequilibrium best reflect some fundamental features of real-world markets. To exemplify this last statement, it is only in disequilibrium that the *data* are not fully known to economic agents even in a probabilistic sense (i.e. that they are, in part, radically ignorant), that there is uncertainty and not only risk, that there is room for creative decision-making and entrepreneurship, that there are pure profits and losses, and so on. These features of reality do not exist, by definition, in equilibrium.

Although there is at present no generally accepted economic theory of markets in disequilibrium, there are economists, belonging to what is often called the ‘modern Austrian’ school of economics, who aim at producing such a theory. Their approach is to view market activity as a continuing *process* of changes and adjustments

to changes, and not as something that occurs within a fully adjusted *state*. Given that the analysis of the disequilibrium informational role of prices requires focusing on the disequilibrium market process and not on equilibrium states, it is to their work that this book turns for a theoretical framework (which henceforth will be described as a 'market-process' approach).²

As mentioned above, the current equilibrium perspective sees prices as summaries of information, as 'signalling' devices, or as sources of information, for uninformed optimizing individuals. This book will argue that, from a market-process perspective, prices are not only more or less effective signals or summaries of information. Disequilibrium prices also turn out to be sources of profit opportunities that stimulate the discovery and exploitation of previously unnoticed alternatives by radically ignorant individuals. Disequilibrium prices are therefore an indispensable aid for individuals coping with the problem of finding what economists usually call the *data*, the information which economic theory all too often assumes economic agents to know before making their decisions. (This is a problem Don Lavoie has termed the 'knowledge problem'.) In fact, it is profits that stimulate the discovery of what and how much to produce, how to produce it, at what prices to buy and sell, what organizational forms to adopt, and so on. As these opportunities are discovered and exploited by entrepreneurs, those profits are whittled away (and other profit opportunities are created). Thus disequilibrium prices provide through profits a feedback mechanism for their own correction that makes them a more sophisticated informational device than they may seem when concentrating only on their equilibrium role. It is because of this feedback mechanism, it will also be argued, that prices in reality may also perform relatively well some of the other informational roles usually attributed to equilibrium prices.

From a disequilibrium perspective, then, the economics of information, because it generally concentrates on equilibrium states, misses some crucial aspects of the knowledge problem that have to be solved by any economic system, aspects that are present only in disequilibrium.

An alternative approach to informational issues in economics is provided by bounded rationality theory. This approach, usually associated with the work of Herbert Simon, has a very particular view of the knowledge problem. This view, while provocative and no doubt important, in a sense also tends to underestimate the

problem—perhaps surprisingly, given its expressed dissatisfaction with the assumption of optimizing individual behaviour underlying even the economics of information. This underestimation, it will be argued, seems to be due to its emphasis on a special notion of complexity and on a theory of cognition that has been seriously questioned by some specialists in the cognitive sciences. Although these notions are still being debated and need more detailed study, probably by disciplines other than economics, it will be argued that they apparently have not allowed this approach to perceive fully, let alone appreciate, the role of prices as catalysts of a competitive discovery procedure. In this respect, it will be shown that bounded rationality theory, though it has been asking many of the right questions, has not provided an understanding of prices and markets that is significantly different from that of more conventional economic theory.

While considering the different approaches to the informational role of prices, other issues arise from emphasizing a more radical type of ignorance and the entrepreneurial activities of individuals. Specifically, some consideration will be given to the issue of why prices in reality (i.e. disequilibrium prices) may be able to perform reasonably well the informational roles usually ascribed only to equilibrium prices. There will also be some suggestions for a more appropriate normative standard, given the limited relevance of traditional welfare economics for judging how different economic arrangements cope with radical ignorance. Finally, an entrepreneurial perspective on organizations and hierarchies will be outlined briefly.

WHY A MARKET-PROCESS PERSPECTIVE IS IMPORTANT

The concern of economists with the problems caused by ignorance and uncertainty, and with the different ways of overcoming them, is helping to remedy what was a serious deficiency in economic theory. However, the fertility of approaches such as that of the economics of information and of search theory may lead economists into believing that all problems of knowledge in an economy are being studied. It will be argued that these approaches, because they treat knowledge as just another scarce good, do not fully encompass the knowledge problem faced in reality and in fact underestimate it. If they were to

mislead economists into overlooking a more radical type of ignorance, they would have obstructed a fuller comprehension of prices and markets. (Another problem, briefly touched on here, is that these approaches to information allow the continued use of normative criteria of questionable validity for judging real markets.)

A similar risk is caused by the bounded rationality approach. For a long time, this approach has been critical of more standard economic theory, focusing precisely on its neglect of people's cognitive limitations. To the extent that bounded rationality becomes perceived as the only alternative to an economics of information framework, it could also cause an underestimation of the problems caused by radical ignorance. It would also, therefore, stand in the way of a full understanding or adequate appreciation of the informational role of entrepreneurial competition and market prices.

In this regard, the present work attempts to present a disequilibrium view of the market process in which radical ignorance and entrepreneurial discovery play a central role. This view emphasizes rivalrous competition as the market's way of attempting to overcome, or at least deal as well as possible with, the knowledge problem. It sees prices performing a crucial informational role in this competitive process, a process for which there appears to be no room in the other approaches. At the same time, it does not deny the informational roles of prices that are generally being studied within an equilibrium framework.

Admittedly, most of the work in the disequilibrium perspective of markets has not been presented in the mathematically formalized fashion that characterizes most economic theory. On this old methodological issue the position taken here is, on the one hand, that, although not mathematically formalized, there is available already a body of theory that is valuable for the study of markets in disequilibrium and that should not be discarded merely because of its form of presentation; but also, on the other hand, that as more economists become interested in analysing disequilibrium, mathematical studies that do not assume away its more crucial characteristics may well become useful aids to a further understanding of market phenomena. (Some initial efforts in this direction are cited in later chapters.)

OUTLINE OF THE BOOK

The argument begins in chapter 2 by indicating the dissatisfaction some economists have expressed with the exclusive concentration on the analysis of equilibrium states of much economic theory. The main problem with focusing only on equilibrium, this chapter tries to indicate, is that the defining characteristic of equilibrium is the absence of the knowledge problem. Such an approach is therefore unable to say much about the ways in which this problem is tackled by different economic systems, undoubtedly a matter of great importance not to be neglected by economics. The chapter then shows some of the problems encountered by attempts to analyse disequilibrium within the usual theoretical framework of economics. Finally, in search of a disequilibrium approach to markets, it turns to the writings of economists who have attempted to provide one. The chapter therefore examines some key concepts of the theory of the market process found in the writings of Ludwig von Mises, Friedrich A. Hayek, and Israel M. Kirzner, the so-called modern Austrian economists.

Using this disequilibrium perspective as a theoretical framework, the following chapters analyse different approaches to the study of the informational role of prices. For such an informational role to be necessary, individuals must be seen as having limited knowledge, a view these approaches share. What distinguishes them is their respective interpretations of the nature of this limitation.

Chapter 3 examines an economics of information approach to prices. The analysis of the informational efficiency of prices by Sanford J. Grossman and Joseph E. Stiglitz is chosen as representative of this approach, although some work by other economists is also briefly examined. The chapter considers difficulties caused by confining the analysis to equilibrium states and distinguishes three different informational roles of prices (as surrogates for knowledge, as sources for the inference of knowledge, and as sources of profit opportunities that stimulate the discovery of knowledge). It also analyses the frequent description of prices as sufficient statistics and points out some of the ambiguities that arise in the economics of information approach regarding what normative standard to adopt for evaluating different situations.

Chapter 4 analyses an alternative view of the cognitive problem faced by individuals, Simon's bounded rationality approach. The chapter proceeds by describing some of Simon's criticisms of equilibrium economics and his view of the knowledge problem. It

then considers some of the ways Simon believes individuals get round this problem, namely, through ‘satisficing’ instead of optimizing, and by relying on prices as summaries of information and on organizations. Because Simon thinks he is adopting Hayek’s approach to markets, his view of the knowledge problem and his consequent understanding of prices are contrasted with those arising from a disequilibrium view of markets. This contrast is completed by an examination of writings of modern Austrian economists, indicating similarities to and differences from bounded rationality theory.

What could appear to be a third perspective on markets in the context of informational limitations, the work of Richard R. Nelson, is examined in chapter 5. The chapter describes his criticism of welfare economics and of some arguments in favour of the market system, and his view of the knowledge problem and of the relative advantages of markets and central planning. This view is analysed and found to be similar, in important respects, to that of Simon. Although Nelson has always been explicit about the influence Simon’s work has had on his, which would render this finding trivial, the conclusion is that on this issue Nelson is so close to Simon as to be susceptible to the same criticisms. Evaluations of economic systems caused by different interpretations of the knowledge problem are then contrasted, and some consideration is given to the importance of change for the knowledge problem and to the problem of co-ordination of activities in a market economy. Finally, the chapter tries to clarify discussions regarding the relative advantages of markets and ‘hierarchies’ by providing an outline of a market-process approach to organizations.

Chapter 6 provides a summary and some suggestions for further research.

Chapter 2

A theory of the market process

Most economic theory today is almost exclusively concerned with equilibrium states. Research efforts are largely devoted to proving the existence of equilibrium for specific models, proving its stability, and comparing the equilibria for different sets of *data* (comparative statics). As the theory becomes more sophisticated, so do the models and their equilibria. Thus there are not only static equilibria but also a variety of ‘dynamic’ and ‘stochastic’ equilibria, among others.

However dissimilar different areas of economics are in other respects, they almost all rely heavily on the notion of equilibrium. And to practically all these areas can be applied Franklin Fisher’s remark about microeconomic theory. According to Fisher, in microeconomics

very little is said about the dynamics of the process that leads an equilibrium to be established in the first place or by which the system adjusts to a new equilibrium when the old one is displaced by a parameter shift. Attention is centered on the equilibria themselves..., and points of non-equilibrium are discussed by showing that the system cannot remain at such points.

(Fisher 1983:3)

This is true of most theory, whether it is concerned with static equilibrium, equilibrium ‘paths,’ ‘sequential’ equilibria, or any other equilibrium state.

THE DEFINING CHARACTERISTIC OF EQUILIBRIUM

In spite of the centrality of equilibrium to economic theory, most economists have been using what Hahn (1984a:8) describes as a 'sloppy equilibrium concept'.¹ Although equilibrium has frequently been associated with situations of market-clearing,² the essential property of equilibrium, as Hayek (1937) pointed out, is the perfect co-ordination of individual plans. The confusion may have arisen because this wider definition will often include situations of market-clearing. However, it is the perfect co-ordination of plans that allows the theorist to imagine that in such a state no further change will occur, thus allowing him to speak of equilibrium.

Frank Hahn (1984a:44), who describes himself as a neoclassical economist, has, in a Hayekian vein, described equilibria as 'those states in which the intended actions of rational economic agents are mutually consistent and can therefore be implemented'.³ The stochastic equilibrium notion he has in mind would be consistent with 'short enough and rare enough episodes of uncleared markets,' although it 'implies almost the missing traditional complement that markets are cleared' (Hahn 1984a:60). Stiglitz (1987: 28) has also recently pointed out some limitations in the traditional notion of equilibrium. According to him,

Traditional theory has taken the equality of supply and demand to be part of the definition of equilibrium. This, I think, is wrong.⁴

And, he adds,

the equality of demand and supply should not be taken as a definition of equilibrium, but rather as a consequence following from more primitive behavioral postulates.

For Stiglitz, equilibrium is better defined as 'a state where no economic agents have an incentive to change their behavior'. Such a state can occur only when there remain no unexploited profit opportunities. As long as any opportunity continues to exist there is the possibility that the agents will discover it and disturb the existing situation (which therefore is not properly described as in equilibrium). In other words, equilibrium occurs only once agents have discovered all the available opportunities and once they know they have done so. (As will be indicated below, this situation is compatible with the

existence of optimal ignorance.) The perfect co-ordination of individual plans mentioned by Hayek is made theoretically possible by assuming that individuals have acquired the necessary knowledge to achieve it.

The essential point, then, is that the defining characteristic of equilibrium is not the equality of quantities supplied and demanded but, instead, the knowledge (and exploitation) by the trading agents of all profitable opportunities, that is, 'perfect' knowledge. As Hayek (1937:42) put it, the assumption of 'correct foresight' is not a precondition for the achievement of equilibrium but, instead, its *defining characteristic*.

WHY IS DISEQUILIBRIUM IMPORTANT?

This informational characteristic of equilibrium serves to explain why equilibrium does not provide an adequate framework for studying how an economic system solves the knowledge problem involved in discovering profit opportunities: in equilibrium the problem is already solved. To the extent that many observed market phenomena arise as a consequence of this knowledge problem, an economics interested exclusively in equilibrium would never be able to explain them satisfactorily.⁵ Specifically, for the purposes of this study, such an economics is unable to say much about any informational role that prices may perform in disequilibrium. An economic theory of disequilibrium is necessary for this task.

Many economists appear to be reticent about studying disequilibrium situations, both because they believe that most economic phenomena of interest will, sooner or later, be accommodated within an equilibrium framework, and because they fear that a concern with disequilibrium is synonymous with the abandonment of rigorous economic theorizing. However, some have started to take a different attitude and to point out that much is missed by neglecting the study of disequilibrium. Because their remarks to this effect do not seem to have received much attention, they will be quoted here at some length. One example is provided by Frank Hahn's (1984:4) comments regarding the 'danger' of considering 'nothing but equilibrium' and the 'foolishness' of claiming 'that all theory should be equilibrium theory':

What is plain is that by narrowing our viewpoint in this manner we shall remove a great deal of interest and importance from scrutiny.

For instance, imposing the axiom that the economy is at every instant in competitive equilibrium simply removes the actual operation of the invisible hand from the analysis. By postulating that all perceived Pareto-improving moves are instantly carried out all problems of co-ordination between agents are ruled out. Economic theory thus narrowly constructed makes many important discussions impossible.

Similarly, Fisher (1983:7) thinks ‘disequilibrium questions cannot be avoided’ and that

If disequilibrium effects are in fact unimportant we need to prove that they are. If such effects are important, then the way in which we tend to think about the theory of value needs to be revised. Interest must then center not on equilibrium itself but on disequilibrium adjustment. Different economies cannot then be studied as though their future were determined solely by tastes, technology, and initial endowments with adjustment but a transient matter.

He concludes that

The issues involved in disequilibrium analysis are too important to economics to be avoided. They must be faced head on rather than assumed away in the course of a desire to do what economists do best—analyze equilibrium without regard for the foundations on which such analysis must rest.⁶

(*ibid.*: 217–18)

TOWARDS A DISEQUILIBRIUM THEORY OF MARKETS

Of course, the step from emphasizing the importance of disequilibrium to providing a disequilibrium theory is a large one and may require some important changes in the theoretical apparatus of economics. Attempts to model disequilibrium have found that some characteristic elements of equilibrium theory are inadequate for the task. This is particularly so with respect to the behaviour of economic agents. For example, Fisher says that

it is a mistake to ground disequilibrium theory in the equilibrium

behavior of agents. Rather, the theory of the household and the firm must be reformulated and extended where necessary to allow agents to perceive that the economy is not in equilibrium and to act on that perception. Without this, we cannot hope to provide a theory of what happens when arbitrage opportunities appear, for the essence of the appearance of such opportunities is that agents see and act on them.

He points out that

Agents in the standard theory react to given prices and take no account either of the fact that prices may change or of the possibility that they may not be able to complete their own transactions. So long as the plans which agents make are compatible, this presents no difficulty; in equilibrium the equilibrium assumptions of agents are fulfilled. If we are to deal with disequilibrium, however, this will not be the case, and we must start at the level of individual agents.
(1983:11)

Related considerations lead Hey (1981:201) to state that ‘if one wants to model genuine disequilibrium, then the optimising model of individual choice must be abandoned’.

Israel M. Kirzner (1973) has described the main problem with the economic agent of equilibrium theory, whom he terms the Robbinsian (for Lionel Robbins) maximizer. The neoclassical agent maximizes either utility or profits (or some other magnitude in less frequent versions). The facts the agent uses for such a calculation (technologies, resource availabilities, prices, probabilities of occurrences of events, and so on) are presumed to be somehow given to him and, most frequently, also to be correct. How these facts were perceived, or what the agent would do if they were to turn out to be wrong, is beyond the scope of the maximizer. For example, the plans of a set of such agents, based on incorrect facts, would produce surpluses and shortages that, strictly speaking, could persist indefinitely: there is no indication of how, if at all, these agents’ perceptions would be modified under such circumstances. As T.W. Schultz (1975:829) puts it, ‘determining precisely what people do who are not in equilibrium is not one of the notable achievements of economics’.

Kirzner describes this characteristic by saying that the standard maximizer operates within a *given* means-ends framework. When there is disequilibrium ‘the Robbinsian framework suggests that the

unsuccessful plans will be discarded or revised, but we are unable to say much more than this'. The problem is that 'Robbinsian theory only applies after a person is confronted with opportunities; for it does not explain how that person learns about opportunities in the first place' (Kirzner 1983:6–7). It does not say anything about how agents *discover* unexploited opportunities.

These remarks are not made unaware of the fact that newer versions of this Robbinsian agent are sometimes assumed to possess 'learning functions' allowing them to modify their behaviour. But the methods most frequently used to model this 'learning' still require that the agents already have a great amount of given knowledge. Although an analysis of the limitations of this approach is not possible here, it has been said that to use it

there must be some aspects of the world that the decision-maker takes as given.... The question then arises: how does the decision-maker know that what he takes as given is, in fact, true? Indeed, once we have entertained the idea that the 'given' may possibly turn out not to be true, then an even more worrying question arises: what happens if some evidence is generated which suggests that the 'given' is, in fact, not true? ... In other words, what happens if the decision-maker is *surprised*? It seems to us that the Bayesian approach rules out the possibility of surprise... This seems a rather alarming deficiency.

(Hey 1981:99; emphasis in original)

In fact, some authors are sceptical about the possibility of ever modelling 'learning' formally for a reasonably realistic world. Gordon and Hynes (1970:377) state that

a formal decision process for this learning is not possible in a world where the underlying stochastic process is not stable. It is true that the response sellers make to new *data* can, ex post, be described as the rational response to subjective prior distributions. However, since there is not sufficient information to accumulate relative frequencies, these subjective estimates will depend, in part at least, on 'judgment', will differ among rational persons confronted with the same measurable *data*, and will also alter from period to period in an unpredictable manner on the basis of information external to the individual's own sampling experience.⁷

However insuperable these difficulties in the way of a

disequilibrium theory of markets may seem, there are economists who have been working in this direction. Although most of their work has, for a variety of reasons, not been formalized mathematically, it constitutes a body of theory that is useful for the examination of the informational role of prices in the following chapters. The remainder of this chapter is therefore devoted to the description of the main aspects of this disequilibrium approach.

THE AUSTRIAN VIEW OF THE MARKET PROCESS

This disequilibrium view of markets is sometimes termed a ‘market-process’ approach and is associated with the writings of Ludwig von Mises, Friedrich A. Hayek, and Israel M. Kirzner, among others. These authors are often grouped as belonging to the modern Austrian school of economics.

Although Kirzner (1973, 1983, 1985a) has most explicitly and consistently articulated the Austrian view of markets, he has repeatedly stated that much of his work is a development of arguments that were more or less explicit in the writings of Mises and Hayek. This section will draw mostly from the writings of these two latter authors for a description of the market-process approach because, although on some of the topics there are differences—of greater or lesser significance—among contemporary Austrian economists, all acknowledge their intellectual debt to Mises and Hayek.

The knowledge problem

Mises’s 1920 critique of the socialist economy already displayed a concern with problems of information that did not appear so prominently in other economists’ work. His whole argument against the central planning of an economy was based on the lack of information that would make the planner’s attempt to improve on, or even mimic, the results of capitalism impossible. As Mises (1949:692) put it some years later, if one were to attribute omniscience to the planner,

then one could not help concluding that the infallible state was in a position to succeed in the conduct of production activities better than erring individuals. It would avoid all those errors that often

frustrate the actions of entrepreneurs and capitalists. There would no longer be malinvestment or squandering of scarce factors of production; wealth would multiply. The ‘anarchy’ of production appears wasteful when contrasted with the planning of the omniscient state.

But, Mises affirmed, to assume such an omniscience and infallibility on the part of the planner ‘would be nothing short of idiocy’ (1949:696). What had to be emphasized was ‘that the market is peopled by men who are not omniscient and have only a more or less defective knowledge of prevailing conditions’ (ibid.: 379). However, this was not the direction in which economics developed, as the assumption of perfect knowledge became an increasingly common ingredient of economic theorizing. This development was noted with growing dissatisfaction by Hayek, who dealt with these specific issues even more explicitly than Mises.

Hayek (1945:91) considered ‘the unavoidable imperfection of man’s knowledge and the consequent need for a process by which knowledge is constantly communicated and acquired...an essential part of the phenomena with which we have to deal’. For him

to assume all the knowledge to be given to a single mind in the same manner in which we assume it to be given to us as the explaining economists is to assume the problem away and to disregard everything that is important and significant in the real world.

(ibid.)

Like Mises, he stated that ‘if anyone really knew all about what economic theory calls the *data*, competition would indeed be a very wasteful method of securing adjustment to these facts’ (1968:179).

In other words, economists in the Austrian tradition emphasize that a crucial aspect of the economic problem is one of knowledge.⁸ The problem of what can be demanded and produced, of what resources and technologies are available, and so on, is one that must be solved before any optimizing procedure can be carried out. Hayek (1945:77–8) pointed out that

The economic problem of society is...not merely a problem of how to allocate ‘given’ resources—if ‘given’ is taken to mean given to a single mind which deliberately solves the problem set by these ‘*data*.’ It is rather a problem of how to secure the best use of

resources known to any of the members of society, for ends whose relative importance only these individuals know. Or, to put it briefly, it is a problem of the utilization of knowledge which is not given to anyone in its totality.

Furthermore, the problem is not only the use of existing dispersed information but also the discovery of knowledge which no individual in the economy yet has (Bartley 1985:31).

Prices and information

It is mainly Hayek who is nowadays, justly, credited with pointing out most explicitly the service that prices perform with respect to this knowledge problem. Until he wrote his classic 1945 article, 'The use of knowledge in society', most economists explained prices almost exclusively as useful incentives for adjusting individual plans to scarcity. According to Hayek, however, market prices also perform an informational role. The point can be conveyed by describing his often cited tin example. Assuming that one source of supply has disappeared, the resulting rise in the price of tin leads its consumers to economize on it and other suppliers to increase their output. The price is not only an incentive for this economizing to occur: the adjustment also happens without most of the people who are carrying out the necessary adjustments 'knowing anything at all about the original cause of these changes' (Hayek 1945:86). The real function of prices, Hayek said, is to communicate information.

Competition as a discovery procedure

Although, as shown in later chapters, Hayek's point has generally been interpreted in terms of an efficient use of existing dispersed information, and although this may have been his central concern in his 1945 article, in later years he came to see the problem also as one of discovering previously unknown information. According to Hayek, what has to be discovered is, among other things, 'which goods are scarce goods, or which things are goods' (1968:181), the prices, quantities or qualities of the goods to be produced and sold, the lowest cost at which the commodity can be produced, and even 'the most effective size of the individual firm' (1979:78). Contrasting with the

impression sometimes given that the only fact to discover is the appropriate configuration of prices and quantities, for Hayek (1946:101) the solution of the economic problem had become ‘a voyage of exploration into the unknown, an attempt to discover new ways of doing things better than they have been done before’. He had come to view prices as carrying out two informational tasks: he described the price system as

a sort of discovery procedure which both makes the utilization of more facts possible than any other known system, and which provides the incentive for constant discovery of new facts which improve adaptation to the ever-changing circumstances of the world in which we live.

(1968:236)

It is clear that, for him, the problem to be solved by an economic system

is not merely a task of utilizing information about particular concrete facts which the individuals already possess, but one of using their abilities of discovering such facts as will be relevant to their purposes in the particular situation.

(1979:190, n.)

If competition, for market-process economists, is a ‘discovery procedure’,⁹ at least two questions can be asked: (1) How and why does this discovery take place in a market economy? (2) What is the role of prices in this procedure? These questions are tackled in the following sections.

Entrepreneurship and profits

The entrepreneur

As long as useful information regarding economic opportunities remains undiscovered, better courses of action will be left unpursued by individuals in an economy. This is the same as saying that as long as there is ignorance in the economy¹⁰ there will be profit opportunities: there will be trades available at more convenient terms, there will be arbitrage opportunities, there will be different goods to

produce, better technologies to use, more efficient organizational forms to adopt, and so on. The individuals who, spurred by the profit opportunities, discover and adopt the better courses of action the modern Austrians term the entrepreneurs.¹¹

The term 'entrepreneur' refers to those individuals who are, to use Kirzner's terminology, 'alert' to the existence of previously unknown profit opportunities. ('Entrepreneurs' are, of course, an abstraction of economic theory, such as 'resource owners' and 'consumers'. Strictly speaking, all individuals are entrepreneurial to some extent.) Entrepreneurship is, according to Mises, understanding 'there is a discrepancy between what is done and what could be done' (1949:336), being 'shrewd', quick of apprehension and far-sighted (328). To say that they possess this ability is not to say that these agents have perfect knowledge: their advantage is only a relative one. They

earn profit not because they are clever in performing their tasks, but because they are more clever or less clumsy than other people are. They are not infallible and often blunder.

(1974:114)

The Austrian entrepreneur may be said to include the Robbinsian maximizer, but the former agent also has the ability to perceive the ends-means framework, and to revise it when his plans are disappointed. In this view, this perceptive ability constitutes the entrepreneurial component of the agent's action. Given he has (correctly or incorrectly) perceived the facts of the market, this agent also optimizes.¹² But the act of perception is crucial; the remainder is merely a problem of mechanical computation (in the sense that, in principle, a computer could solve it).

It should be noticed that the Robbinsian approach does not make the entrepreneurial task unnecessary. Instead, it presupposes implicitly that it has been done elsewhere, and that it has been done perfectly, allowing the agent to optimize with respect to correct information.

Finally, it is worth pointing out that the entrepreneur may be difficult to incorporate into standard economic theory as it is currently conceived because, whereas Robbinsian maximizing lends itself to being formalized mathematically, entrepreneurial behaviour seems much less deterministic and is much less understood. In fact, it may never be amenable to such a representation, except under very restrictive simplifications.¹³ These difficulties may help explain the relative neglect of entrepreneurship by standard theory.

Profits as incentives

The role of the price system in the entrepreneurial discovery procedure is the provision of profit opportunities that spur the discovery of better alternatives. As long as some ignorance persists in the market, there is a constellation of disequilibrium prices. By definition, disequilibrium prices provide pecuniary profit opportunities. And these opportunities attract the attention of alert entrepreneurs.

Of course, as human action in reality cannot be instantaneous, the prices here referred to are not current prices but, rather, expected prices (even if they only refer to the following minute).¹⁴ It is thus, for example, that profit opportunities can appear for goods that have not been traded, or even produced, previously at all. But the important point is that it is prices that translate the situations of ignorance in the market into profit opportunities and thus provide the incentive for their elimination.

Economic theory has generally interpreted profits as incentives to action in a market economy. But, with their perception of the knowledge problem, market-process economists make a distinction between two types of incentives, a distinction that sometimes turns out to be important.

Incentives, in the standard sense, are rewards that encourage the agents' adoption of certain courses of action already perceived and known by them. Profits, from such a point of view, make it worth while for the agent to engage in these actions. These actions, however, were known to him before, but were not worth the costs involved without the reward. This type of incentive is undoubtedly important, but it is not the only one. Profit incentives, in the sense emphasized by market-process economists, are rewards that encourage the discovery of 'opportunities that have until now been perceived by no one at all' (Kirzner 1985a:29; emphasis removed).¹⁵ If this second type of incentive were to be absent, the problem, as Lavoie has put it, would not be

that people will be insufficiently motivated to do the right things but, more fundamentally, that they will not know what the right things to do *are*, even if they passionately wanted to do them.

(1985b:21; emphasis in original).

The differences between these types of incentives turn out to be

important in discussions concerning the relative merits of different economic systems.

This distinction also exemplifies the possibilities of confusion caused by the fact that the market-process approach often attaches to terms and concepts meanings which are slightly different from those of standard economics. To reduce the risk of such confusion happening here, some ambiguous notions will be clarified in the remaining sections of this chapter. In particular, what role market-process economists attribute to the notion of equilibrium in economic theory and some differences in their approach to ignorance will be considered. Also, brief mention will be made of some of their reservations regarding the normative standards often used in economics.

The role of equilibrium

The perfect (or sometimes only optimal) knowledge feature of equilibrium is the reason why market-process economists do not consider equilibrium an adequate framework for studying how an economic system solves the knowledge problem. Their complaint is that equilibrium theory assumes

that the *data* for the different individuals are fully adjusted to each other, while the problem which requires explanation is the nature of the process by which the *data* are thus adjusted.

(Hayek 1946:94)

All this was also clear to Mises several years earlier when he stated, with slightly older terminology, that

under stationary conditions there no longer exists a problem for economic calculation to solve. The essential function of economic calculation has by hypothesis already been performed.

(Mises 1936:120)

In such a situation there is no room, or need, for entrepreneurial behaviour (Mises 1949:702). Others have remarked that the entrepreneur has ‘virtually disappeared from the theoretical literature’ (Baumol 1968:64), and that the ‘received theory of competition gives the impression that there is no need for entrepreneurship’ (Leibenstein 1968:72). The almost exclusive concentration of most economics on

equilibrium states may provide a good explanation for this neglect. It certainly helps to explain why the market-process view of competition differs from the standard one. For Hayek ‘what the theory of perfect competition discusses has little claim to be called “competition” at all’ because it ‘throughout assumes that state of affairs already to exist which, according to the truer view of the older theory, the process of competition tends to bring about (or to approximate)’ ((1946:92).¹⁶ This confusion is due to

the absurdity of the usual procedure of starting the analysis with a situation in which all the facts are supposed to be known. This is a state of affairs which economic theory curiously calls ‘perfect competition.’ It leaves no room whatever for the activity called competition, which is presumed to have already done its task.

(Hayek 1968:182)

It is because of these differences that Austrian economists claim to view the market as a process and that they attribute to equilibrium economists the view of the market as a state. For the Austrians real-world market activity is a disequilibrium process in which the facts are not fully known, and in which, therefore, profit opportunities abound. Entrepreneurs are constantly engaged in discovering these opportunities and correcting their mistakes, without equilibrium ever being achieved. This, of course, raises the question of what role, if any, is attributed to the notion of equilibrium in a market-process approach.

The traditional market-process approach does not deny the usefulness of equilibrium for economic theorizing, although there have been some recent moves in this direction. However, equilibrium is only an ‘auxiliary tool’, an ‘imaginary construction’ that is ‘an indispensable tool of economic reasoning’ but that ‘has...no counterpart in reality’ (Mises 1949:701–2). Mises was aware that this construction ‘cannot even be thought through consistently to its ultimate logical consequences’ but did not feel this detracts from its usefulness. He argued that the equilibrium state is useful as a ‘negative description’ of the market process: it is a state that would suspend the motion of the market process, a state in which there would no longer be the restlessness and activity perceived as characteristic of the market (ibid.: 355).

Although it may seem paradoxical, something is learnt about the market process by imagining a state in which it would have no function. But, for Mises, this negative description is ‘merely

auxiliary': the fundamental part of economics is the 'positive description', which shows how the price discrepancies that characterize disequilibrium provide profit opportunities for entrepreneurs who, in the process of successfully exploiting them, improve the co-ordination of the economy, without, in reality at least, ever reaching equilibrium. From such a perspective, the concentration on the study of equilibrium states distracts attention from the crucial problem. Hence Mises's provocative statements that 'the mathematical description of various states of equilibrium is mere play. The problem is the analysis of the market process' (Mises 1949:356), and that the 'problems of process analysis' are 'the only economic problems that matter' (ibid.). In this view, the equilibrium approach has led not only to a neglect but, even worse, also to a misunderstanding of crucial aspects of the market.¹⁷

For thinkers such as Mises and Hayek an equilibrium state is not only a useful, though unrealistic, imaginary construction. For them it is also a description of a state towards which real markets 'tend'. This has become an increasingly controversial idea among market-process economists,¹⁸ and will be only briefly described here. For Mises, it is the activity of entrepreneurs, bent on discovering and exploiting profit opportunities, that produces a 'tendency' towards equilibrium (which is why, although markets are viewed as always in disequilibrium, they are at the same time not viewed as chaotic). This tendency would be fully realized were it not that the facts are constantly changing in reality (Mises 1949: 337–8).¹⁹ Markets do not achieve a state of co-ordination; instead, market competition is a co-ordinating process. It is this entrepreneurial process that is deemed worthy of attention, and not a state in which people are already assumed to know all that is worth knowing.

Ignorance and the economics of information

Starting perhaps with George Stigler's 1961 article, the problems of information in an economy have attracted much attention. Since that essay, the field known as the economics of information has grown very rapidly. As Machlup (1984:13) put it with characteristic thoroughness, this new specialization studies

the complexities that may arise from the fact that information, new or old, may be inordinately uncertain, incomplete, partial, biased,

misleading, costly, available to some but not to others, or giving rise to expectations, warranted or unwarranted, of various future developments.

Most, if not all, work in this field, however, continues to be done primarily within an equilibrium framework. Knowledge is treated as a costly commodity that, like all other commodities, must be economized on. Therefore, the resulting equilibria entail not perfect but only optimal knowledge: agents will have deliberately uneradicated ignorance. Such ignorance remains uneradicated because the benefits of additional information do not compensate for the costs of acquiring it. However, this approach has some problems, or at least limitations, in that to make such decisions correctly—as must be the case in equilibrium—agents must know beforehand, among other things, what they are ignorant of and the costs and benefits of the knowledge they could acquire; that is, *they must know what it is they do not know*. This is obviously still a strong assumption—at best only slightly weaker than the assumption of perfect knowledge. The attempt to treat this required initial knowledge also as deliberately acquired knowledge leads to an infinitely regressing argument. This infinite regress means that the origin of some initial knowledge is always left unexplained by the theory.²⁰

The market-process approach, in Kirzner's work, distinguishes this optimal ignorance from what can be termed 'sheer' ignorance.²¹ For the economics of information, ignorance seems to signify almost exclusively the result of a deliberate decision by an optimizing individual, after weighing the costs and benefits of an additional 'unit', not to acquire further information. In contrast, when market-process economists refer to ignorance and error and to the need for a discovery process, they have in mind pure error. In this case agents have failed to fully perceive the alternatives available to them not because they have correctly judged the effort unprofitable, but because they either have not noticed the alternatives or have not noticed the profitability of searching for them.²² This ignorance is due to the individual's lack of alertness and not to his optimal response to costly information. Of course, this emphasis on sheer ignorance does not imply the non-existence or lack of importance of optimal ignorance. The point is only that sheer ignorance should not be ignored or assumed away in economic analysis. Recognition of its existence is important for a proper understanding of market processes.

Some writers, notably Hayek and Lavoie, have emphasized a different difficulty regarding knowledge. Their concern is that much useful knowledge is inarticulate or inarticulable (or ‘tacit’), an idea influenced by the work of Michael Polanyi.²³ In their argument, the market is a process for discovering and conveying such knowledge.²⁴ Although this approach seems perfectly compatible with Kirzner’s, a comparison of both approaches will not be attempted here.

Not all economists have noticed the limitations of the economics of information for an understanding of market processes. Because of their interpretation of ignorance exclusively in terms of costly information, some still believe that

although the problem of decentralized co-ordination of economic activity in an environment of transaction and information costs is complicated, there is certainly no reason why maximization techniques cannot and should not be used.... We just must assume a richer informational background under which individual maximizing decisions take place.

(Klein 1975:1307–8)

A disequilibrium view of normative economics

Another implication of the market-process approach is that the usual welfare standards—such as Pareto optimality—that operate (implicitly) from the point of view of an omniscient observer (or planner) become irrelevant for evaluating actual markets once the latter are viewed as always in disequilibrium. From such a point of view, to use Fisher’s words,

welfare comparisons of equilibria would be largely irrelevant since what would matter would be the comparison of the relatively ‘transient’ behavior of alternative systems including alternative forms of market organization.²⁵

(Fisher 1983:9)

Fisher’s position, it should be noted, is not exactly the same as that of market-process economists. For the latter, markets never achieve full equilibrium, so that the market process is certainly not seen as ‘relatively transient behavior’.

Although the above citation puts the problem in terms of equilibrium and disequilibrium, which is, of course, perfectly acceptable, it is important to understand that what this really means is that to judge reality from an omniscient perspective becomes irrelevant once it is seen that there is no human being in possession of such omniscience. Hayek has expressed the complaint clearly:

To use as a standard by which we measure the actual achievement of competition the hypothetical arrangements made by an omniscient dictator comes naturally to the economist whose analysis must proceed on the fictitious assumption that he knows all the facts which determine the order of the market. But it does not provide us with a valid test which can meaningfully be applied to the achievements of practical policy. The test should not be the degree of approach towards an unachievable result, but should be whether the results of a given policy exceed or fall short of the results of other available procedures.

That standard for judging the performance of competition, in other words, must not be the arrangements which would be made by somebody who had complete knowledge of all the facts.

(Hayek 1979:67)

This market-process position must be carefully distinguished from the important criticisms of standard welfare analysis offered by Harold Demsetz. In a 1969 article Demsetz criticized ‘much public policy economics’ for implicitly presenting ‘the relevant choice as between an ideal norm and an existing “imperfect” institutional arrangement’. According to him,

This nirvana approach differs considerably from a comparative institution approach in which the relevant choice is between alternative real institutional arrangements. In practice, those who adopt the nirvana viewpoint seek to discover discrepancies between the ideal and the real and if discrepancies are found, they deduce that the real is inefficient. Users of the comparative institution approach attempt to assess which alternative real institutional arrangement seems best able to cope with the economic problem; practitioners of this approach may use an ideal norm to provide standards from which divergences are assessed for

all practical alternatives of interest and select as efficient that alternative which seems most likely to minimize the divergence.

(Demsetz 1969:1)

Demsetz identifies several logical fallacies frequently committed in standard welfare analysis, two of which are of interest here:

1. The habit of ‘invoking an unexamined alternative’ as soon as a discrepancy between the ideal and the real is detected. This, which he terms ‘the grass is always greener fallacy’ (ibid.), usually takes the form of calling for government intervention as soon as any ‘market failure’ is found. The fallacy consists in ignoring that showing that actual markets are not as efficient as ideal ones is not by itself proof that regulating or eliminating such markets, or even eliminating the whole market system, will lead to greater efficiency.²⁶

2. The analyst often judges some market outcomes non-optimal because certain costs are neglected by him. This, which Demsetz calls ‘the fallacy of the free lunch’ (Demsetz 1969:4), frequently appears with respect to the problems supposedly caused by the ‘absence of complete markets’ (neglecting that the cost of establishing certain markets may exceed the benefits), and by incomplete information and incomplete risk-shifting (neglecting their costliness). Equating this type of incompleteness to non-optimality is, according to Demsetz, equivalent to denying ‘scarcity is relevant to optimality’, obviously a difficult position for an economist to hold.

With respect to this second fallacy, Demsetz’s conclusion at the time was that

modern analysis has yet to describe efficiency in a world where indivisibilities are present and knowledge is costly to produce. To say that private enterprise is inefficient because indivisibilities and imperfect knowledge are part of life, or because people are susceptible to the human weaknesses subsumed in the term moral hazards, or because marketing commodity-options is not costless, or because persons are risk-averse, is to say little more than that the competitive equilibrium would be different if these were not the facts of life. But, if they are the facts of life, if, that is, they cannot be erased from life at zero cost, then truly efficient institutions will yield different long-run equilibrium conditions than those now used to describe the ideal norm.
(ibid.: 19)

Demsetz’s is, undoubtedly, an important contribution. However, it

should be noticed that it is a contribution that does not require the abandonment of an equilibrium approach to economics. What he can be read as saying is that equilibria must be constructed without omitting relevant costs, and that they must not be compared to non-existent ideal alternatives. Significant as it is, nothing in this position suggests any need for the study of disequilibrium market processes, nor are its insights derived from adopting such a perspective.

The adoption of a market-process perspective, it can be argued, yields an additional insight, one that will only be dealt with briefly in this book. To view the world in disequilibrium terms, as this perspective does, leads to an acceptance of the possible existence of many of the problems and inefficiencies pointed out by critics of the market, but also suggests the as yet undeveloped possibility of a normative standard that is not the Pareto-optimal state and that does not focus so exclusively on allocational outcomes of the market process. This standard would try to establish which institutional arrangements are more likely to stimulate the discovery of, among other things, these problems and inefficiencies and of possible solutions to them.

How does this differ from Demsetz's position? Briefly, in the entrepreneurial or market-process view, the market system can be said to be, at any time, chock-full of regrettable inefficiencies and mistakes (many of which will be in the process of being entrepreneurially discovered and corrected), inefficiencies and mistakes that cannot be explained away by resorting to neglected costs but that are due only to ignorance.²⁷ The Demsetzian perspective, by contrast, risks degenerating into a Panglossian view that states that whatever is is best (i.e. is perfectly efficient once all the relevant costs are taken into consideration). The entrepreneurial perspective can view the market process as imperfect, as seen through the eyes of some hypothetical omniscient being. But, from such a perspective, it is not reasonable simply to assume the existence of a non-market entity in possession of all the knowledge necessary for solving these imperfections, and thus to conclude that the market is inefficient. The analysis should, more appropriately, consider which social arrangement has means for the discovery of such knowledge. Viewed in the light of such a standard, the market, as a preliminary conclusion, has at least one major advantage over other systems: through its translation of countless inefficiencies and mistakes into pecuniary profit opportunities, it alone seems to have the capacity to mobilize and awaken the entrepreneurial alertness of

market participants. It thus promotes the discovery of these inefficiencies and of their solutions.²⁸

Although market-process economists may be accused of not having fully developed an alternative normative standard, they certainly do not attribute to free markets the achievement of anything like Pareto optimality, as has sometimes been suggested. Mises (1949:705), for example, states that

We do not assert that the capitalist mode of economic calculation guarantees the absolutely best solution of the allocation of factors of production.... What the operation of a market not sabotaged by the interference of compulsion and coercion can bring about is merely the best solution accessible to the human mind under the given state of technological knowledge and the intellectual abilities of the age's shrewdest men. As soon as any man discovers a discrepancy between the real state of production and a realizable better state, the profit motive pushes him toward the utmost effort to realize his plans. The sale of his products will show whether he was right or wrong in his anticipations.

And, again,

There are [in the market economy] disadvantages caused by inadequate foresight. It would be a universal boon if every man and all the members of the market society would always foresee future conditions correctly and in time and act accordingly. If this were the case, retrospection would establish that no particle of capital and labor was wasted for the satisfaction of wants which now are considered as less urgent than some other unsatisfied wants. However, man is not omniscient.

(*ibid.*: 665)

The development of a more appropriate normative standard for judging disequilibrium market processes is only one among several research projects which still remain to be done. Even so, market-process theory in its present state serves to shed light on the informational role of prices and to examine other theoretical approaches to this matter, tasks to be undertaken in the following chapters.

Chapter 3

Equilibrium prices and information

The previous chapter mentioned the almost explosive growth of the economics of information. This explosion has led to a renewed interest in the writings on related issues of some market-process economists, particularly Hayek's. One of his themes that has gained wide acceptance in economics is that market prices, in reflecting relative scarcities more or less faithfully, perform some type of informative function. According to Kreps (1988:114), 'the notion that prices contain and convey information is standard doctrine among economists'.¹ However, because most work in the economics of information has been done within an equilibrium framework, Hayek's arguments have not always been fully understood and some of their implications have been missed.

This chapter will consider interpretations of Hayek's arguments regarding the informational role of prices, all of which have in common an emphasis on equilibrium. The work of Sanford J. Grossman and Joseph E. Stiglitz will serve as an organizing principle, although the remarks that follow are also applicable to other authors. Some of these authors will be mentioned when their arguments add something significant to the issue under consideration.

Many economists interpret Hayek as saying that prices are sufficient statistics, in the sense that they are signals that convey all relevant information to traders at low cost. Grossman and Stiglitz have, both singly and in collaboration, dealt quite critically with this version of Hayek's arguments.² They claim to show that his argument is not correct for situations with costly information, where it is most important, and that prices are 'informationally inefficient'. With regard to this last point, Streit (1984:392) argues that Grossman and Stiglitz's proof also 'serves as an argument against Fama's widely used proposition that in efficient speculative markets at any time

prices fully reflect all available information'.³ The first task will be to summarize their arguments.⁴

'HAYEK'S' ARGUMENT

Grossman (1976:574) says that

Hayek (1945) argues that the essence of a competitive price system is that when a commodity becomes scarce its price rises and this induces people to consume less of the commodity and to invest more in the production of the commodity. Individuals need not know why the price has risen, the fact that there is a higher price induces them to counteract the scarcity in an efficient way.

This refers to the transmission of knowledge from informed to uninformed individuals. Grossman (1981:556) also finds in Hayek the 'idea that a fundamental role of competitive prices is the aggregation of information'. Hayek's view of dispersed information (or 'diverse' information, as Grossman and Stiglitz call it) is claimed to be that

each trader knows something about his own customers and neighbourhood. No one knows everything about the economy. Each individual's little piece of information gets aggregated and transmitted to others via trading. The final competitive allocations are as if an invisible hand with all the economy's information allocated resources. However, a planner without all of that information could not have done as well.⁵

(Grossman 1981:555)

(The distinction Grossman and Stiglitz make between the role of prices as transmitters and their role as aggregators of information is not important for present purposes.)

Grossman and Stiglitz find these ideas provocative but feel that Hayek presented them in a vague fashion. They therefore try to present his arguments more 'rigorously'. After doing so, they find Hayek's conclusions difficult to sustain.

A brief digression. Before proceeding with the argument, it is necessary to mention another body of literature, pioneered by the work of Leonid Hurwicz, that has interpreted Hayek's arguments similarly.⁶ The claim here is to have shown that, as long as the usual

assumptions that Hurwicz describes as belonging to a ‘classical environment’ hold (i.e. absence of externalities, public goods, increasing returns, indivisibilities, and so on), the perfectly competitive price mechanism is ‘informationally best’. It is ‘informationally best’ in the sense that ‘it uses the minimum number of variables for transmitting information between economic units’ (Hurwicz 1984:421).⁷ But this conclusion, attributed to Hayek, is found to be invalid for ‘non-classical’ environments that are believed to be more realistic.

This literature then attempts the theoretical design of ‘informationally decentralized’ resource allocation mechanisms combining planning and markets that may achieve more efficient results than perfectly competitive markets for such ‘non-classical’ environments. It takes as given ‘that there is initial dispersion of information, with each economic unit processing only partial knowledge of the environment’ and

that it is impossible to transfer this information to other units in such a way that at some stage of the process some one unit would be, through messages received from others, in possession of complete information concerning [the environment] or concerning the proposed actions of all the other units,

i.e. ‘that it is impossible through communication to centralize dispersed information’ (Hurwicz 1972:301; emphasis removed).⁸

What is relevant for the purposes of this book, however, is these authors’ interpretation of Hayek’s argument. It turns out that, because it is similar to that of other economists considered here, this particular approach need not be examined separately. Many of the arguments in this chapter will, with some adjustments, be applicable to it.

GROSSMAN AND STIGLITZ’S ARGUMENT

The ‘informational efficiency’ of prices

Grossman and Stiglitz (1976:246) point out, as an introduction to their work, that

although the price system is conventionally praised as an efficient way of transmitting the information required to arrive at a Pareto

optimal allocation of resources, the context in which the price system is usually discussed is not one in which the informational efficiency of the price system can be properly evaluated. Questions of how the price system leads the economy to respond to a new situation, how it conveys information from informed individuals to uninformed individuals, and how it aggregates the different information of different individuals, are never directly attacked.

Theirs is an attempt to remedy this perceived deficiency. As a result, Grossman claims to have ‘formalized Hayek’s contention that prices are aggregators of information’ and to have proved that ‘if prices are sufficient statistics, the competitive economy where traders have diverse information generates allocations that cannot be improved upon by a central planner with all the information’ (ibid.: 252). However, the ‘competitive economy’ is usually assumed to have costless information, and Grossman and Stiglitz think that the informational role of prices is of interest only when information is costly.⁹ Like most information-as-a-commodity economists, they seem to believe such a context is the only alternative to the full-knowledge situation found in the perfectly competitive model. But it is precisely when information is costly, they argue, that prices cannot be sufficient statistics and that, therefore, Hayek’s arguments about the achievements of the price system run into trouble. They consider different possible cases:

Fully informative prices

If, as they interpret Hayek to be saying, prices fully reflect all available information, that is, if the ‘price system is a perfect aggregator of information’, no equilibrium exists, according to Grossman (1976:574), when information is costly. ‘A perfect competitive market will break down because no equilibrium exists where information collectors earn a return on their information, and no equilibrium exists where no one collects information.’ In other words, if market prices reflect all necessary information, there is no incentive for anyone to engage in the costly activity of acquiring it because each trader could do equally well by observing only the price, instead of purchasing the information (ibid.: 581–2). But then, of course, prices will not reflect information because it is not rewarding for anybody to collect it.

On the other hand, Grossman and Stiglitz argue, the situation in which no one collects information is not an equilibrium either: in this case a given individual will find it profitable to gather costly information because, owing to the price-taking assumption of the perfectly competitive model, he believes his activity will not affect the equilibrium price. (Therefore, his information will not become freely available to other traders, allowing him an advantage.) However, as soon as many individuals start collecting information, the equilibrium price is affected and aggregates their information perfectly. This, again, provides an incentive for individuals to stop gathering costly information and to obtain it costlessly from the price.¹⁰ As a result, Grossman argues, a breakdown of markets occurs ‘when price systems reveal too much information’ (ibid.: 574).¹¹

The source of the paradoxes of the price system described by Grossman and Stiglitz is a problem of externalities. As Grossman (1981:557) has pointed out, when prices are taken as *sources* of information they create ‘an externality by which a given individual’s information gets transmitted to all other traders’. In the case of costly information, uninformed traders will free-ride on the information-gathering activity of informed traders. The latter, unable to reap the full rewards of their (costly) activity, will tend, as in standard externality analysis, to seek information in a non-optimal fashion.

‘Noisy’ prices

Another possibility is that prices do not aggregate information perfectly (i.e. they are, to use Grossman and Stiglitz’s terminology, ‘noisy’). As exemplified below, it is not possible for individuals to obtain all the necessary information from such prices. In this case, Grossman and Stiglitz argue, it may become worth while for traders to engage in costly information-gathering activities, and an equilibrium is possible. Still, in this case—the only one capable of sustaining an equilibrium—Hayek’s argument, as interpreted by Grossman and Stiglitz, breaks down: when the price system is ‘noisy’, ‘some traders want very much to know why prices are, for example, unusually high. It is not enough for traders to observe only prices’ (Grossman 1976:585). Hayek’s argument that a higher price is sufficient to lead agents to react in an efficient way appears to be wrong.

An example. The two cases considered by Grossman and Stiglitz can perhaps be made clearer by means of an example they provide (Grossman and Stiglitz 1976:246–7).¹² Suppose there is a risky security with an uncertain return but that some information about this return can be obtained at a cost. The demand for this security by informed traders will then depend on its current price and on their acquired information about its return. The demand by uninformed individuals will depend only on its price. The example assumes there is a fixed stock of the security, and no other source of changes in its price. Uninformed individuals will then be able to infer costlessly from, say, a higher price that there is information (possessed by informed individuals) about a higher return. This price system, Grossman and Stiglitz argue, conveys information from informed to uninformed individuals.

The situation with a fully informative price is not an equilibrium because no trader finds it necessary to spend resources seeking information. But then the price will not be informative! The situation with uninformative prices is not an equilibrium either: in this situation it is profitable for any trader to seek information about the return on the security. Each trader thinks this will give him an advantage over others because he believes his informed activity will not modify the price and convey his information to them. However, when many traders behave in this way the price starts to reflect their information, making their costly information-gathering unattractive, as at the beginning of the example. Therefore no equilibrium exists.

The example is then modified slightly by introducing some randomness, for example, in the stock of the risky security. Then its price may be higher because the informed individuals have increased their demand in the expectation of a higher return, or because the supply has fallen. ‘The price system conveys some information, but does not transmit all the information from the informed to the uninformed: on average, when the price is high, the return is high...but the price is a *noisy signal*’ (Grossman and Stiglitz 1976:247; emphasis in original). It is not the same to observe the price as to obtain information about the return on the security. Under these circumstances there is a return to the information-gathering activity, and there may exist an equilibrium in which an optimal amount takes place. (Grossman and Stiglitz present a case in which there is an optimal fraction of the population becoming informed.)

This only situation capable of sustaining an equilibrium, Grossman and Stiglitz argue, violates Hayek's description of the price system because these prices are not fully informative: some individuals make economic decisions without being fully informed. (And Hayek is believed to be arguing that in a competitive market individuals achieve allocations identical to those that could be achieved by an invisible hand with all the information in the economy.)¹³

Grossman and Stiglitz's argument applies not only to the transmission of information through prices. They also make a similar point for the aggregation of information by prices in market situations in which individuals have different information: they show that, if the price aggregated their information perfectly, the individuals would no longer base their demands on their private information but only on the price. But then a paradox occurs: 'If all traders ignore their own information how does the information get into the price?' (Grossman 1976:582). Again, Grossman and Stiglitz show, the paradox might not appear if the price system were 'noisy,' i.e., if it failed to aggregate information perfectly. However, this condition for the existence of an equilibrium would (once more) violate Hayek's description of the achievements of the price system.

The case of 'costless' information

Although Grossman and Stiglitz devote very little space to it, they do mention the case of what they describe as costless information. It is mentioned here because it might seem to contradict the description of these authors as information-as-a-commodity economists: this latter approach tends to suggest that, without costly information, there would be no informational problem. It turns out that what Grossman means by situations of costless information are situations in which there is information that is costly for any arbitrary individual to acquire but that was costless to its possessor. (This is not, then, a pure case of costless information.) He gives the example of a shoe seller who, in the course of his business, 'learns a little about the demand for shoes, for free, from his customers'. This would be one of those frequent cases in which information is 'produced complementary with the production, and distribution of other commodities'. According to Grossman (1981:556),

In this case, when [the market price] is a sufficient statistic, no disincentives to acquire [the private information] are generated (since it is free), but yet every trader is being given information by [the price] which he could not obtain otherwise, except at great cost. This is the sense in which a Rational Expectations equilibrium in a complete market captures Hayek's idea that a fundamental role of competitive prices is the aggregation of information.

Grossman obviously thinks this represents another instance of Hayek's argument. Although the critical comments below will also apply to this case, it should perhaps be pointed out that the example is not totally convincing even within an equilibrium framework.¹⁴

Welfare appraisal and the planning debate

As mentioned before, Hayek (like Mises before him) saw the informational characteristics of prices as providing arguments for the market system in the socialist calculation debate of the 1930s. Obviously, then, a successful questioning of the informational efficiency of prices is likely to be relevant to the comparison between alternative economic systems. Therefore, before proceeding to analyse their arguments critically, it is interesting to note briefly what Grossman and Stiglitz have to say about the normative appraisal of the price system, and about the implications of this for the comparison of market systems with centrally planned regimes.

Regarding the normative issue, Grossman and Stiglitz find the evaluation of the efficiency of the market situations considered above to be 'a subtle and difficult question. It is not obvious what the appropriate comparisons ought to be' (Grossman and Stiglitz 1976:251). They say that 'although it is easy to show that the market solution is not, in general, efficient, it is difficult to ascertain whether there is too little or too much information acquisition'.¹⁵

An evaluative approach they consider briefly is the comparison of the decentralized process of the market with a centralized process. This, they argue, 'was the central question of the Lange-Lerner-Taylor-Hayek debate' (ibid.: 251–2). The result that only markets with noise will exist in equilibrium and, therefore, that prices will not be perfect aggregators of information suggests to Grossman and

Stiglitz that ‘a central planner with all the information can improve on the competitive equilibrium’. Therefore, they conclude,

the Lange-Lerner-Taylor-Hayek debate comes down to the fundamental distinction between economies where: (1) prices and hence allocations are the outcome of a competitive arbitrage process which will, of necessity, be imperfect because of the costs of arbitrage..., and (2) economies where prices and hence allocations are the outcome of a centralized allocative mechanism which will, of necessity, be imperfect because of the costs of monitoring bureaucrats.

The advantages of the price system, unlike what Hayek seemed to believe, are not so unambiguous, a comparison of the costs of each system being required. In fact, they argue, a determination of the relative efficiency of these systems is not possible ‘without more knowledge of the costs of operating a centralized informational mechanism’ (ibid.).

Having described Grossman and Stiglitz’s arguments, this chapter will now proceed to analyse them critically. Their approach to the informational role of prices raises several interesting issues, so, to avoid making the argument too confusing, it will be presented in separate subsections.

A MARKET-PROCESS PERSPECTIVE

Equilibrium versus disequilibrium

There is an important difference in these theoretical approaches to the analysis of prices and information. Grossman and Stiglitz analyse Hayek’s ideas from an equilibrium perspective. At least in this respect, then, their approach differs from that of modern Austrian economists. For the latter, it was argued, market prices are always in disequilibrium, which is when they perform their main informational role. Market prices, from this perspective, are not merely an efficient way of aggregating and transmitting information already known by someone but, more important, they are the stimulators of a discovery process. The reason why this difference of approach is important is that there is no room for such an informational role in a framework, such as Grossman and Stiglitz’s, that confines itself to the analysis of

equilibrium. As already pointed out, the ignorance (i.e. ‘sheer’, as opposed to ‘optimal’, ignorance) that is dissipated through (successful) entrepreneurial discovery cannot exist in equilibrium—equilibrium excludes it by definition.

In spite of this fundamental difference, Grossman and Stiglitz’s argument leads to some insights and some clarifications which are important also for a disequilibrium perspective. (Of course, their arguments should be of particular interest—or concern—to those economists who have agreed with Hayek’s position regarding the informational characteristics of prices while interpreting it from an equilibrium perspective.)

Grossman and Stiglitz analyse prices from the perspective of the economics of information, a perspective in which information-gathering is costly. For example, they consider arbitrage activity to be costly and, therefore, point out that it requires a reward if it is to be undertaken.¹⁶ For them, arbitrage is an equilibrium activity.¹⁷ They then notice that perfectly arbitrated prices, in the usual sense, leave no margin to reward this activity and therefore conclude that ‘the assumptions that all markets, including that for information, are always in equilibrium and always perfectly arbitrated are inconsistent when arbitrage is costly’ (Grossman and Stiglitz 1980:393). It seems clear to Grossman and Stiglitz that it is the assumption that all markets are always perfectly arbitrated that must be given up and, implicitly, not the other one. When this is done, equilibrium prices can no longer be described as perfect transmitters or aggregators of information as, they argue, Hayek does.

While Grossman and Stiglitz think they have found a deficiency in Hayek’s argument through their equilibrium modelling, other authors draw very different conclusions from their work. Streit (1984:393), for example, claims that their argument ‘is revealing because it throws some light on the inadequacy of abstracting from time and on the related limitations of temporary equilibrium analysis’. For example, the information externality that Grossman and Stiglitz find seems to be largely due to their exclusion of disequilibrium trading from their model. According to Streit,

the informational externality created via trading can only endanger the existence of a speculative market if an informed trader has no chance to trade before his new informative situation has become general.... [A] trader cannot avoid announcing his divergent beliefs if he wants to trade correspondingly. However,

the smaller the market weight of his transaction demand and the less he has to share the informational advantage with others, the greater his chance to trade at a price which is false in the light of his beliefs and to re-trade at a profit if his beliefs turn out to be correct.

(ibid.: 393–4)

In a similar vein, Radner (1979:656), commenting on the type of paradox that leads Grossman and Stiglitz to assert the non-existence of informationally efficient equilibrium, says that ‘to examine this question more carefully one needs a model that reflects the dynamics of market adjustment and price formation’.

In the example above of a risky security with an uncertain return, assuming the stock fixed, any information purchased by some (informed) traders about, for example, a higher return was conveyed to other (uninformed) traders, for free, through the change in the price of the security. As the model confines itself to equilibrium situations, the price of the security is assumed to jump instantaneously to its new, higher level. This, as Streit argues, does not allow the informed traders to profit from their expensive information; not a realistic result. It also leaves unexplained the process by which the new information gets incorporated into the price: one would expect the price increase to occur as a consequence of the profitable trading of the informed individuals. (The size of the profits they make before other traders become aware of the opportunity will depend, of course, on the trading strategy they adopt.) This has led to the comment that

Grossman’s proposition does not seem to capture the original idea that individuals react to their individual informations [*sic*] and *therefore* the equilibrium price reflects some aggregate of [them]. Instead, it must be presumed that the ‘auctioneer’ somehow happens to know the vector [of individual informations], in which case perfect aggregation of information through the price induces agents to disregard their individual information and therefore is consistent with market clearing.

(Hellwig 1980:478; emphasis in original)

Furthermore, in Grossman and Stiglitz’s setting it is not clear of what use the information on the higher returns can be to uninformed traders if they have to derive it from a price that has already fully discounted

those returns, hence allowing no profits to be made. Several of Grossman and Stiglitz's results may therefore not resist critical examination.

One, perhaps minor, point that deserves brief mention is that Grossman and Stiglitz use equilibrium theory because of an explicit dissatisfaction with disequilibrium analysis. When explaining that they have chosen to describe an economy affected by different shocks—which requires new information to be transmitted to trading agents—by a ‘stationary stochastic process’ they say, ‘others have described this as a disequilibrium situation, but have been unable to say much about it’ (Grossman and Stiglitz 1976: 248). Unfortunately, they do not specify who these others are and what Grossman and Stiglitz would have wanted them to say, although there is reason to suspect that this is a reflection of the methodological differences regarding the role of mathematical formalism mentioned in the introduction. What Grossman and Stiglitz do instead is to construct a model with ‘an equilibrium degree of disequilibrium: prices reflect the information of informed individuals (arbitrageurs) but only partially, so that those who expend resources to obtain information do receive compensation’ (1980:393).¹⁸ It is with such a model that they analyse Hayek's argument.

Two informational roles of equilibrium prices

The writings of Grossman and Stiglitz serve to highlight a significant difference between Hayek and other authors regarding the informational role attributed to prices. Hayek was saying that agents, by economizing with respect to market prices, respond to events they are not—or, rather, need not be—aware of. (An individual engaged in central planning, Hayek's argument went, would have to know about these events.) Although Grossman and Stiglitz seem to understand this, their work is often concerned with a different point, apparently without them noticing it. They analyse situations in which individuals infer information from market prices, in the Rational Expectations fashion (Grossman 1981:557). In other words, individuals facing market prices do not act *as if* they knew the relevant information but, instead, they obtain information from prices. This interpretation is confirmed by Grossman's example in which

consumers are buying land and are uncertain about the harvest on land. There the utility he [*sic*] currently gets out of land depends on his information about the future income flows the land gives. He would like to know the information other traders possess about the flows to better compute his expected utility.

(*ibid.*: 556)

In Grossman and Stiglitz's models the agents are assumed not to know what the value of the good is to them, so they need to obtain this information. In these particular models, they get the information from prices. A higher price of land, for example, could indicate to the potential buyer that future income flows from land are now expected by other traders to be higher. This could perhaps increase his interest in it. Grossman and Stiglitz's work is another instance of the cases in which the agents' beliefs about the quality of what is being traded depend on its price.¹⁹ Aside from these cases, there is also the literature on rational expectations, in which agents are often assumed to obtain information from observed prices to form their expectations of future occurrences. Most of this work, however, is really concerned with an informational role of prices different from that emphasized by Hayek: in his view, a higher price of land simply leads consumers to economize on its use because (probably unknown to them) it has become more valuable for some other purpose. His main point, in fact, is that prices *make it unnecessary* that individuals become informed of large numbers of facts. Prices in Hayek's argument are more appropriately described as knowledge *surrogates*, while in the approach exemplified by Grossman and Stiglitz they are sources for the inference of knowledge.

This difference serves to explain why, whereas Hayek could present his argument using relatively standard theory at the time, it is nowadays sometimes felt that contemporary theory is inadequate for the task. Grossman (1981:554) complains that in the usual economic models

no one learns anything from prices. People are constrained by prices (often in just the right way so that individual rationality is transformed into collective rationality); however, they are not informed by prices in the classical Walrasian or Marshallian models.

Dasgupta (1980:115) expresses a similar view when he says

prices in the Fundamental [Welfare] Theorem do not actually *transmit* information in any sense. They clear markets and enable a separation between individuals in such a manner that they do not need to act strategically. Hayek is suggesting that prices do something else in addition: that they may be a vehicle for making public all the private bits of information. [Emphasis in original.]

In the Walrasian or Marshallian models there is certainly no continuing activity of inference of information, or ‘learning’: people have already learnt what is worth knowing through their successful exploitation of disequilibrium profit opportunities. However, Hayek was not dealing with this type of situation. Hayek’s point *is* reflected in ‘classical Walrasian or Marshallian models’ precisely by the fact that their equilibrium prices transform ‘individual rationality’ into ‘collective rationality’ without it being necessary that these individuals know much about the situation. His claim is not that prices ‘transmit’ or ‘make public’ all the ‘private bits of information’. (Of course, neither does he deny that this may sometimes happen.)

In his article in honour of Grossman, Kreps shows an awareness of the distinction made here when he refers to a ‘classic (Walrasian) notion of “information communication” by prices, that should be distinguished from the information communication function of prices in a rational expectations equilibrium’ (115). However, he does not draw from this distinction any specific implication for Grossman’s argument, particularly for his supposed refutation of Hayek’s view of market prices.

The information Hayek finds reflected in equilibrium prices refers only to relative scarcities. The information disseminated in Grossman and Stiglitz’s framework is about the quality (in a broad sense) of the traded good. This latter case, in which traders obtain information about the good from its price, appears to be of less generality than Hayek’s. (Although their argument is of more generality in assuming that agents do not have perfect knowledge about the quality of goods, it is of less generality in postulating an important role of prices as a source of such information.)

Whatever the case may be, a large body of literature, both in microeconomics and in macroeconomics, relies on the notion of inferring knowledge from prices, frequently—and mistakenly—citing Hayek’s 1945 article as its origin. Of course, there is no need to deny the possible existence of this informational role of prices:

individuals in a Hayekian world could attempt to infer information from prices if it were profitable for them to do so. Nevertheless, it is a different informational function. And to the extent that it is with the inference kind of activity that Grossman and Stiglitz are dealing their analysis becomes largely irrelevant with respect to Hayek's argument.

A question remains regarding whether or not the individuals in general equilibrium need the information provided by prices as surrogates. Although the contrary is often argued, it has been said that in the competitive model '*individuals do not have perfect information*. Instead they have costless market information about exchange opportunities as conveyed by prices and, of course, it must be assumed that they know the equilibrium set of prices' (Veljanovski 1982:63–4; emphasis in original).²⁰ It is this type of consideration that leads Coddington (1975:154, n.) to remark that

the result that in equilibrium market prices are perfect-knowledge surrogates is something of a swindle or, at best, a piece of conceptual conjuring. This is so because all the epistemic problems have to be solved in reaching equilibrium (e.g. by the Walrasian auctioneer). The reason that market prices 'reflect' everything that traders need to know about the markets is because—somehow—they have been rigged to do so.

The last section of this chapter will argue that Hayek's argument, here captured by an equilibrium approach, may also remain valid from a market-process perspective that addresses Coddington's concern with disequilibrium. However, Coddington's remarks do not alter the fact that there are two informational roles of equilibrium prices to be distinguished. Much of the literature has not only missed the entrepreneurial, discovery role of disequilibrium prices but it is also confusing the roles of prices as knowledge surrogates and as sources of information. However, this is not a deficiency of all the research concerned with prices and information: other work, including that of Hurwicz mentioned above, deals more directly with the role of prices as surrogates, although still to the exclusion of all entrepreneurial aspects. At any rate, the distinction made here should be kept in mind in the discussion of Grossman and Stiglitz's writings that follows.

The informational role of disequilibrium prices

Grossman and Stiglitz's argument is that prices cannot be perfect transmitters of information because they cannot be fully arbitrated if they are to provide a reward for the (equilibrium) activity of obtaining information, such as arbitrage. On the other hand, when viewing markets in disequilibrium terms, it is true that prices cannot be taken 'literally', as price-takers are supposed to take them, because out of equilibrium they provide wrong information about relative scarcities. Although for Grossman and Stiglitz this means that prices will not be informationally efficient, which in a sense is true, Kirzner argues that disequilibrium prices perform a more important informational role.

Market prices, which are imperfect knowledge surrogates if taken as equilibrium prices, provide rewards for those agents who notice unexploited opportunities. In fact, 'it is the very inadequacies that cloud the manner in which these price-summaries express existing knowledge, that create the market incentives for their modification' (Kirzner 1984b:202). The profits resulting from exploiting price disparities spark the entrepreneurial discovery of new, previously unthought of knowledge. The claim is not that price-taking agents, by deciding on the basis of known prices, act as if they know more than they actually do, but rather that profits lead them to find out about better available courses of action. As Kirzner (1984b:205) puts it, 'the social function served by market prices is captured far more significantly by the concept of discovery, than by that of communication'. (Perhaps it would help to stress that it is really profits, rather than prices directly, that perform this informational role.)

This informational role is crucial because, as will be argued later on, the extent to which prices may reflect information, both as surrogates and in the Grossman and Stiglitz sense, will be a result of the disequilibrium rivalrous bids and offers of entrepreneurs.²¹

It is well known that most economists have neglected the disequilibrium informational role of prices. What is more interesting is that even a particularly sympathetic reader of Hayek such as Thomas Sowell, in his book *Knowledge and Decisions*, described by Kirzner (1984b:202) as 'the most extensive and wide-ranging development of the implications of the Hayekian insights', has been almost exclusively concerned with the role of equilibrium

prices as summaries or surrogates of knowledge rather than with their disequilibrium role as stimulators of entrepreneurial discovery.²² (In Sowell's argument, as opposed to Grossman and Stiglitz's, prices perform the role satisfactorily, although this, of course, has something to do with what standard of efficiency is used.)

Although acting as summaries is an important function of prices, Sowell seems to be unaware of the entrepreneurial one, or to take it for granted. In fact, he could frequently be read as believing that market prices are not too different from equilibrium prices. Thus he says, 'nobody needs to know the whole story in order for the economy to convey the relevant information through prices and secure *the same adjustments as if everyone had known*' (Sowell 1980: 75; emphasis added), that '*accurate* prices resulting from voluntary exchange permit the economy to achieve *optimal* performance' (79; emphasis added), that 'knowledge transmitted by low prices (wages) is generally accurate knowledge' (174), and that with respect to changing technology, tastes, etc., 'price changes are virtually instantaneous' (216). These are statements an entrepreneurial perspective would not endorse without qualifications. However, this chapter will later argue that most of Hayek's, and Sowell's, argument can be reinterpreted from an entrepreneurial perspective and remain valid.

Kirzner's emphasis is more on the discovery of knowledge than on its summarization or its transmission. But the question could be raised: could this discovery role perhaps be accommodated within the approach of Grossman and Stiglitz? It is, after all, their argument that imperfectly arbitrated prices provide incentives for information-gathering activities; in other words, it is because of price disparities that the production of new knowledge becomes economically viable. Therefore, even in Grossman and Stiglitz's framework prices perform another informational role, although it is not highlighted: imperfectly arbitrated prices, as in market-process theory, lead to the appearance of new information. Of course, as has already been indicated, Grossman and Stiglitz refer implicitly to production of information-as-a-commodity, not to genuine discovery in the market-process sense. A consideration of some differing implications of these approaches will show that this distinction matters here.

The apparent similarity of this implication of Grossman and Stiglitz's approach to the modern Austrian argument could mean, for example, that they would share the latter's conclusions regarding the

informational advantages of market prices over other alternatives. A comparison of, for example, the performance of markets with that of a centrally planned economy, both of them under conditions of dispersed, costly knowledge, would show the market providing pecuniary incentives for information-gathering activities, incentives that would not seem available under central planning. Within Grossman and Stiglitz's framework, there would appear to be no rewards for costly information-gathering under a central planning system without prices and profit-motivated arbitrageurs.

However, it is here that the information-as-a-commodity approach makes a difference. From this perspective there might be no theoretical reason why an incentive system could not be devised for a centrally planned economy to achieve similar, if not better, results than a market economy. After all, given the way ignorance is treated in the economics of information, the knowledge that could possibly be found out and is worth knowing must be, as the previous chapter described, already known, at least probabilistically, by the deciding agent (in this case, the central planner or the designer of this economic system). There would be nothing in this approach, it would seem, preventing the design of an incentive scheme that would lead to an optimal amount of information-gathering.²³ Then, the relative efficiency of the two economic systems, as Grossman and Stiglitz argued in a quotation above, could be established only after determining their respective informational costs, an empirical matter not answerable by pure theory.

On the other hand, the type of ignorance considered by market-process economists cannot be overcome easily by a central planner. The problem is not that the planner will have no incentive to engage in activities (including information-gathering) that he knows would be desirable (which may, of course, also be the case). It is that there will be nothing (such as pecuniary profits in a market economy) to spur his entrepreneurial discovery of what the desirable activities are. In Bartley's (1985:31) words,

the problem is not only how to utilize uncommon *existing* dispersed knowledge, but also how to elicit implicit and not yet fathomed knowledge... Competition not only makes the best use of *existing* dispersed knowledge, but also generates new knowledge which *none* of the participants in the process yet possesses. [Emphasis in original.]²⁴

In a market-process perspective profit incentives are not rewards to encourage the adoption by agents of certain courses of action already perceived and known, in at least a vague way, by the designer of the incentive system. Market profits are, as argued in the previous chapter, incentives that encourage the discovery of 'opportunities that have until now been perceived by no one at all' (Kirzner 1985a:29).

The conclusion, then, is that although there could be some room for another informational role of prices within Grossman and Stiglitz's theoretical framework, their treatment of the ignorance to be dispelled in terms of the economics of information is likely to produce an undervaluation of the achievement of prices in this role. From such a perspective, prices would turn out to be a (probably imperfect) mechanism for rewarding the production of knowledge, a mechanism that could be replicated, or even surpassed, by some other alternative. From the market-process perspective, on the other hand, the crucial problem is not merely one of efficiently deploying knowledge existing somewhere in the economy but also one of prompting entrepreneurial discovery of previously unknown knowledge. And this is something for which there appears to be no good substitute for market prices.

To recapitulate, this chapter has hereto identified three informational roles of prices: (1) prices may inform the actions of an individual (without informing him), (2) prices may be 'read' by individuals to infer some information, and, lastly, (3) price disparities may provide rewards for the 'production', or discovery, of new knowledge.

Prices as 'sufficient statistics'

Grossman and Stiglitz interpret Hayek as saying that prices in a market economy serve as sufficient statistics, that knowledge of prices is sufficient for agents to act so as to achieve an efficient economic allocation.²⁵ They are not alone in this interpretation. Already, in 1957, Koopmans (1957:22–3, n.), after citing Hayek's 1945 article, pointed out that

there is a striking similarity between the summarization of supply and preference *data* through prices and the notions of sufficient and efficient statistical estimation procedures, proposed by R.A.Fisher

as devices for the 'reduction' of *data* and widely adopted and further developed by statisticians. The basic idea there is to find a set of numbers which adequately summarizes a much more detailed body of information for the purposes of a certain class of decisions.²⁶

More recently, in an attempt to outline the market-process position, Loasby (1982:114–15) has described it as arguing, among other things, that

a decentralized economy needs only to establish a set of prices. Each economic agent can then apply his own specific knowledge of resources and of technology and his own particular pattern of preferences to those prices: there is *no* need for information about these matters to be communicated. What is more, as F. A. Hayek, for example, has emphasized, *no one* needs to know why the price of some particular commodity is whatever it is. A particular material may rise in price because its supply is becoming exhausted, because new uses have been found for it, or because existing uses are becoming more popular. The cause does not matter: whatever that may be, the consequence is that more effort should now be devoted to ways of increasing the supply, using it more effectively, or replacing it by some alternative. But in a pure market economy no one needs to be instructed to do any of these things. The increased price provides the *only* signal needed, and anyone who has the knowledge, or the particular pattern of preferences, to contribute in any of these ways will do so.²⁷

But, Loasby realizes, this interpretation leaves one question unanswered: how are these prices to be arrived at? Agents in this version of Hayek's story act as price *takers*. What is needed is a 'theory of price setting', a theory which Loasby believes Kirzner provides in his work on entrepreneurship (1982:115). Loasby sees this lack as a problem, in contrast to Grossman and Stiglitz, who do not seem worried about it,²⁸ because he is not concerned only with equilibrium states but also with the processes by which they might be arrived at.

Loasby's interpretation has been criticized by Garrison (1982). Garrison quotes from Hayek's tin example to show that Hayek did not envisage an economy in which nobody possessed the relevant

information. Instead, he argues, Hayek distinguished between two types of agents: what modern Austrian economists would call entrepreneurs, and what can perhaps be described as price-taking individuals with only local knowledge. He quotes from Hayek:

If only *some of* [the tin users] know directly of the new demand, and switch resources over to it, and if *people who are aware* of the new gap thus created in turn fill it from still other sources, the effect will rapidly spread throughout the whole economic system...and all this without the *great majority* of those instrumental in bringing about these substitutions knowing anything at all about the original cause of these changes.

(133; emphasis added)

In this interpretation Hayek was saying, in 1945, that prices are sufficient statistics only for the ‘great majority’ of agents, that only these individuals make the necessary adjustments, responding to price changes without ‘knowing anything at all about the original cause’. The prices they react to are being set by alert entrepreneurs discovering and exploiting profitable opportunities.

This version of Hayek may be interpreted in two ways. Hayek may be resorting to the heuristic device of isolating economic functions for the purposes of analysis and thus speak of entrepreneurs, price-taking consumers and price-taking resource owners as separate agents. This is perfectly legitimate. Alternately, he may be taking this division of tasks as an accurate description of reality. This second interpretation may be objectionable.

Although Grossman and Stiglitz’s criticisms are partially obscured by their emphasis on knowledge inference,²⁹ it is possible to think, even for Hayek’s tin example, of cases in which most (or all?) traders would like to know more than just the price change. For example, is this change transitory, requiring only a temporary reduction in the consumption of tin, or is it permanent, justifying, for example, the adoption of new production techniques and machinery? Price-taking individuals would perhaps not need to answer this type of question if there were ‘complete markets’ (or at least, given the costs involved in setting up markets, ‘optimally complete’ markets) and, consequently, enough futures prices. But it is doubtful whether Hayek would have wanted to make this assumption.

It is probably more realistic to say, with Mises (1949:252), that ‘in any real and living economy every actor is always an entrepreneur’. The idea of prices as sufficient statistics has to be used, if at all, in a more limited sense, as in a recent statement by Hayek where he says

the price mechanism operates as a medium of communicating knowledge which brings it about that the facts which become known to some, through the effects of their actions on prices, are made to influence the decision of others.

(1979:125)

The point then is not that it is enough (sufficient) for an individual to know prices, in addition to his preferences, resources, and technologies, to act correctly—the target of Grossman and Stiglitz’s criticism. Instead, it is that prices *reduce the amount* of detail that he needs to know to do so.³⁰ As O’Driscoll and Rizzo (1985:39) put it, ‘the crucial point is that, overall, more information is conveyed through a market price system than without one.’ The effectiveness of prices in this role will depend crucially on the existence of a rivalrous competitive process, an issue considered below.

There is another interpretation of Hayek to consider which at first appears also to treat prices as sufficient statistics. Vernon L. Smith (1982a) has conducted experimental tests regarding ‘the institutional and technical conditions necessary to achieve a competitive equilibrium (C.E.)’.³¹ He says:

one view, which has commanded a modest following since the classic work of Adam Smith, suggests that the attainment of C.E. allocations do [*sic*] not require *any* individual participant to have knowledge of the circumstances of other agents, or to have an understanding either of the market as an allocation system or of his/her role in promoting ‘an end which was no part of his intention’.

(1982a:166; emphasis added)

Smith singles out Hayek as the economist who has presented this position ‘more strongly and more influentially in recent decades’ (1982a:166). Smith decides to test what he therefore terms the ‘Hayek hypothesis’, which affirms that ‘strict privacy together with the trading rules of a market institution are sufficient to produce competitive market outcomes at or near 100% efficiency’ (167). By ‘strict privacy’ he means ‘each buyer in a market knows only his/her

own valuation of units of a commodity, and each seller knows only his/her own cost of the units that might be sold' (ibid.).

In addition to the 'extreme case of "little" knowledge' Smith describes two 'contrary hypotheses [that] have formed the core of main stream economic thought concerning the conditions for competitive equilibrium': the 'price taking hypothesis', which states that large numbers of buyers and sellers, producing price-taking behaviour on their part, are an essential feature of a competitive market, and the 'complete knowledge hypothesis', which asserts that 'competitive allocations require perfectly "foreseen" conditions of supply and demand' (ibid.).

Smith is interested in testing the 'Hayek hypothesis', among other reasons, because 'the vast majority of economists in the main stream of British and American economic thought have not accepted, indeed have been openly sceptical of Hayek's claim that decentralized markets are able to function with such an extreme economy of information' (ibid.). In fact, 'Hayek's claims concerning the price system as an economizer of information, must be classified as an "outrageous" hypothesis contrary to what the common sense of most scholars had led them to expect' (1982a: 177).

Interestingly enough, Smith reports that his tests are favourable to the 'Hayek hypothesis' (176). However, in his experiments, unlike in the previous interpretations of Hayek,

each agent is *not* in a price taking environment. The environment is one of multilateral negotiation in which each agent is as much a *price maker* (who actively announces bids or offers) as a price taker (who accepts bids or offers).

(169)

This separates Smith's version of Hayek from others. It is significantly different, because what he tests is whether agents with 'strict privacy' and knowledge of the trades taking place in the experimental market can achieve fairly quickly the theoretically determined equilibrium price. Smith is testing whether *equilibration* occurs, while the 'sufficient statistic' argument is concerned about whether or not 'strict privacy' and knowledge of the equilibrium price lead traders to produce a Pareto-optimal outcome.

How should Smith's results be interpreted? The conditions established in his experimental markets make it possible that some kind of entrepreneurial process was taking place. Though it may be

true that individuals need not start with more knowledge than that encompassed by ‘strict privacy,’ any systematic (as opposed to purely coincidental) equilibrating tendency that may be found either in real markets, or in experimental settings that try to mirror some of their characteristics, must occur because the individuals are able to form increasingly correct expectations of the others’ plans. This ability is what Kirzner calls entrepreneurship. The individuals’ expectations will be partially based on entrepreneurial interpretation of the experiences in the market process. A better understanding of how the equilibrating results were achieved in Smith’s experiments could perhaps be obtained by examining more closely the conditions under which the tests were conducted.

The important point for present purposes is that Smith’s experiments are not testing the price-taking ‘sufficient statistics’ argument criticized before. Although his results appear to falsify the hypotheses which say that large numbers of agents or omniscience are necessary conditions for competitive outcomes, they are not a refutation of the disequilibrium explanation of the market process.³² Furthermore, without such an explanation his results are quite mysterious: there is no reason why ‘price-making’ agents remaining within the confines of ‘strict privacy’ should ever converge quite systematically to competitive equilibrium prices, or why Hayek should want to affirm something so ‘outrageous’. As Garrison (1982:133) has put it,

Hayek called our attention to the marvel of the market economy functioning as it does on the basis of such little knowledge; he did not insist on a miracle in which the economy functions in the total absence of knowledge.

The entrepreneurial ability highlighted by Kirzner needs further study, but it helps make intelligible the rather systematic nature of observed market processes.

Grossman and Stiglitz and the normative standard

Grossman and Stiglitz’s work exemplifies some of the current ambiguity regarding which standard to use for evaluating economic systems and situations. Like much of the literature in the field, they use the standard of Pareto efficiency. Therefore, after presenting their

arguments, described above, they judge the price system 'informationally inefficient', even though they assume away several typical 'imperfections' usually attributed to markets in reality (such as, for example 'incomplete' markets). Since Arrow's (1962) paper it has been common to argue that, because information has attributes of a public good and there are increasing returns to its uses, a perfectly competitive economy will underinvest in its production. On the other hand, Jack Hirshleifer (1971) has pointed out that, if secrecy is possible, there may be overinvestment in information-gathering activities. Arrow (1984:143) offers an example of this possibility:

each firm may secretly get the same information, either on nature or on each other, although it would of course consume less of society's resources if they were [*sic*] collected once and disseminated to all.

An awareness of these issues leads Grossman and Stiglitz (1976: 251) to affirm that 'although it is easy to show that the market solution is not, in general, efficient, it is difficult to ascertain whether there is too little or too much information acquisition'.

Stiglitz seems to have taken some of Demsetz's criticisms of the usual welfare approach (presented in chapter 2) into account when arguing recently that 'obviously, economies with perfect information are likely to function better than economies with imperfect information: That is an irrelevant comparison'. The relevant question, he states, is whether the market is 'constrained Pareto efficient, taking into account the imperfections of information and costs of obtaining more information' (Stiglitz 1987:13–14). However, this only takes care of what Demsetz terms the 'fallacy of the free lunch' —in this case, judging a situation inefficient because it does not include complete information, even though information may be costly. But Stiglitz's (1987:14), and others', conclusion that such economies are 'essentially never constrained Pareto efficient' suggests that they have in mind a better alternative that is not made explicit. This seems to be a variant of Demsetz's 'the grass is always greener' fallacy which, in this case as in most welfare analysis, consists of assuming implicitly the existence of an alternative institution (1) that can costlessly (or at least with advantage with respect to markets) obtain the necessary knowledge and (2) that will have the motivation to improve on market outcomes. Although he rejects it explicitly, Stiglitz reintroduces implicitly the perfect

knowledge standard in the form of an at least highly informed government when he concludes that welfare improvements can in many cases be achieved through appropriate combinations of taxes, subsidies, and centralization of decision-making (1987:14).

At a later point, when in a comparison of government planning and market allocation he assumes that the government may have no more information than private individuals, Stiglitz does not argue why even this should be an acceptable assumption. Even from a perspective of costly information, it may turn out to be economically impossible to put in a government's hands as much information as exists in a decentralized form in a market economy. The assumption becomes even less acceptable for a market-process perspective: Stiglitz does not specify for government any mechanism that will replace the discovery procedure provided by prices and competition to individuals in a market economy.

The ambiguities regarding which standard to use persist over all Grossman and Stiglitz's work, as in much of the literature on the economics of information. These ambiguities become particularly noticeable when, on the one hand, Grossman (1981:555; emphasis added), in his description of Hayek's argument, says, 'a planner *without* all of that information could not have done as well', and, on the other, Grossman and Stiglitz (1976:252; emphasis added), after arguing about the informational inefficiency of prices, conclude that 'in this case a central planner *with* all the information can improve on the competitive equilibrium'.³³

Of course, most of these comments can be made from an equilibrium, somewhat Demsetzian, perspective that still accepts some variant of Pareto efficiency as its norm. As the previous chapter stated, a disequilibrium, market-process approach will adopt a different normative standard. Streit (1984:394) seems to have it in mind when he says:

As in other cases of socially useful competition, inefficiency from a static point of view can be the source of dynamic efficiency. And from this perspective of discovery and adaptation, the basically static verdict of Pareto non-optimality carries little weight if any.³⁴

From this perspective, the hope expressed by Rothschild (1973: 1304) that the economic profession will start 'to develop standards of efficiency and equity and to begin to ask what sorts of institutional

arrangements seem to operate most effectively in the face of imperfect information' has not yet been fulfilled.³⁵

DISEQUILIBRIUM PRICES AND INFORMATION

This chapter has drawn a rather sharp difference between the informational role attributed to prices by an equilibrium interpretation of Hayek's 1945 article and the market-process view. The former, it argued, consists of the claim that price-taking agents, guided solely by market prices, would make correct decisions. The market-process view argues that 'incorrect' (i.e. disequilibrium) market prices provide incentives, in the form of pecuniary profit opportunities, to the discovery of better alternatives by entrepreneurial agents. It was also briefly mentioned that adoption of the market-process view does not imply the complete denial of the other role and that, therefore, arguments such as those of Thomas Sowell in his book *Knowledge and Decisions*, which rely extensively on it, still retain much validity from a market-process perspective. This point will now be argued in more detail.

The concern of many economists, as shown, is that market prices in reality are affected by several factors that make them inefficient conveyors of information for price-taking agents. These factors include diverse deviations from the theoretical model of perfect competition, mostly those under the label of externalities. On the other hand, for modern Austrian economists the market, which they view as a disequilibrium process, will always have 'wrong' prices—that is, prices that lead to losses and profits because they are not perfectly adjusted to each other. Such prices, as was mentioned before, will also be inefficient conveyors of information if they are to guide price-taking (as opposed to entrepreneurial) agents.³⁶ It is for the case of purely price-taking agents that Coddington (1975:154) can rightly say that 'in a world where outof-equilibrium trading is binding rather than hypothetical there is no presumption that market prices are very serviceable as knowledge surrogates; indeed, they are generally misinformation surrogates'.

Therefore, market-process economists would have to see two problems with prices as conveyors of information in the equilibrium sense: not only would, for example, externalities make them imperfect, but also the fact that market prices are never fully adjusted to each other. (Because in this view an equilibrium state is never

reached in reality.) How could market-process economists then accept also this informational role of prices?³⁷ To answer this question, another one has to be answered first, namely: given that the standard informational role of prices is based on price-taking agents, to what extent can a market-process view accept this notion it has generally considered inadequate for analysing markets? An analogy presented by Lavoie is useful for answering this question. According to Lavoie (1985b:83–4; emphasis added),

market participants are not and could not be price takers any more than scientists could be theory takers. In both cases a background of *unquestioned* prices or theories is relied upon subsidiarily by the entrepreneur or scientist, but the focus of the activity is on *disagreeing* with certain market prices or scientific theories. Entrepreneurs (or scientists) actively disagree with existing prices (or theories) and commit themselves to their own projects (or ideas) by bidding prices up or down (or by criticizing or elaborating existing theories).

Say that an entrepreneur were to notice that a sum of resource costs is such as to allow him a profit, given his expected price for the final good and that getting ready for its production requires some time. To undertake such a project he would have to believe that those prices are relatively correct—they would have to be, in Lavoie’s terminology, his ‘unquestioned’ prices.

If, on the contrary, the entrepreneur knew, for example, that one of the resources was mistakenly underpriced in the particular market he was examining, and that at the price for that resource prevailing elsewhere (or in the future) his project would not be profitable, the following could happen. (1) The entrepreneur would not start his lengthy project (which would be likely to turn out to be unprofitable by the time he was ready to start, because other entrepreneurs would perhaps engage in arbitrage activities that would increase the price of his resource). (2) He would engage in arbitrage activities with that resource.³⁸ By implication, then, if he does start with his project it is because he believes the observed prices are quite reliable. If asked, he would most likely not claim that the prices are equilibrium prices but, rather, only that he can ‘see nothing wrong with them’, that he knows of no more convenient prices.

As Lavoie’s analogy indicates, a scientist will rely on innumerable theories in his research. If asked, the scientist would surely not feel

able to affirm that all of them are unquestionably true but only, like the entrepreneur above, that he can 'see nothing wrong with them', that he can think of no better theories, and will therefore rely on them for the time being. Would it be reasonable to describe this scientist as a 'theory-taker'? It is in this sense that something similar to standard theory's price-taking agents is acceptable to market-process theory. (Of course, in reality nobody will be exclusively a price-taker: while some prices will be taken as 'unquestioned', there will always be some the individual will 'disagree' with.)

The next step in the argument is to show some consequences for a disequilibrium view of markets of the existence of these 'price-accepting' agents. Say that an individual notices that apples are traded in his area at \$3 and decides to purchase some. If, unnoticed by him, apples are being traded at a second location at, say, \$7, his decision would not bring about the best result (most probably that apples be shifted from the first location to the other one). Another individual noticing only the apple price in the \$7 market would make a similar mistake. In this way a 'price-accepting' agent (that is, an individual who cannot 'see anything wrong' with the existing prices and takes them as *data* for his decisions) economizing with respect to the observed apple prices will not receive adequate information. Therefore, the market-process approach cannot accept the standard informational role of prices without additional explanation; disequilibrium prices cannot, without further qualifications, be said to perform effectively the informational role attributed to them in the equilibrium models of standard economic theory.

The inefficacy of prices as knowledge surrogates shown above is due to entrepreneurial error. And errors of this kind are most likely occurring constantly in a world in which agents are not omniscient. But this means there may be a problem if, as argued, there are at any moment individuals taking these erroneous prices as unquestioned information. If the errors in these prices were always of an extreme type (i.e. if most entrepreneurs were systematically missing very profitable opportunities), markets peopled with 'price-accepting' agents would turn out to be quite chaotic, which, although not an inconceivable outcome, is not what has been generally observed. In this last regard, as Hayek (1941:27, n. 2) said,

it should be remembered that nearly the whole of economic science is based on the empirical observation that prices 'tend' to

correspond to costs of production, and that it was this observation which led to the construction of a hypothetical state in which this 'tendency' was fully realized.³⁹

Some may interpret the observed regularities as the result of pure coincidence (an approach leaving little or no room for—or, rather, need of—a science of economics), or in terms of equilibrium states (as many economists, at least implicitly, do). The Mises-Hayek view, on the other hand, sees them as the outcome of a 'co-ordinating', entrepreneurial market process. Underlying such an interpretation is what Garrison (1982) has termed a 'middle-ground' view of the world. In this view the entrepreneurial ability of agents—i.e. their ability to discover opportunities—and the pace of change of the 'facts', both exogenous and endogenous, is such that some co-ordinating action is possible, although not the attainment of full equilibrium.⁴⁰ It is from such a 'middle-ground' view, which implies that prices, although not in equilibrium, are not radically 'wrong', that the standard interpretation of the informational role of prices is partly acceptable.

In this way many of Sowell's arguments that rely on this role need not be rejected, although a market-process approach will emphasize that it is not the only informational role of prices: prices are not only conveyors of information in the standard sense—imperfect conveyors,⁴¹ in fact, given that they are disequilibrium prices—they also contain the incentives to the correction of their imperfection. That is, by providing profit opportunities, they provide information about their incorrectness and rewards for its removal. While an equilibrium economist may want to argue that, even if prices are informationally inefficient (in the standard sense), they are still the best alternative in a world that is not Nirvana, to use Demsetz's words, from a market-process perspective prices are not only an imperfect but 'least bad' information transmission system: they are also sophisticated informational devices, with a feedback mechanism (profits) that induces their correction by entrepreneurial agents. This correction is never fully achieved in reality, but the degree of order observed in markets is, in the market-process view, to a large extent due to the degree of success of entrepreneurs in responding to this feedback.

The argument above does not say that, in one way or another,

entrepreneurs come to know the equilibrium value of prices and trade at disequilibrium until this value is reached. As Mises says:

The activities of the entrepreneurs or of any other actors on the economic scene are not guided by consideration of any such things as equilibrium prices...The entrepreneurs take into account anticipated future prices not...equilibrium prices. They discover discrepancies between the height of the prices of the complementary factors of production and the anticipated future prices, and they are intent upon taking advantage of such discrepancies.

(Mises 1949:329)

Neither the theorists, nor the capitalists and entrepreneurs, nor the consumers, are in a position to form, on the ground of their familiarity with present conditions, an opinion about the height of such an equilibrium price. There is no need for such an opinion. What impels a man toward change and innovation is not the vision of equilibrium prices, but the anticipation of the height of the prices of a limited number of articles as they will prevail on the date at which he plans to sell.

(ibid.: 711)

In other words, whatever degree of co-ordination is achieved in the economy will be an unintended consequence of the profit-seeking activity of entrepreneurs. As Streit (1983:8) has very aptly put it,

the favourable 'performance' of the market would be the unintended outcome of numerous decisions of the many participants who traded precisely because they considered various prices to be inappropriate and unjustified, not least in the light of the information available to them, and who intended to profit from the mistakes. The rationale for active market participation differs completely from the observable market result.⁴²

To illustrate the extent to which the standard interpretation of the role of prices is compatible with the disequilibrium view of markets, Hayek's tin example can be explained from a market-process angle. This story would start with some entrepreneur noticing (rightly or wrongly) a new opportunity for the use of tin, or the loss of one of its sources of supply. In either case he bids up the current price of tin to

take it away from present buyers,⁴³ in the first case perhaps to sell it at a profit to new users (or to use it profitably himself), in the second perhaps to resell it profitably in the (near or distant) future when the new scarcity of tin becomes noticeable. Of course, for this (or these) entrepreneur(s) it *does* matter which of these two causes has made tin more scarce. But previous buyers of tin will very probably not need to know so many details about these causes once they face the higher price (although, as suggested earlier, they would probably like to know some things, such as the expected duration of the price increase, to decide what adjustments to make). Faced with the higher price, the previous buyers may adjust in somewhat ‘Robbinsian’ fashion by adopting some previously known (but less profitable) alternative, or entrepreneurially, by discovering a previously unknown alternative that has become profitable with the new price of tin.

The economy of knowledge with which the system works is due not so much to the fact that information is summarized in prices but, rather, to a division of entrepreneurial labour caused by the fact that each individual ‘disagrees’ only with a few prices while ‘accepting’ all others. This happens because each individual’s alertness allows him to discover only some of the many price discrepancies existing in a disequilibrium market, inevitably leaving some for other entrepreneurs and, of course, some unexploited. Each entrepreneur will concentrate on exploiting the price differences he has noticed, while accepting other prices unquestioningly. This entrepreneurial activity may bring the existence of a profit opportunity to the attention of some less alert entrepreneurs, whose competition will tend to whittle the profits away. In this way the process of adjustment takes place without any ‘need for the great majority of [the users of tin] even to know where the more urgent need has arisen, or in favour of what other needs they ought to husband the supply’. The essence of Hayek’s argument is preserved without having to rely on price-taking agents reacting to mysteriously modified prices.

SUMMARY

This chapter has examined some interpretations of the informational role of prices fairly representative of the work currently being done in the economics of information. The main results of this examination have been:

1. The discovery that prices may perform three different informational roles: prices may make it possible for individuals to make decisions as if they possessed much more knowledge than they really do (Hayek), prices may serve as devices from which individuals can infer knowledge (Grossman, Stiglitz),⁴⁴ and prices, when in disequilibrium, provide profit opportunities that spark an entrepreneurial discovery process which produces previously unthought-of knowledge (market-process approach).

2. Several limitations of the standard analysis of the informational role of prices have been pointed out. It has been shown that some economists confuse the roles of prices as surrogates and as sources of inference, attributing both to Hayek, when it is quite clear that he had in mind only the first. This confusion makes it necessary to re-examine the relevance of conclusions reached without making these distinctions. Also, most economists have remained generally unaware of the discovery role of prices, probably because of their concentration on the analysis of equilibrium states, which leave no room for such a role. Having found deficiencies in the way the price system performs the other two roles—usually judging it from an ideal normative perspective that excludes ‘sheer’ ignorance by assumption—some of them have tended to think that any theoretical case for markets based on informational considerations is on shaky grounds. How alternative systems can cope with the problems of sheer ignorance and of stimulating entrepreneurial discovery, however, still has not been adequately addressed by most economists.

3. Finally, the extent to which a disequilibrium approach can consistently accept informational roles of prices other than that of sparking entrepreneurial discovery has been considered. As economists from a market-process perspective emphasize particularly the role that is being neglected, it was necessary to point out that this does not imply denying that prices may also serve as knowledge surrogates. The last section of the chapter showed how this acceptance can be justified theoretically in an entrepreneurial framework. The same applies to the possibility that prices may serve to infer information from them, although this alternative was not analysed separately.

Market-process economists have not been the only ones to express dissatisfaction with the way in which most economic theory has dealt with the cognitive problems faced by human beings. Some other

critics of the omniscient (or optimally ignorant) agent of equilibrium theory are frequently grouped as bounded rationality theorists. It is to an examination of their approach that the next chapter will be devoted.

‘Bounded rationality’ and the price system

Like market-process economists, Herbert A. Simon sees the ignorance produced by the complexity of the task faced by a planner as the main obstacle to comprehensive central planning of an economy. However, Simon has a different view of this ignorance. He bases it on certain limitations of the human mind that he describes as ‘bounded rationality’. The advantage of markets, from this point of view, is that they permit the decentralization of activities and that they generate prices that summarize information. They thus allow human beings with necessarily limited minds to make use of a larger amount of knowledge than would otherwise be possible. A comparison of this interpretation with the market-process approach is useful, given that there is a risk that the differences between them will not be noticed.¹

From Simon’s writings will be sought (1) an understanding of the idea of bounded rationality, which, in spite of its being frequently referred to, has never been examined in much detail by economists; (2) a brief view of the idea of ‘satisficing’, for which Simon is most known among economists; and (3) a bounded rationality view of the market system.

HERBERT A. SIMON

Simon, winner of the 1978 Nobel Memorial Prize in Economics (although economics is not his primary field), has devoted a considerable amount of effort to the study of human rationality. He has emerged as a severe critic of the standard description in mainstream economics of individuals as omniscient, perfectly rational maximizers. Ando (1979:92) has pointed out that Simon’s contribution, from the point of view of economics, has been

to work out the implications of recognizing the limitation of man to comprehend, to adapt to, and to affect his environment completely and instantaneously, as largely assumed in classical economics.

Simon's writings have had, and continue to have, a substantial impact on different areas of economics. His view of human rationality seems to be increasingly perceived as the only alternative to the perfectly optimizing version of mainstream theory. Economists such as Roy Radner, Richard Nelson, Sidney Winter, Oliver Williamson, Brian Loasby, to name a few, have at some point relied on Simon's ideas.² These ideas bear some resemblance to those held by market-process economists, although, it will be argued, important differences appear on closer examination.

Simon's interests have covered a wide range of issues, but the focus here will be on those ideas that help understand his view of markets. Unfortunately, Simon has never drawn a picture of this aspect of his thought that is detailed enough for present purposes. Ando (1979:87) has said that, because many of Simon's findings have been derived from laboratory experiments, 'the task of extracting implications about market...economic behavior from such micro findings...will be extremely difficult, and will not be accomplished overnight'. Therefore, an attempt will be made to put together Simon's view on these matters from writings he has produced over a span of about thirty years, a task which will require relatively abundant quoting from them.

BOUNDED RATIONALITY

Simon (1957:202) criticizes standard economic theory for 'assuming that rational choice is choice among objectively given alternatives with objectively given consequences that reflect accurately all the complexities of the real world' and for 'seeking to erect a theory of human choice on the unrealistic assumptions of virtual omniscience'. He states that the theory must consider 'situations where the alternatives of choice are not given in advance, but must be discovered; where the means-ends connection between choices and consequences are imperfectly known' (Simon 1958:393). These views are similar to those of modern Austrian economists.³

There are more points in common with market-process economists: Simon (1959:314) has also shown some interest in the analysis of disequilibrium and of the processes of adjustment to equilibrium.⁴ In more recent writings he has pointed out the need for what he calls a 'Schumpeterian component' in economic theory, because in the language of mainstream economics 'there is no room for a concept like "initiative"':

In the Walrasian picture, there is at all times a fixed set of markets, each with its supply function and demand function and a price fluctuating around the equilibrium value. Markets are neither created nor destroyed. In an economy where actions must be positively motivated (let me call it a Schumpeterian economy), commodities are produced only after someone is motivated to consider producing them. Investments are made only when someone is motivated to pay attention to a potential investment opportunity and decides to invest. Job slots are created only when employers attend to the need for more workers and decide to try to employ them.⁵

(1984:53)

It was probably this type of statement that led McNulty (1968: 656) to point to Simon's work as suggesting 'a promising, if not yet altogether satisfying, new dimension to microeconomics' that could lead to a 'reformulation and expansion of the concept of competition'. This reformulation could make the concept of competition, 'once again, what it was at the hands of Adam Smith: a disequilibrium, behavioral concept which is meaningful and relevant in terms of the contemporary pattern of economic life'. Interestingly, market-process economists also hold this rivalrous notion of competition favoured by McNulty.

With so much in common, major differences between Simon and market-process economists only start to appear on closer examination of his thought. For Simon, what prevents man from being the perfectly optimizing creature of equilibrium economics is that he suffers from bounded rationality. As he states in an often quoted definition:

the capacity of the human mind for formulating and solving complex problems is very small compared with the size of the problems whose solution is required for objectively rational

behavior in the real world—or even for a reasonable approximation to such objective rationality.⁶

(Simon 1957:198)

Although Simon has stated that ‘these limits are not physiological and fixed’ (ibid.: 200), descriptions like the following suggest the opposite:

Every human organism lives in an environment that generates millions of bits of new information each second, but the bottle-neck of the perceptual apparatus certainly does not admit more than 1,000 bits per second, and probably much less. Equally significant omissions occur in the processing that takes place when information reaches the brain.

(Simon 1959:306)

Williamson (1975:21–2) has described these limitations as taking the form of ‘rate and storage limits on the powers of individuals to receive, store, retrieve, and process information without error’.⁷

According to Simon, the world within which human beings have to make decisions and choices is very complex compared to their mental capacities, if these decisions and choices are to be objectively, or ‘substantively’, rational.⁸ This complexity of the world consists of the fact that

in any realistic description of the environment of a human decision maker, the variables and information to which he might attend (and to which he must attend to satisfy the strict requirements of rationality) are innumerable.⁹

(Simon 1984:47–8)

That Simon thinks the number of variables is so large that the agent cannot attend to them all—and this turns out to be important later when contrasting Simon with market-process economists—becomes clear in his comments about the validity of equilibrium analysis. He says, about the standard requirement that the agent take into account all the ‘objectively’ relevant variables when deciding, that ‘whether that is easy or hard to do—or even possible—depends on *how many* relevant variables there are in the world’ (Simon 1984:47; emphasis added). Neoclassical analysis, he argues, is appropriate to ‘simple, slow-moving situations, where the actor has a single, operational

goal'. A more detailed analysis becomes necessary 'as the complexity of the environment increases, or its speed of change' (Simon 1959:313). The traditional approach seems acceptable to Simon in simple, unchanging situations 'where it might be assumed that additional *computational time* or *power* could not change the outcome' (1978b:457; emphasis added). For him,

classical economics was highly successful in handling small-maze problems without depending on psychology. Labor relations, imperfect competition, uncertainty, and long-run dynamics encase the decision maker in a much larger maze than those considered in classical short-run static theory.

(Simon 1963:343)

The same emphasis appears in his criticism of search theory as a remedy for the deficiencies of neoclassical man:

...the new theories do nothing to alleviate the computational complexities facing the decision maker...but simply magnify and multiply them. Now he needs to compute not merely the shapes of his supply and demand curves, but, in addition, the costs and benefits of computing these shapes to greater accuracy as well.

(Simon 1979:485)

Simon uses the example of chess to illustrate the type of limitation humans face. It is known that chess has at least one optimal (i.e. winning) strategy. However, to find it, it would be necessary to consider a huge set of possible strategies. The number of strategies in this set, although finite, is unmanageably large even for the most powerful computers. Therefore, Simon argues, it has not (yet) been possible to find the optimal one. And this is also what happens in reality: the large number of variables to be taken into account makes it computationally impossible for individuals to achieve optimality (Simon 1972:417).¹⁰

The 'information-rich' world

In Simon's view, human beings in a complex world are always in danger of being saturated by the many variables that are there to be known.¹¹ Information 'consumes the attention of its recipients' (Simon 1971:173). Therefore, he describes this as an 'information-

rich' world (Simon 1970:167). In such a world 'a wealth of information creates a poverty of attention' for individuals with bounded rationality (Simon 1971:173). The problem for individuals is not the acquisition of information. Instead, it is to possess arrangements that will act as 'buffers' against irrelevant information that would distract their scarce attention.¹² (These arrangements, it can be added, need not be deliberately designed.) The problem becomes one of conserving and effectively allocating this scarce element (Simon 1971:174). (As will be seen, this allocation, in Simon's perspective, cannot be put in the usual terms of equating the marginal costs and benefits of directing attention to a specific area.) Simon's idea of an 'information-rich' world is not without problems, but it provides another example of the difficulties he believes are faced by individuals with bounded rationality.¹³

The consequences of bounded rationality

Simon argues that in reality it is generally not possible for human beings to optimize as neoclassical theory assumes they do. Given the complexity of man's environment and his limited information-processing capabilities, 'no procedure that he can carry out with his information processing equipment will enable him to discover the optimal solution' (Simon 1976:430). Simon's point is not merely that agents do not seem to be optimizing. He says that, because of their known computational limitations, they cannot be doing so in most real-world situations. Without this last argument his position would be open to the suggestion that individuals may be making what Baumol and Quandt (1964) have called 'optimally imperfect' decisions.¹⁴

Although modern computers have, in Simon's (1976:430) view, enlarged the area within which man's computational capacity matches the complexity of the world, this area is still very small.¹⁵ Therefore, Simon (1959:259–60) argues, human beings, who act quite reasonably (or successfully) in reality, must be engaging in something else than (neoclassical) optimizing. Simon's (1955:243) suggestion is that one should observe how human actors actually proceed in reality to obtain an alternative explanation of their reasonable behaviour in a complex world. (This is the 'behavioural' element in his work.)

A sensible way in which bounded-rationality individuals could deal with a complex world, Simon suggests, is through the adoption of simplifying devices. In his observations, for example, he finds that

agents build simplified models ('heroic abstractions') to deal with the situations in which they find themselves. This makes Simon (1956:278) adopt the familiar distinction between subjective and objective rationality: he defines the former as 'behavior that is rational, given the perceptual and evaluational premises of the subject', and the latter as referring to 'the rationality of the perceptions themselves (i.e. whether or not the situation as perceived is the "real" situation).' Although individual behaviour may be subjectively rational, he argues, the complexity of the world imposes a need for simplification that makes it very unlikely that it will be objectively rational.¹⁶

An agent can be said to be optimizing with respect to his simplified model, but 'such behavior is not even approximately optimal with respect to the real world' (Simon 1957:199). This lack of optimality can apparently occur in two respects: (1) the actor will not be able to encompass the whole set of relevant variables, and (2) given the limited amount of information that he can handle, there is no assurance that he will take into account the most profitable portion of reality (as a neoclassical agent who had to allocate his limited attention optimally would).¹⁷ To emphasize further that he is not talking about optimal simplifications by individuals, Simon objects to their being described as 'approximations' or the agents' perceptive mechanisms as 'filters':

The term 'approximation' implies that the subjective world of the decision-maker resembles the external environment closely, but lacks, perhaps, some fineness of detail. In actual fact the perceived world is fantastically different from the 'real' world. The differences involve both omissions and distortions, and arise in both perception and inference. The sins of omission in perception are more important than the sins of commission. The decision-maker's model of the world encompasses only a minute fraction of all the relevant characteristics of the real environment, and his inferences extract only a minute fraction of all the information that is present even in his model.

Regarding the term 'filter', the type of filtering that actually occurs is

an active process involving attention to a very small part of the whole and exclusion, *from the outset*, of almost all that is not within the scope of attention.

(Simon 1959:306–7; emphasis added)

Thus the simplification is not—and cannot be—done, as is assumed in an optimizing approach, after evaluating all the potentially available information, however vaguely the latter might be supposed to be known.

SOME SIMPLIFYING DEVICES

Two of the devices Simon identifies that reduce the complexity of decision problems, making them manageable for individuals, will be considered here: (1) at the individual level, the procedure Simon has named ‘satisficing’ (arguably his most renowned contribution to economics) and (2), at the social level, ‘the combined use of markets and administrative hierarchies.’ The idea of ‘satisficing’, not crucial for present purposes, will be briefly described to help understand bounded rationality.

Satisficing and aspiration levels

‘Satisficing’ simplifies the agents’ choice problem by having them look for an alternative that is satisfactory, or good enough, rather than best, or optimal.¹⁸ Instead of having the individual scan through all the available alternatives, satisficing requires him only to search until he finds one that, in Simon’s words, meets his ‘aspiration level’.

As an example of how this procedure is supposed to operate, Simon (1978a:468) provides the following:

If needles are distributed randomly in a haystack of size, H , with an average density of distribution, d , then to find the sharpest needle in the stack, we have to search the entire stack, and the search time will vary with H . Search time will be linear with size, which does not seem too bad until we remember that the haystack of life is essentially infinite.

If we are satisfied with any needle, however (after all, they are all sharp enough to sew with), then the amount of search to find one will vary with d —that is, will be independent of the size of the stack.

According to Simon, man will satisfice at two decision stages:

1. When deciding whether to search for alternatives. The agent will not search for alternatives as long as his present one meets his

aspiration level. It is only when the current course of action ceases to be satisfactory (either because its pay-off has diminished or because the aspiration level has increased) that a search process starts.

2. When selecting an alternative: as already described, the agent stops his search as soon as he finds a satisfactory alternative.

This theory places a lot of weight on the idea of an 'aspiration level', an idea that Simon draws from psychology. However, he is not precise about how this aspiration level is formed. He defines aspirations as 'expectations—adjusted in the long run to realities—of the result that can reasonably be attained'. These expectations cannot be obtained by individuals from an optimization procedure:

They are not formed on the basis of detailed evaluation of alternative courses of action. Indeed, their principal usefulness lies in the fact that they remove the necessity for such evaluations until the failures of existing programs indicate the need for innovation.
(Simon 1958:399)

About changes in aspiration levels, Simon suggests that

as the individual, in his exploration of alternatives, finds it *easy* to discover satisfactory alternatives, his aspiration level rises; as he finds it *difficult* to discover satisfactory alternatives, his aspiration level falls....

(Simon 1955:253; emphasis in original)

Apparently, the individual will sometimes be 'satisfied' with his present alternatives but sometimes, for no specified reason, his aspiration level will change, and he will engage in a new search. This leaves many questions unanswered. It is not clear, for example, how the aspiration level becomes 'adjusted in the long run to realities'. Even Simon has admitted that much more knowledge is needed about this notion, crucial to his theory.¹⁹ As long as it is not provided, it is understandable that most economists—particularly those trying to build 'formal' models of human behaviour—have not found the idea of satisficing very convincing. This in spite of Simon's belief that 'models of satisficing behavior are richer than models of maximizing behavior, because they treat not only of equilibrium but of the method of reaching it as well' (Simon 1959:297).

Of more interest for present purposes is the simplifying device

Simon calls the ‘combined use of markets and hierarchies’. Its description sheds light on Simon’s views of markets and prices.

Markets and hierarchies

The combined use of markets and hierarchies is, according to Simon, a simplifying device used at the social level. It is the analysis of this aspect of his thought that has been the motivation for the previous sections.

Simon shares with Richard Nelson, whose writings the next chapter will examine, and with other authors, a use of the term ‘hierarchy’ that does not distinguish clearly (1) between market and non-market hierarchies, and (2) between ‘appropriately decentralized’ hierarchies (i.e. hierarchical organizations that take into account the individual’s bounded rationality) and markets. The importance of making these distinctions will be shown later, in the next chapter. However, a lack of these distinctions frequently leads these authors to speak of ‘markets versus hierarchies’ without making it explicit whether they mean ‘market exchange versus firm’ (or ‘internal organization’ as Williamson (1975) calls the latter)²⁰ or ‘markets versus government planning’. Thus it is not clear how controversial some of Simon’s arguments are from a market-process perspective: after all, no market-process economist has ever denied the usefulness of firms and other market-created organizations. Even so, this difficulty with Simon’s terminology does not render a comparison of both theoretical perspectives impossible. This is because, whether he is discussing the advantages of firms versus market exchange or of government planning versus the market system, both discussions involve a view of what the advantages of price-mediated transactions are. And it is this view that will be contrasted with the market-process view of prices.²¹ However, Simon’s use of terminology should be kept in mind.

In the early 1960s, Simon (1962a:69) described the price mechanism as ‘just one—although an exceedingly important one—of the means that humans can and do use to make rational decisions in the face of uncertainty and complexity’. He believed that what he called ‘the great plan versus no plan debate’ depended on empirical comparisons between price mechanisms and planning mechanisms: ‘what costs they impose of information gathering and computing; how stably and rapidly they adjust the system to environmental

change' (Simon 1962a:70). Studies inside business firms, he argued, were 'already calling into question beliefs that allocation through markets simplifies information processing as compared with centralized allocative processes' (ibid.). He held that, in firms,

with modern computing equipment, the solution-finding process is handled centrally as readily as it could be through decentralized price mechanisms. The decentralized procedures simply do not yield the savings in information-transmitting cost usually claimed for them....

(ibid.: 61)

Two decades later, Simon engaged in slightly more detailed consideration of markets. He then stated that 'the social function of markets is to coordinate the decisions and behavior of multitudes of individual economic actors...' (Simon 1981:37). Markets are only one component of a spectrum of methods of co-ordination that includes 'statistical averaging, markets, bargaining, hierarchy, and voting' (ibid.: 38).²²

Simon's view of the market is, he says, not that of general equilibrium theory, which is

a dazzling piece of machinery that combines the optimizing choices of a host of substantively rational economic actors into a collective decision that is Pareto optimal for...society.

(Simon 1981:43)

The Pareto-optimal result is usually obtained under the assumption that the trading agents are perfectly (or at least optimally) informed about available opportunities, and that they are fully capable of maximizing their utility or profits. However, Simon (1981:39) points out, people are satisficers, and 'markets populated by consumers and producers who satisfice instead of optimizing do not meet the conditions on which the theorems rest'.

According to Simon (1981:41), 'no one has characterized market mechanisms better than Friedrich von Hayek...', so he adopts what he understands Hayek's view of the market to be, a view that does not claim optimality properties for it. In Simon's (1981: 39) view, the market manages to bring into 'patterned order' the 'productive efforts and consumption activities of a great population', although this pattern will not necessarily have any optimal properties. Simon

believes that it is this Hayekian view that best describes the markets of reality (43).

What is, in Simon's view, the achievement of the market? This is where considerations of bounded rationality enter the scene:

even without assumptions of perfect competition and perfect rationality markets provide a way of restricting how much we need to know about everyone else's business in order to do our own. The market mechanism may provide a way to reach tolerable arrangements in a society even if optimality is beyond reach.

(Simon 1983:89)

Markets constitute 'one of the mechanisms that enable human beings having limited information and computational capacity to operate more or less intelligently' (ibid.). To illustrate his point, Simon provides an example showing how a buyer needs to know only the price of a good to make his purchasing decision; no knowledge of how the good is made, of the supplier's problems, etc., is necessary. Markets and prices turn out to be

extremely powerful mechanisms in modern societies for helping each of us to make decisions without having to learn a whole lot of detail about other people who may be involved. All the relevant information is summed up in the price we have to pay in order to make the transaction.

(ibid.: 88)

In Simon's interpretation, Hayek's defence of the market does not rest on its achievement of optimal states. Instead, its defence is based on an awareness of the computational limits of human beings:

Market processes commend themselves primarily because they avoid placing on a central planning mechanism a burden of calculation that such a mechanism, however well buttressed by the largest computers, could not sustain. They conserve information and calculation by making it possible to assign decisions to the actors who are most likely to possess the information (most of it local in origin) that is relevant to those decisions.²³

(Simon 1981:41)

For Simon (1981:49, 51), markets are only one of the 'mechanisms for

distributing computational functions through a social system' that 'magnify the computing capabilities of individual human beings and enhance the possibilities of their collective survival'. Their 'chief competitor' is the hierarchical organization.²⁴ Markets and hierarchies are both

social schemes that facilitate coordinated behavior, at the same time conserving the critical scarce resource of human ability to handle complexity and great masses of information.

(Simon 1981:60–1)

Apparently, in Simon's view, the same advantage can be obtained from both types of organization:

matters of fact can be determined at the particular loci in an organization that are best equipped by skill and information to determine them, and they can then be communicated to 'collecting points' where all the facts relevant to a specific issue can be put together and a decision reached. Only a small part of the source knowledge and information and expertise need be present at the collecting points, and these points can themselves be numerous and dispersed through the organization.

(1981:49)

If this is the case, what differences exist between the two types of organization? In a somewhat puzzling statement, Simon mentions as a difference between markets and hierarchies that 'none of the theorems of optimality in allocation of resources...can be proved for hierarchic decision-making processes'. Therefore, he argues (1981:49–50), 'hierarchies sometimes resort to internal markets or schemes of "shadow prices" in order to come closer to allocative optimality'. However, these statements are not compatible with Simon's view of markets, which, as he stated, is devoid of optimality claims. Simon therefore leaves unanswered the question of what the advantages of price-mediated trading might be, particularly with respect to an *appropriately decentralized* hierarchical organization, an issue that the next chapter will consider in more detail.

When considering the case for hierarchies as opposed to markets as a form of organization, Simon mentions that they may make possible the internalization of externalities. In addition, they may be useful under certain situations of uncertainty. He says:

If what is uncertain is a multitude of facts about conditions in individual markets, then decentralized pricing will appear attractive; if the uncertainty is global, infusing major events that will affect many parts of the organization in the same direction, then it may be advantageous to centralize the making of assumptions about the future and to instruct the decentralized units to use these assumptions in their decisions.

(Simon 1981:51)

Given that, as he argues, at least one of these elements is almost always present, and given his use of terms, Simon may be stating that individuals in a market system will generally find it advantageous to choose a combination of price-mediated transactions and firms. Alternatively, he may be endorsing—with almost ‘neoclassical’ arguments—a combination of markets and government planning as the most desirable form of social organization.²⁵

A MARKET-PROCESS PERSPECTIVE

Complexity and the knowledge problem

The description of Simon’s ideas and the previous discussion of the market-process approach suggest that the type of ignorance Simon sees as due to complexity and the ignorance to be overcome by Kirzner’s entrepreneurial discovery are different. Whereas Simon is dissatisfied with the omniscient agent of equilibrium economics only when it is applied to complex situations (i.e. situations with too many elements to be considered for an optimal result),²⁶ Kirzner’s entrepreneur is necessary even in the simplest situations.

For example, Simon’s approach might not object to a neoclassical (equilibrium) description of the exchange expected to take place between two potential traders in an imaginary isolated island as unrealistic: the information would not tax the computational capacities of these agents excessively.²⁷ A market-process economist, on the other hand, would point out that there is a knowledge problem to be overcome by the islanders’ entrepreneurial discovery of the profitable trading opportunity. The opportunity might not be perceived by the agents, and so it may not occur to them to trade at the most appropriate terms, or, for that matter, to trade at all. (That

the knowledge problem may seem trivial in such a setting shows how much people's entrepreneurial abilities are often taken for granted.)

The difference arises because of the rather special views of perception and ignorance held by Simon. His arguments seem based on the idea that an individual's limited computational and perceptive capacity is the only barrier between his knowledge and 'objective facts'. And these limitations he describes in very physical terms. At the risk of oversimplifying, Simon seems to say there are many 'facts' to be seen, but that, because of their bounded rationality, individuals can see only some at a time (and that it would take them too long to scan all of them before making a decision). As pointed out above, Simon stresses '*omission* in perception', that the individual's perception 'encompasses only a minute *fraction* of all the relevant characteristics of the real environment...' (1959:306; emphasis added), as the problem. This enables him to imply that with the aid of computers man gets (even if only slightly) closer to (neoclassical) optimizing²⁸ and that he may achieve objective rationality in 'simple problem situations', in which case the neoclassical agent may be appropriate.

In the market-process approach, on the other hand, facts, even if they are few and simple, have to be noticed, discovered, by alert, active agents. What complexity does is increase the likelihood that instances of 'sheer' ignorance will happen, making a discovery process even more necessary. It is not that the number of facts will become so unmanageably large as to saturate a human mind, but that it becomes much more probable that many facts will not be noticed at all. In this view, the facts do not 'hit' the individuals 'in the face' while they act as passive receptacles of knowledge. Nor are these facts there simply to be seen by anyone who merely 'scans' the environment. As Kirzner (1983:29) has put it,

The entrepreneurial alertness with which the individual is endowed does not refer to a passive vulnerability to the impressions impinging on his consciousness during experience in the manner of a piece of film exposed to light; it refers to the human propensity to sniff out opportunities lurking around the corner.

Of course, Simon does not say human beings act in a purely passive manner: he describes satisficing agents as actively searching their environment for a satisfactory alternative. However, although it is not explicitly stated, this search must be

occurring within an already perceived (discovered) environment. How it was perceived is left unexplained. Simon (1957:253–4) suggests also that it is possible that the agent's failure to find a satisfactory option among perceived alternatives may lead him to search for additional ones. This suggestion reveals a need for a process of genuine discovery of these alternatives, such as that described by an entrepreneurial perspective. Yet this process is absent in Simon's framework.

When Simon describes the individual as 'searching' for new alternatives he only avoids the problem of discovery, leaving it unsolved: to search for something one must first perceive, even if only dimly, some environment that will then be searched. (This is implicitly recognized by standard search theory when the agents are assumed to be endowed with at least probabilistic knowledge regarding the area to be searched.) Entrepreneurial alertness is necessary for the agent to first discover what his environment is like. Simon's lack of awareness of this cognitive problem has also been noticed by critics of his work in the area of 'Artificial Intelligence'.²⁹

Also, whereas in Simon's view individuals 'scan' rather mechanically all possible facts in search of the most satisfactory alternative, the market-process view argues that entrepreneurial discovery does not happen in a purely random manner. According to Kirzner (1983:29), 'the notion of human action gives us...the recognition that people possess a propensity to discover what is useful to them'. Admittedly,

we know very little about the precise way in which pure profit opportunities attract entrepreneurial attention. But there can be little doubt about the powerful magnetism which such opportunities exert.

(Kirzner 1984a:415)

This, interestingly enough, is also an issue that has arisen in the area of Artificial Intelligence when referring to

the contrast between the ability of human beings to 'zero in' on relevant features of their environment while ignoring myriad irrelevancies and the enormous and admitted difficulty of artificial intelligence in determining what is relevant when the environment

presented to a digital computer has not, in some way, been artificially constrained.³⁰

(Dreyfus 1979:xii)

An advantage of the price system, for an entrepreneurial perspective, is, as already explained, that it translates unknown facts into attractive profit opportunities for entrepreneurial individuals who otherwise might not have found the discovery of such knowledge to be in their interest. Profits thus stimulate innumerable individuals to 'zero in', even if only fallibly, on knowledge that has gone unnoticed up to that moment.

Some of the cognitive issues that separate Simon from market-process economists are difficult and still have to be solved. However, given Simon's assimilation of human beings to digital information processors, it is suggestive that in the last decade or so something called 'knowledge representation' is considered 'the most central problem confronting artificial intelligence' (Cercone and McCalla 1987:xxi).³¹ In recent work in this area the following difficulties are mentioned:

In addition to the problems of acquiring a knowledge base, an intelligent agent must use knowledge to advance its goals. This requires being able to perceive what's happening in the world in order to act appropriately. It is necessary for the agent to perceive that it is in a situation in which knowledge is applicable, and to find the knowledge that is relevant to the situation. It is necessary to use knowledge to support the perception of what is the same and what has changed from a previously known state of the world. Among other things, knowledge will be used to perceive new individual entities that are present but formerly unknown and to identify new perceptions with preexisting concepts of known individuals where appropriate.

(Woods 1987:55–6)

The ways in which human beings handle these problems (however imperfectly when seen from an omniscient perspective) are emphasized by an entrepreneurial perspective and used for understanding markets. But they have not been included adequately in Simon's framework.

The distinctions made between the approaches of Simon and of market-process economists are not meant to deny the existence of

computational limits in human beings that could make some (or even most) situations too complex for them to handle without simplifying devices. As already stated, market-process economists emphasize a different problem, one that Simon's approach seems to assume as solved. And this difference helps explain the divergences in the interpretations of the role and achievements of markets that will now be considered.

On Simon's view of the market

From a perspective emphasizing bounded rationality what is of interest is the consideration of mechanisms that will allow individuals to make the best possible ('reasonable') decisions given their limited computational capacities. Proposals for comprehensive central planning appear to be, from this point of view, completely oblivious of these human limitations: no central planner could handle successfully the enormous quantity of facts required for planning a complex economy.

For someone concerned about ways to make complex problems manageable for individuals with bounded rationality, one alternative is to have them disregard some information (and thus have them satisfice instead of optimize). Another alternative is to reduce the amount of information they need to handle. In other words, to look for ways of summarizing information so that as much of it as possible can be used. The market system appears then, from such a point of view, as one ingenious system for achieving this end. As Simon (1983:88) describes it in the passages quoted above, prices act as summaries that contain 'all the relevant information'.

Nevertheless, a bounded-rationality theorist could find puzzling why anyone would believe that relying solely on the price system is the best way of coping with complexity. (Given the equation of the term 'market' with price-mediated trading, this is what market advocates are interpreted to be doing.) After all, Simon (1962a:69) argues, the price mechanism is 'just one—although an exceedingly important one—of the means that humans can and do use to make rational decisions in the face of...complexity'. Why not take advantage of other mechanisms capable of helping man cope with complexity? If these other mechanisms are also market alternatives—such as firms, contracts, and different institutional arrangements—nothing controversial is being said for an

entrepreneurial perspective. Furthermore, the conception of the market of market-process economists certainly includes the possibility of adopting these, and many other, alternative arrangements.

But a bounded-rationality approach might not see any theoretical reason for, say, Hayek's refusal to rely on government to improve on the result of the operation of the market, *as long as the task assigned to the planner does not become overwhelming for his computational capacities*. An emphasis on bounded rationality does not suggest that government planning is unlikely to be as effective as the market process: it only reminds the system designer that he should not place on a 'planning mechanism a burden of calculation that such a mechanism, however well buttressed by the largest computers, could not sustain' (Simon 1981:41). It has also been taken to imply that as technological progress continues expanding man's computational capacities the possibility of successful central planning becomes greater (although, as indicated before, it is not clear that this conclusion follows). This point has been made by Loasby (1985:12):

what prevents [Simon] from the advocacy of any kind of system of central planning is his continuing conviction that all our technical advances still leave far too much beyond the bounds of our rationality for that to be a sensible way of organizing our affairs.

Therefore, there would not necessarily be an inconsistency were someone to emphasize bounded rationality, and to adopt simultaneously a quite standard argument for government intervention as a remedy for externalities and certain types of uncertainty. The central planner, from Simon's perspective, is ignorant only in the sense that, because of the complexity of the problem he faces, he cannot take some facts into account at all. But given a simpler task, such as smaller interventions in a market economy, most of his knowledge problem would presumably vanish, enabling the planner to make a beneficial contribution.³² (Of course, it can still be argued that any intervention in the market, because of its secondary effects on the economy, is too complex for a planner to carry out optimally. The bounded-rationality theorist could reply to this that the planner would also have to be a satisficer.)

It is also of interest that, for Simon to be correct in saying that prices convey *all* the necessary information, he must still be thinking of equilibrium prices—in spite of his recognition that non-

maximizing agents will not achieve an (optimal) equilibrium—because they are the only ones that may fulfil this requirement.³³ This suggests that he may not yet have noticed some important modifications in the analysis of the market that his approach entails.³⁴

What Simon would probably really want to argue is not that prices contain all the information but, rather, that they contain only enough to generally allow reasonable behaviour to take place successfully. This changes matters only slightly. It is true that in his vision markets achieve only market-clearing, in the sense of equating quantities demanded and supplied. But this, even if not a full equilibrium state, still requires a significant amount of co-ordination to have been, somehow, achieved between demanders and suppliers. Simon emphasizes only the role of prices as information ‘summaries’, and does not explain the way in which they have come to embody such knowledge. As Kirzner (1984a:415; emphasis in original) has put it,

to make the assumption that markets are close to equilibrium is essentially...*to beg* (rather than to overcome) the Hayekian problem of dispersed knowledge.... One does not ‘solve’ the problem of dispersed knowledge by *postulating* prices that will smoothly generate dovetailing decisions.

The market-process view of the market goes beyond Simon’s. As pointed out, in the market-process perspective the economic problem involves the discovery of what the facts are. Market prices are stimulators of a competitive discovery procedure, not merely summarizers of information that economize on the limited computational capacities of economic agents. The main function of market (i.e. disequilibrium) prices emphasized by market-process economists is the provision of profit opportunities to spur the discovery of new knowledge by entrepreneurial individuals. It is not primarily the conveyance of relatively accurate information to satisficing price-takers.

From this perspective, the main problem with central planning — both of the comprehensive type and of the type intended merely to supplement or modify market activity—is not so much that the planner would not be able to handle the large number of facts required by the planning task (although this may also be the case). It is rather that he will not be able to discover, and thus make use of, as much knowledge as the entrepreneurial competitive process: participants in a central-planning regime face no personal profit

opportunities to encourage their discovery of new, previously unknown alternatives as alert entrepreneurs spurred by market profits do.³⁵

The market-process position does not claim that full co-ordination of individual plans actually ever occurs. Austrian economists argue that market activity is better understood as a (disequilibrium) process of discovery and constant change rather than as a state of perfect adjustment. On the other hand, Simon sometimes seems to hold a rather static view in which markets achieve a loose (satisficing?) equilibrium in which they generally clear without reaching optimality. This difference appears to stem from the dissimilar views held of ignorance and discovery, and from the difference between interpreting prices as only 'information-saving' devices and interpreting them as also part of an entrepreneurial discovery procedure. It also stems from the fact that market-process economists have used their views to develop a fuller theory of markets than Simon yet has.

MARKET-PROCESS ECONOMISTS AND COMPLEXITY

For Simon the main barrier preventing individuals from achieving omniscience is the excessively large number of facts they have to handle with their limited information-processing capacities. From the market-process point of view, complexity only increases the likelihood that useful knowledge will go unperceived, thus making entrepreneurial abilities even more necessary than under hypothetical simple situations. This entrepreneurial perspective provides a subtler elaboration of the problems posed by complexity. But this elaboration was not perfectly clear to Austrian economists from the very beginning.

From Mises's first writings, through the socialist calculation debate, and till the present day, Austrians have been experiencing an improved understanding of their own views of the market process. In their initial writings they were not wholly aware of the exact nature of the knowledge problem faced by a central planner, aside from noticing that improving on the results of a market economy seemed too complex for any human being. It was mostly in response to their critics that they became more aware of the cognitive role of prices, of competition as a discovery procedure, and of the crucial role of entrepreneurship in a market economy, thus producing the theoretical

framework described in chapter 2 (Kirzner 1988). And this sharpening of the market-process perspective continues to the present day.³⁶ In fact, the specific relationship between complexity and the knowledge problem has probably not been considered before as explicitly as here, where the bounded-rationality interpretation has made some clarification necessary.

This section considers some statements drawn from Mises, Hayek, and Don Lavoie: (1) to clarify further the distinction between ‘Simon’s’ ignorance and ‘sheer’ ignorance by pointing out a few inconsistencies in this respect remaining in market-process writings, and, as a by-product, (2) to show writings that could lead readers to see no difference between bounded-rationality and market-process theories.³⁷

In his 1920 critique of socialism, Mises (1920:102) argued that the problem facing any attempt to plan an economy centrally is that in the present economic system

no single man can ever master all the possibilities of production, innumerable as they are, as to be in a position to make straightaway evident judgments of value without the aid of some system of computation.

Emphasizing ‘the oppressive plenitude of economic potentialities’ (101), he says that

in the narrow confines of a closed household economy, it is possible throughout to review the process of production from beginning to end, and to judge all the time whether one or another mode of procedure yields more consumable goods. This, however, is no longer possible in the incomparably more involved circumstances of our own social economy.... The human mind cannot orientate itself properly among the bewildering mass of intermediate products and potentialities of production without such aid [i.e. economic calculation].

(103)

This statement is suggestive of Simon’s approach, because it says there is no knowledge problem under the simple conditions of a household where the production process can be ‘reviewed from beginning to end’. As already mentioned above, entrepreneurial discovery is necessary also in this situation, although it becomes even

more important under complex conditions. Mises had probably not yet become fully aware of the nature of the knowledge problem in this first round of the debate.

Also, although describing the human mind as 'bewildered' and disoriented may suggest that it faces too many facts, rather than that it is unaware of them, this interpretation is not the only possible one. Mises had in mind the situation faced by a planner dealing with an economy left over from a previous capitalist system. In this case, he was pointing out, the planner would suddenly have to make innumerable decisions just to keep the economy functioning. Were Mises to have considered the situation of a planner who had to start a socialist economy from scratch, it would have been easier to see that the problem would be that the planner would most likely remain ignorant of the 'bewildering mass of intermediate products and potentialities of production' discovered in a market economy. However, that was not a very relevant argument for his confrontation with socialist reformers who wanted to overthrow an existing capitalist system.

Some decades later, in his book *Human Action*, when considering the problems faced by a central planner, Mises mentions

the *embarrassing multitude* of producers' goods and the *infinite variety* of procedures that can be resorted to for manufacturing definite consumers' goods. The most advantageous location of each industry and the optimum size of each plant and of each piece of equipment must be determined. One must determine what kind of mechanical power should be employed in each of them, and which of the various formulas for the production of this energy should be applied.... Each case offers special conditions and requires an individual solution appropriate to these special *data*. The *number of elements* with which the director's decision has to deal is much greater than would be indicated by a merely technological description of the available producers' goods in terms of physics and chemistry.... The director does not simply have to deal with coal as such, but with thousands and thousands of pits already in operation in various places, and with the possibilities for digging new pits, with the various methods of mining in each of them, with the different qualities of the coal in various deposits, with the various methods for utilizing the coal for the production of heat, power, and a great number of derivatives.

(Mises 1949:698–9; emphasis added)

Again Mises could be interpreted as suggested above, although it is more likely that he was exemplifying how many things a central planner would have to discover entrepreneurially. This interpretation is strengthened by the fact that in this same book Mises emphasized that the problem faced by a planner was that he would lack the necessary knowledge, not that the task would be too complicated for him. This being the real problem to be overcome, argued Mises,

there is...no need to stress the point that the fabulous number of equations which one would have to solve each day anew for a practical utilization of the method would make the whole idea absurd even if it were really a reasonable substitute for the market's economic calculation.

(Mises 1949:715)

The market-socialists mistakenly interpreted the Mises-Hayek argument as relying on a computation problem (i.e. that the problem was the enormous number of equations a planner would have to solve) (Lavoie 1985a: esp. 80, 90–2). In responding, it was so clear to Mises that this was not the main problem that he added to the last edition of his book, as a footnote to his statement above, that ‘therefore the construction of electronic computers does not affect our problem’ (Mises 1949:715). This interpretation of Mises is reinforced by the fact that his statements appear in the book in which, as Kirzner (1988:7) puts it, ‘Mises emphasized the importance of seeing the market as an entrepreneurial process with unsurpassable clarity’.

Although it can be argued that Mises never became fully aware of the implications of his understanding of the market process, it is clear that his views were much improved and better articulated by the end of the 1940s. And, in this regard, Hayek's development was not very different.

Hayek, in some of his criticisms of socialism, also refers with some ambiguity to the ‘complex conditions of a large modern society’ as the main problem for a central planner. For Hayek

It is...[the] fact that one central authority has to solve the economic problem of distributing a limited amount of resources between a *practically infinite number* of competing purposes that constitutes the problem of socialism as a method.

(1935a:130–1; emphasis added)

Lavoie (1985a:21) argues that it was largely Hayek's (and Robbins's)

comments 'about the computational difficulties of the equation-solving approach [that] were responsible for misleading interpretations of their arguments'.

Hayek adopts a position very similar to Simon's in the following passages of his book *The Road to Serfdom*. He says:

There would be no difficulty about efficient control or planning were conditions so simple that a single person or board could effectively survey all the relevant facts. It is only as the factors which have to be taken into account become so numerous that it is impossible to gain a synoptic view of them that decentralization becomes imperative.

(1944:48–9)

The point which is so important is the basic fact that it is impossible for any man to survey more than a limited field, to be aware of the urgency of more than a limited number of needs.

(59)

In both paragraphs Hayek uses the word 'survey' rather than 'discover'. This is similar to Herbert Simon's preference for the word 'search.' The problem with these terms is that they suggest that 'facts' are readily visible to anyone who cares to look at them, if only they weren't so many and so complex. This ambiguity in some of Hayek's writings has also led other authors, aside from the market-socialists and Simon, to interpret him as being concerned with the overwhelming quantity of facts.³⁸

Hayek then proceeded, during the 1940s, to articulate the market-process view of the informational role of market prices and of the role of competition as a discovery procedure shown in chapter 2. He not only emphasized that much useful knowledge (skills, 'techniques of thought,' etc.) appears in a form that cannot be articulated or conveyed to any other point.³⁹ In his latest writings he also points out that an economic system should be capable of using individuals' abilities 'of discovering such facts as will be relevant to their purposes in the particular situation' (1979:190, n.). That is, it should provide a framework stimulating their entrepreneurial alertness. In this way, it has become increasingly clear that his view of markets points beyond Simon's.

Lavoie (1985a:66), who has provided an account of the socialist calculation debate from a modern Austrian perspective, says at one

point that ‘the whole of a static complex production process could conceivably be surveyed “from beginning to end” because the time and effort necessary for the survey would not render the gathered information obsolete’. However, it is doubtful that, without a framework stimulating entrepreneurship, time and effort would lead to the discovery of much knowledge. Similarly, at another point Lavoie says, when referring to market economies, that

the sheer number of persons whose productive activity has to be mutually coordinated grows until it includes most of the world’s population. Simultaneously, the complexity and variability of this productive activity increases even more dramatically. Whereas it had been possible within traditional society to achieve a more or less complete coordination of actions of members of the community in their production activities according to one established technology, in the Market this becomes quite impossible.

(Lavoie 1985b:37)

The idea that a high degree of co-ordination can be reached more or less completely in a ‘traditional’ society⁴⁰ raises the following question: to what extent is such a statement consistent with a market-process approach that insists that the world be viewed in disequilibrium terms? Surely the claim is not that it is only ‘the Market’ that operates in disequilibrium conditions? Rather, the point is that it is markets that overcome to the largest extent the ignorance that is the cause of disequilibrium. This apparent inconsistency may be due to some ambiguity in the term ‘co-ordination’. Because an attempt at its clarification sheds additional light on the distinction between types of ignorance, a brief digression follows.

The meaning of co-ordination

The term ‘co-ordination’ is used, particularly by modern market-process economists, as a substitute for the term ‘equilibrium’, to refer to the co-ordination of individual plans and actions.⁴¹ But there is some ambiguity in its use that may lead to confusion. To overcome it, a distinction will be made between ‘co-ordination of knowledge’ and ‘co-ordination of actions’.

The notion of co-ordination of actions seems quite clear. In the case of, say, a centrally planned, hierarchical organization such as

an army, it would refer to the mutual compatibility of the individual actions for the achievement of the superior's ends. Were this organization to grow in the number of its members or in the diversity of its tasks, the planner might reach a point in which he was overwhelmed by the complexity of achieving such co-ordination. Consequently, some individuals could receive contradictory orders, others would perhaps realize that they could not comply with their orders because the prerequisite tasks had not been carried out,⁴² some individuals would get in each other's way, and so on. But as long as this is not the case, a large measure of successful co-ordination, one of the characteristics of an efficient army, will be possible.

The notion of co-ordination of *knowledge* is somewhat subtler. An example provided by Kirzner may serve to illustrate it. Say that a mutually beneficial exchange opportunity between individuals A and B exists, but is not exploited because they are not aware of it. In this example,

by A's not buying B's apples, and by B's not selling them to A, each party is, because of ignorance of the other's 'existence,' acting as if the other did not in fact exist. A knows his own taste and assets; B knows *his*. But because the *bits of knowledge* are not coordinated, the *actions* taken by A and B are uncoordinated.

(Kirzner 1973:216; emphasis added)

In this case, lack of co-ordination of knowledge is the cause of discoordination of actions. How can this be solved? According to Kirzner,

the rule is simple and obvious: coordination of information ensures coordination of action. As soon as a single mind becomes aware of the situations and attitudes of two separate individuals between whom exist the conditions for mutually beneficial exchange, so that he perceives the opportunity so presented—as soon, that is, as the previously isolated pieces of information have become coordinated in the mind of a single human being—we are assured of action to coordinate the decisions, plans, and actions of the individuals concerned.

(ibid.: 219)

In the example above, as soon as either A or B becomes aware of the other's tastes, or as soon as a third individual becomes aware of A's

and B's tastes, there is co-ordination of knowledge; a bridge, so to speak, has appeared between the previously isolated pieces of information. (This bridge is due to entrepreneurship.) As soon as this disco-ordination is noticed, moves towards its elimination are assured because doing so is profitable, and co-ordination of action follows. Thus, for example, in a market, there would finally be (were full co-ordination to be achieved) no excess demands or supplies at the new price. However, although co-ordination of knowledge may be a sufficient condition for co-ordination of actions, it is not a necessary one in the usual meaning of 'co-ordination of actions'.

To understand how co-ordination of action does not imply there is also co-ordination of knowledge, the army example is again useful. Say the commander is ordering certain strategic manoeuvres, and there is among his soldiers one individual who, were he consulted, would prove to be a highly capable strategist. But the commander is unaware of his existence and therefore does not make use of his talents. There will not then be co-ordination of knowledge. However, in the sense described initially, there could still be (a visible) co-ordination of actions: the soldiers could act in a harmonious way towards the achievement of the commander's goal.

There are thus two types of disco-ordination that could occur in this example. (1) There could be disco-ordination of *knowledge* because there remain isolated bits of information that have not yet been 'bridged' by a single mind: in this case, not making use of the soldier's talent for strategy. (2) There could be disco-ordination only of *actions* because, for example, the number of men and different tasks have become overwhelming for a single commander's mind, given the present organization of this army. The first is an instance of 'sheer' ignorance, while the second corresponds to Simon's type. This also serves as an additional example of how the knowledge problem is related to the number of facts to be known: in this case, the larger the number of soldiers the more likely the commander is not to notice the individuals' abilities that may be useful to him.⁴³

Although this use of the term 'co-ordination of actions' conforms to its habitual use, Kirzner's example suggests that he prefers to confine its use exclusively to situations in which there is also co-ordination of knowledge. Thus when, say, A trades with B, unaware that he could trade even more profitably with C, who is also unaware of this opportunity, Kirzner would not describe this

situation as one of co-ordination of actions, while the habitual use probably would.

These different uses may be partly responsible for some confusion in an internal discussion among market-process economists: while Kirzner has frequently described successful entrepreneurial activity as a co-ordinating force in the economy (in the sense that it has built more bridges uniting previously isolated bits of knowledge), other economists have considered it as sometimes being disco-ordinating (in the sense that some individuals' plans are disrupted by the new situation). For example, if A discovers the opportunity to trade with C and exploits it, Kirzner will describe this as a co-ordinating activity because it has dispelled some ignorance from the economy (and has added none). Those with the alternative interpretation of the terms may consider this activity more ambiguous: although A and C have co-ordinated their activities, B's plans have been disco-ordinated by the entrepreneurial discovery.⁴⁴

However, Kirzner's argument is more appropriate to a market-process perspective, even if a different terminology may be desirable: what A's entrepreneurial discovery has achieved is the removal of ignorance, one step in overcoming the knowledge problem.⁴⁵ No additional ignorance was caused by A's change. Even if B's previous plans cannot now be carried out, the trade between A and B was a mistake that has now been corrected. It was a mistake not only on A's and C's part, but even on B's, who could have profitably arbitrated between them.⁴⁶

Of course, Kirzner's position need not deny that mistaken entrepreneurial activity (which is inevitable as long as men are not omniscient) is disco-ordinating. But it is not on entrepreneurial mistakes that the market-process approach relies for its explanation of the observed orderly working of markets. Also, from the market-process perspective, the inevitable entrepreneurial mistakes can only be reasonably held against the market system by proposing an alternative regime, capable of achieving similar results while committing fewer errors. Critics of the market have not yet shown such an alternative to exist.

With the distinction between co-ordination of knowledge and co-ordination of actions in mind, the case of simple situations or less complex societies can be seen in a different light. It may be true that in these cases co-ordination of actions may turn out to be simple and not worthy of much scientific analysis. But the

coordination of knowledge (and thus the better co-ordination of actions it would lead to) is not assured because the situation is simpler. As argued before, facts, even if few and simple, still have to be entrepreneurially discovered. 'Traditional' societies that (by definition) do not rely on an extensive network of markets may achieve a co-ordination of actions. But they will be, almost certainly, unco-ordinated as regards knowledge, because they do not provide the institutional framework to encourage individuals to discover as many available opportunities as possible. (After all, these societies were not *so* simple as to make the problem of entrepreneurial discovery completely trivial.) In fact, the stable, unchanging conditions attributed to such societies may not be the cause of an active entrepreneurial discovery process becoming relatively unnecessary. Instead, they are more likely the consequence of the absence of an institutional framework that provides incentives to entrepreneurship.

For example, the agricultural economy of feudal society may have achieved 'a more or less complete co-ordination of actions of members of the community in their production activities'. But it, almost certainly, did not achieve even an approximate coordination of knowledge (or a tendency towards it), because it provided little stimulus for anybody to discover the numerous better procedures and things to be produced that surely existed potentially at the time. Without such discoveries, those societies would, not surprisingly, remain relatively stable and unchanging. ('Stagnant' would perhaps be a better descriptive term.)

The importance of the distinction between these types of coordination is also that for someone emphasizing exclusively the coordination of actions in the habitual meaning, a decentralized entrepreneurial procedure becomes necessary only in complex and changing situations and societies in which centralized alternatives would require much time and effort. For those emphasizing 'coordination of knowledge', entrepreneurship is important in all situations, even in the simplest 'Robinson Crusoe' economies. Without the entrepreneurial element, even two individuals in a solitary island will be—barring the most unlikely coincidence—unaware of the better courses of action available to them, regardless of how much time and effort they have at their disposal.

Conclusions

Several market-process economists have made statements similar to those of Herbert Simon. However, these statements either were made in the early stages of the development of the market-process perspective, or were isolated sentences better interpreted in entrepreneurial terms, given their occurrence in contexts showing an understanding of the dynamic nature of market processes. These isolated quotes only show how the entrepreneurial perspective has been, and must still be, progressively developed and refined. Although some statements by market-process economists may have been somewhat ambiguous, as the entrepreneurial perspective becomes better articulated it becomes increasingly clear that it has gone in important respects beyond the bounded-rationality approach.

FROM SIMON TO NELSON

Simon has concentrated more on other issues and has not really attempted to provide a full theoretical picture of how markets work. This deficiency leads him to oscillate, as shown, between an equilibrium view of markets and an ill-defined alternative. Why then, it may be asked, has so much space been devoted to his ideas? Isn't Simon's view of the role of prices in a market economy almost identical to that of the equilibrium economists considered in the previous chapter and, thus, subject to the same criticisms? The justification for the present efforts has several aspects to it. First, if there is a similarity on this matter between Simon's ideas and the equilibrium approach, this is a finding in itself, given that it is one of the areas in which he is perceived to be offering an alternative view. In fact, this finding would serve as an antidote to the tendency, noted above, of some economists to overlook how far market-process economists have gone beyond Simon's work in presenting a theory of markets.

Also, the consideration of Simon's ideas has drawn attention to interesting problems of perception and discovery, and has highlighted additional differences between considering prices as part of a competitive discovery procedure or merely as a means of summarizing large numbers of facts.

Finally, the understanding of bounded rationality gained from the preceding pages will be useful—in fact, indispensable—for the task the following chapter undertakes: an analysis of the writings of Richard Nelson. In these writings a more detailed and more dynamic approach to the analysis of markets and central planning can be found.

Chapter 5

Change, responsiveness and co-ordination

Richard R. Nelson has pointed out that some economists' assessment of the market system and their belief in its superiority to central planning are not supported by economic theory.¹ Their arguments, he says, are made for a context of unpredictably changing resources, preferences, and technologies, rather than for the unchanging ('static') world of the standard economic approach. Nelson's argument is of interest because he engages in a sophisticated analysis of standard welfare economics and of what he believes to be Hayek's arguments in favour of markets. This analysis leads him to assert that a theoretical case for private enterprise not only has not been made but also, perhaps, cannot be made.

Nelson and his frequent collaborator Sidney Winter say that Simon has influenced their work but that, 'much more than the behavioralists', their 'concern has been with economic change' (Nelson and Winter 1982:ix, 35–6). This chapter will show that, in spite of this concern with change, Nelson's argument is fully within Simon's theoretical framework and is therefore subject to the same observations. Does this justify devoting a chapter to Nelson? It does, for the following reasons. (1) Nelson's perspective could be, incorrectly, considered indistinguishable from a market-process approach. Although he has shown similar concerns and interests, this chapter will argue that he has not fully grasped the nature of entrepreneurial competition as a discovery procedure and the role of prices in it. (2) Nelson's concern with unexpected change serves to clarify why change is important to a market-process perspective. (3) Nelson raises the question of the relative advantages of decentralized organizations and markets and thus provides an opportunity for outlining briefly an entrepreneurial view of organizations. The latter sketch will be useful for clarifying some discussions of 'markets versus hierarchies', as promised in the previous chapter.

Before presenting this outline, the chapter first summarizes Nelson's arguments and tries to show how they are fully within Simon's framework despite his emphasis on change. It then analyses his views and explains the importance of change for a market-process perspective.

WELFARE ECONOMICS AND ADVOCACY OF THE MARKET

Nelson starts by pointing out that advocacy of a market regime cannot be based on the 'twin theorems' of welfare economics² because 'everyone realizes that the real conditions do not meet the assumptions needed to produce' them (Nelson 1981:94).³ In fact, he says, some economists have hailed as the main achievement of normative theory its demonstration of the implausibility of any claim that unregulated markets in reality may ever produce optimal results (Nelson and Winter 1982:357–8). In addition, even welfare economics 'does not hold that competition can outperform central planning or any other organizational alternative'. Instead, it only proves that all other alternatives can at best match the results of competitive equilibrium (*ibid.*). The theorems of welfare economics, Nelson notes, do not support the negative reactions of many economists towards government planning and regulation of the economy: 'nothing in these theorems says that planning or regulation cannot be made to work (optimally)' (*ibid.*).

The arguments for the market

In Nelson's view, the arguments presented by market advocates are in fact based on criteria that are not part of conventional welfare economics. He discerns two types of problems emphasized in contemporary discussion about different forms of social organization:

1. Problems of information and computation. Nelson argues that, even conceding the way equilibrium economics poses the economic problem, to achieve the competitive solution 'an enormous amount of disparate *data* needs to be somehow acquired, brought to appropriate places for computation, and combined in a computation task of staggering scope.'

2. Problems of command and control. Once the previous step is somehow solved, a large number of agents (farms, factories, consumers, and so on) have to be both informed about what they should do and somehow ‘motivated or coerced to do the right things’ (Nelson and Winter 1982:358–9).

Nelson argues that many economists believe that the market has advantages over central planning in handling both problems:

Decentralized free enterprise operating in a market environment has been touted as superior to government ownership of firms and central planning on the grounds that the latter solution would involve vastly more information sending and would require the computation of much larger problems.

They also believe that

free enterprise is much more sensitive to demands, more strongly motivated to cater to preferences of consumers, and more likely to produce efficiently than is government bureaucracy.

(Nelson and Winter 1982:359)

Thus economists advocate the market because they hold

an implicit theory of the role of organization in solving big allocation problems, and some conjectures about the way in which different organizational structures perform.

(ibid.)

Nelson points out that a premise underlying these arguments, separating them from contemporary welfare economics, is the implicit or explicit presumption ‘that man’s *data* processing capacity, his ability to understand complex or novel situations, and his ability to gain agreement are bounded,’ the presumption of ‘bounded rationality’ (Nelson 1981:95). Without this limitation, he argues, the difficulties of collecting information for a central agency and computing a solution—the problem of information and computation—would simply disappear, and the problem of command and control would be, at least, much easier to solve.

(Nelson’s definition of bounded rationality tends to obscure the distinction made in chapter 4 between the view of ignorance of Simon and that of market-process economists. But, as he states, and

as will be shown, Nelson's notion of bounded rationality is the same as Simon's.)⁴

Equilibrium and change

As another proof of the irrelevance of modern welfare economics to this debate, Nelson perceptively notices that although its arguments are presented in an equilibrium setting,

most of the discussions of organizational problems explicitly or implicitly assume a dynamic context, in which preferences, resources, and technologies are changing over time in a way that is not fully predictable.⁵

(Nelson and Winter 1982:360)

If the organizational problem were to be considered for an unchanging world, or for one with fully anticipated changes, both the 'information and computation' and the 'command and control' problems could presumably be solved 'once and for all'. Any differences in the costs of solving them for each system 'might not loom large when...amortized over the time horizon of the economic system'.

Under steady-state conditions, [Nelson (1981:99) argues] resources are already basically allocated, and there is not much information flowing in the system to suggest that resource allocation needs to be changed. Big computational machinery is not needed, and the communication channels can be rather thin.

It is in such an unchanging setting, Nelson says, that standard theory usually analyses the organizational problem (Nelson and Winter 1982:360).

Once the same issues are considered for a world in 'economic flux', the nature and magnitude of the organizational problem changes, and any comparison between alternative economic systems has to be concerned with 'accuracy and speed of response to changed conditions' (ibid.). With changing *data* there will be changing optimum allocations, and although an omniscient agent would always know what the latter are, a bounded-rationality agent will not. The problem is that 'individual economic units have to figure what to do at any time with no assurance that what they were doing last time

is the appropriate thing this time' (Nelson 1981: 101). The problem 'involves responding to unforeseen changes in demand and supply conditions, and the challenge and opportunity of innovation' (ibid.: 99).

Nelson believes that many of these issues were dealt with quite explicitly in the Mises-Lange-Hayek socialist calculation debate. His interpretation of this debate helps to better understand his position.

Nelson and the calculation debate

According to Nelson (1981:94), Mises's proposal was 'not that private enterprise was optimal'⁶ but rather that market prices were 'an essential aspect of any decently working information and computation system' (Nelson and Winter 1982:360). Without them, producers would have no economically relevant information to enable them to decide which goods to produce and what combinations of inputs to use; it would be 'administratively infeasible' for a central planner to achieve a 'sensible allocation of resources' (Nelson 1981:94). With prices 'the computation problem simplifies to calculations of profit or cost' (Nelson and Winter 1982:360).

In Nelson's interpretation Lange's reply involved a recognition of the role of prices. Lange produced a solution in which consumers and plant managers would make their decisions on the basis of prices adjusted in response to excess demands and supplies by the central authority, 'in effect, simulating a market' (ibid.: 361). For Nelson, both Mises and Lange were arguing within a 'static frame'. Hayek, on the other hand, was one of the 'important exceptions to the static perspective'. He emphasized that 'the conditions facing the economic system—resources, technologies, and preferences—tend to be variable' and that these variations are 'particular to place and time, and unpredictable' (Nelson 1977:133–4).⁷ Nelson's interpretation of Hayek's argument is quite revealing of his bounded-rationality view of the economic problem and deserves therefore to be quoted in full:

Hayek's objection to the Lange proposal involved an interesting blend of propositions about information computation and about command and control. He argued that Lange's information-flow scheme could not handle the *enormous amount of detailed data* specific to particular times and places. A rise in the demand for

apples in local region X ought to be reflected in a rise in the price of apples in local region X. To channel excess-demand information back to the center for processing and to wait for central determination that prices ought to rise would be *time-consuming* and *cumbersome*. Farmers and grocers in region X would act to increase prices much more rapidly if they themselves had the authority to do so. A consequence is that internal supply would be stimulated and apples would flow into the region from outside *more quickly* under real free enterprise than under Langian socialism. Hayek also argued that private profit-seeking farmers and grocers have greater incentive to take the appropriate output actions in response to changed prices than would bureaucrats occupying the same jobs.

(Nelson and Winter 1982:361; emphasis added)

In Nelson's view, the Lange 'trial-and-error' solution is not an inappropriate response that simply assumes the knowledge problem away (as modern market-process economists, such as Lavoie (1985a:160–72), have argued). For him the problem with Langian socialism is that under conditions of change in the *data* the system becomes 'time-consuming and cumbersome' compared to the market.

NELSON'S POSITION; THE 'RESPONSIVENESS' ISSUE

Nelson believes that from the earliest days of the discipline economists have seen as virtues of the market system 'an ability to generate a variety of innovations, to screen and select from these, and to assure that in the long run most of the gains would accrue to consumers' (Nelson and Winter 1982:361). While proclaiming himself sympathetic towards this tradition and with what he calls 'contemporary informal normative discussion' (in which he includes Hayek), Nelson has some points of disagreement. The main one is that 'advocates of free enterprise have been too facile in arguing the merits of the stylized system in a stylized dynamic environment' (ibid.: 362). Their arguments, he says, are not only not supported by conventional economic theory, but also 'poorly articulated', and more often the result of casual empiricism and prejudice than of careful study (ibid.: 358–9). He believes there is 'little in the way of theoretical justification' for the argument that free enterprise outperforms central planning in a dynamic environment.

According to Nelson (1981:101), recognition of the presence of unexpected change in reality turns the 'organizational problem' into one of 'tracking as closely as possible a moving optimum that is knowable and achievable only by God'. The 'responsiveness' argument, which he attributes to Hayek, states that private enterprise does this best, that it is more responsive to change 'in comparison with the sluggishness of more bureaucratic alternatives' (ibid.: 95). In a world in which the facts are changing unpredictably, 'centralization imposes high information and calculation costs' (ibid.: 101). Central planning would require 'vastly more information sending' and 'the computation of much larger problems' (Nelson and Winter 1982:359).

Nelson (1981:102) finds these limitations of central planning convincing and accepts that 'in a dynamic context firms in a private enterprise system are lighter on their feet than firms in a centralized system'. But, he adds, an examination of the market alternative shows that it has limitations of its own. These limitations have been overlooked by its advocates because of their idealized view of markets and because of their almost exclusive concentration on the deficiencies of central planning.

For Nelson the organizational problem in a changing world consists of the attempt to 'track' as closely as possible a 'moving optimum'. From this view it follows that the speed of response of a system (the virtue attributed to the market) is not the only important quality: the direction or 'quality' of its response also matters. And it is in this respect that he finds the market inferior to central planning.

Very little formal theorizing. Nelson argues, has dealt with the 'questions of how a private enterprise system gets to equilibrium' or has produced an explanation of 'how a decentralized market system tracks a moving target at all, much less how it tracks it "better than another system would"' (ibid.: 102-3). In fact, the market system, 'without a central overview mechanism', has problems of co-ordination of firms' responses to change. These problems occur both within an industry (given a change, 'companies cannot decide what to do unless they know what other firms in the industry are going to do') and between industries (for example, the expansion of an activity requires an expansion of its suppliers of inputs) (ibid.: 103). For Nelson (ibid.: 103-4), 'not only do market-guided responses to changed conditions lack the advantages of central co-ordination' but, also, disequilibrium market failures inflict harms on people such as shortages, or slack and unemployment.

The identification of advantages of the market ('speed' of response) and of central planning ('quality' of response) suggests to Nelson the existence of a trade-off that may provide support for some combination of both systems (such as indicative planning or an 'industrial policy'). In this respect, he says that 'it is no wonder that many countries that have expressed strong general adherence to private enterprise have established mechanisms to try to make industry responses to changed conditions proceed in a more orderly and coordinated fashion, and to guide those processes so as to reduce the extent of income losses and individual hardships' (ibid.: 104).⁸

A MARKET-PROCESS PERSPECTIVE

On poorly articulated and informal theorizing

Nelson's description of the argument of advocates of free enterprise as 'informal' and 'poorly articulated' may be due to his holding the quite generalized view, mentioned before, of what constitutes appropriate theory. In spite of his professed disagreements with equilibrium analysis, Nelson favors a mathematically formalized approach to economics. He therefore wants a 'well-worked-out generally accepted formalism which explains how a decentralized market system tracks a moving target...' (Nelson 1981:102-3).

Nelson says that 'to someone who proposes that private enterprise responds accurately to changed market conditions, the failure of general equilibrium theorists to prove stability, save under very stringent conditions, should be a severe embarrassment' (ibid.). However, the literature dealing with the stability of general equilibrium could only provoke embarrassment if there were reason to believe it has modelled adequately the cognitive abilities of individuals in a market process. But, as shown in chapter 2, sophisticated theorists in this field of inquiry, such as Hahn and Fisher, have expressed great dissatisfaction in this respect. It is an agreed-on fact that no generally accepted formal theory of the disequilibrium behaviour of individuals exists yet. Furthermore, some theorists have argued that such a theory may not even be possible for any reasonably realistic setting.⁹ Nelson thus assigns too much importance to the very limited results obtained by stability analysis.¹⁰

Although there is therefore no reason for market advocates to be 'embarrassed', Nelson's complaint about the absence of a generally

accepted theory of disequilibrium adjustment can only be shared. Of course, most market-process theorists believe they are already well on the way towards such a theory. On the other hand, economists who find only mathematically formalized arguments acceptable will prefer the interesting lines of research suggested by the already cited work of Littlechild, Fisher, and Frydman, among others.

Nelson and the knowledge problem

Nelson (1981:95) believes bounded rationality is the main problem any form of economic organization has to face. At the risk of some oversimplification, his view of the alternative systems can be presented in the following manner: under a centralized regime, the plant managers, faced with new configurations of preferences, resources, and/or technologies, have to convey some information to the central planner, the amount of which will be constrained by the latter's bounded rationality. The planner must then use the limited information he has been able to receive to compute new courses of action that, in turn, have to be conveyed back to the plants.

Because of the planner's limitations, the whole process will take a significant amount of time, and his computed response will not be 'fine-tuned' to the local conditions to which it refers. (Of course, Nelson is aware that by the time the plants receive their new instructions the *data* will have changed again. This means the system will not manage to achieve equilibrium. However, this is not important because, in his view, neither organizational system achieves equilibrium in a world of change, even though private enterprise is faster than planning.) There is also the problem of control by the planner to ensure that the plant managers are both supplying him with the necessary information and complying with his instructions (Nelson 1981:102).

Private firms, on the other hand, faced with the same new 'facts', do not have to convey the information anywhere and have much smaller computational problems to solve. Also, the profit incentive makes them self-controlling. However, from this point of view, the decentralization of decision-making that produces these advantages of private enterprise also gives rise to its main limitation, the inter- and intra-industry co-ordination problems mentioned above.

As with Simon, the differences between Nelson and an entrepreneurial approach hinge to a large extent on the distinction between the ignorance due exclusively to complexity (in the sense seen in chapter 4) and the ignorance that has to be overcome by discovery. This difference leads each approach to adopt a different interpretation of the working and the limitations of alternative economic systems.

The comparison of economic systems

Nelson apparently assumes an arrangement in which firms (or plants) are not saturated by the facts of their local environment. Given that these units (firms or plants) perceive the new facts about their own environment, the alternatives (traditional central planning having been discarded) are a Langian planned regime, with a 'sluggish', but fairly co-ordinated, adjustment process, or private enterprise, producing an agile ('light on its feet') decentralized response with (potentially serious) co-ordination-of-response problems. This seems to suggest that mixed systems that combine government planning and market forces could be preferable.

The entrepreneurial approach, with its emphasis on the need for discovery of knowledge, perceives the alternatives differently. The problem with the Langian 'trial-and-error' solution is not that it is still too complex (i.e. that it doesn't simplify the task enough) for a central planner in a world of change. Instead, it is that it provides a solution only by assuming away the knowledge problem. Like Simon, Nelson seems to hold a theory of perception that assumes that at least the plant managers are 'hit in the face' by new configurations of their local *data* (or that they can easily 'see' them). The main problem is then to convey this information to the planner, solve the large computational problem, and send the results back to the managers. But this assumes too much knowledge to be already somehow given to the different individuals in the planned regime.

Although mistaken courses of action may often be noticed by plant managers (through, say, shortages or surpluses of goods), this does not imply that they will automatically perceive correctly what their circumstances are. As argued before, it is necessary that individuals discover, or 'zero in' on, the previously unknown knowledge. This does not happen automatically but requires something to attract their attention. Pecuniary profits do this in a

market economy. No equivalent mechanism is apparent in Lange's model.

The model proposed by Lange assumes away the knowledge problem in yet another way. Although Lange and—implicitly—Nelson seem to believe that the only unknowns to find out are the appropriate prices and quantities of goods, much more knowledge must also somehow be acquired before even attempting to carry out Lange's scheme. For example, the planner must know the list of the goods to be priced and produced (some should not be produced at all, some have never been produced before, some should be modified), the list of resources to be priced, the economic results of previous price structures, the patterns of consumer demand and resource supply in the period ahead, and so on.

The managers, on the other hand, should at least know about sources of resource supply, technological possibilities (many of which had never been known to be available), and 'the need for and possibility of any number of changes (innovative or otherwise) which changed patterns of tastes, for example, might make worthwhile' (Kirzner 1985a:32). All these things have to be discovered, and neither the Lange proposal nor Nelson's version of it provides any suggestion of how this will happen under central planning.

The main feature of the market system is not that it is 'light on its feet' because it is a decentralized system. It is primarily that prices, through the creation of profit opportunities, spur entrepreneurial discovery of much previously unknown knowledge. It is not so much that the market is quick to respond to perceived change but, rather, that it is better able to lead to the discovery of changed circumstances and of the most appropriate responses to them.

To summarize, for Nelson the extreme alternatives are either a slow, cumbersome planned system or a 'responsive' but not fully co-ordinated private enterprise system. Neither system solves in a fully satisfactory way the problems posed by bounded rationality. From an entrepreneurial perspective, on the other hand, the superiority of the market system over central planning in handling the knowledge problem is unambiguous. (This, of course, does not constitute a complete case for the market. The conclusion is only that the market is unambiguously better than planning as a discovery procedure.)

Furthermore, a full understanding of the knowledge problem faced by any economic system renders a comparison in Nelson's terms rather irrelevant: it doesn't make much sense to compare the 'responsiveness' of two systems to perceived changes once it is

realized that one of them has no mechanism to stimulate such perception.

The knowledge problem and change

To understand the effect of change on the knowledge problem it is useful to imagine a situation without it. The knowledge required by a Lange-type system would still have to be discovered in an imaginary unchanging world. If a centralized planning system has no discovery procedure comparable to that of the market it will be much less able to discover and exploit the most desirable alternatives—i.e. reach an equilibrium—even if the data *were* to remain unchanged.¹¹

This casts some doubt on the appropriateness of assuming that the economic systems are in equilibrium when comparing them under hypothetical conditions of no change. Nelson makes such an assumption explicitly when saying that with unchanging conditions both the ‘information and computation’ and the ‘command and control’ problems could be solved once and for all, with few if any significant differences remaining between central planning and private enterprise.¹² This assumption could, presumably, be justified if there were reason to believe that any system could reach the ‘optimum’ if given enough time. Additional time may be of use under static conditions if what has to be overcome is complexity in Simon’s and Nelson’s sense—particularly if change is believed to be one of the main sources of this complexity. However, additional time would not make much of a difference to a planned regime if its problem was its lack of entrepreneurial incentives and, thus, of a discovery procedure. The planned regime could, presumably, persist indefinitely in a state in which many desirable alternatives remained unnoticed. It can be argued that ‘under static conditions every person would soon find out...everything in his situation and surroundings which affected his conduct’ (Knight 1921:79) only if there are profit incentives to stimulate entrepreneurial discovery.

Although it is easy to agree with Nelson’s statements regarding the irrelevance of standard welfare economics to the debate regarding economic systems, his interpretation of the content of arguments such as Hayek’s cannot be accepted uncritically. Not only ignorance due to the complexity caused, or increased, by change, but also ‘sheer’ ignorance, has to be overcome by any economic system. Change only complicates the knowledge problem even more,

creating a constantly renewed need for the discovery of what the new circumstances are. For Nelson, on the other hand, as also for Simon, change seems to be important mainly because it requires an excessively large amount of information to be continuously moved around and processed by a central planner. From such a point of view, in the absence of change the differences between the economic systems might not be significant. It is therefore not surprising that for Nelson it is Hayek's supposedly stronger emphasis on change rather than Mises's argument that is most damaging to the Langian proposal.

Nelson's 'co-ordination problems'

In pointing out the co-ordination problems that, in his view, a market economy is likely to suffer Nelson appears, perhaps surprisingly, to have a rather 'static' view of firms under private enterprise and to neglect the wide variety of alternatives (such as different types of organizations, vertical or horizontal integration, contracts, and so on) that are available and are, in fact, used in a market economy for the solution of these co-ordination problems. This has led to his approach being labelled 'organizationally static' (Pelikan 1985).¹³

However, the possible occurrence of such co-ordination problems cannot be denied. Although profit opportunities stimulate entrepreneurial alertness, not all of them will be discovered. Also, some courses of action will be pursued because they are mistakenly believed to be profitable. In other words, there will be entrepreneurial mistakes in spite of the strong incentives to avoid them provided by profits and losses. Nelson's 'co-ordination problems' are examples of such mistakes—mistakes that could probably have been avoided by the parties concerned had they been aware of them. But these limitations of the market should not be compared with those of a hypothetical centralized system whose only liability is its 'sluggishness'. Instead, they must be compared with the limitations of a system in which, because of its lack of any discovery procedure, widespread ignorance of the correct courses of action—and, thus, mistakes—will be the rule.

Even if only limited government intervention in the market is being advocated, there is no clear reason why the intervener should be assumed to know which market situations could benefit from his

deliberate co-ordination, what type of co-ordination would be best, and so on. Of course, were one to start by assuming that the planner has superior knowledge of market conditions and opportunities, it could follow that his intervention is desirable. But this assumption is highly objectionable from an entrepreneurial perspective, and no planning proposal has yet explained satisfactorily how it could be justified.

An entrepreneurial view of organizations

Private enterprise, in Nelson's view, has the advantage of making the problems of the transmission of information and of the computation of solutions much more manageable for individuals with bounded rationality. In fact, the advantages Nelson attributes to private enterprise are mostly due to its being highly decentralized. This leads to the question of why the same, or even better, results could not be achieved by decentralizing a planning system. An answer to this question requires a prior clarification of terminology and an outline of an entrepreneurial view of organizations.

Many economists, not only Nelson or Simon, use terms such as 'market' and 'hierarchy' frequently to encompass different things. Therefore, in discussions of the relative merits of 'markets versus hierarchies', several interpretations of the terms 'markets' and 'hierarchies' are possible.¹⁴ First, when talking about 'markets' the reference may be to price-mediated transactions only or *also* to market-generated hierarchies (such as firms). (Market-process economists generally use the term in this latter sense.) Second, an allusion to 'hierarchies' may refer to market or to non-market hierarchies.¹⁵ Finally, 'hierarchies' could be seen as encompassing a range of organizations from highly centralized to decentralized. With these distinctions in mind, the following pairs of comparisons could be meant by 'markets versus hierarchies':

- 1 Price-mediated transactions versus highly centralized non-market hierarchies.
- 2 Price-mediated transactions versus decentralized non-market hierarchies.
- 3 Price-mediated transactions versus highly centralized market hierarchies.

- 4 Price-mediated transactions versus decentralized market hierarchies.
- 5 Several comparisons between market and non-market hierarchies, both in their centralized and in their decentralized versions.

Pairing (1) was, in most interpretations, the subject of the socialist calculation debate. Pairings (3) and (4) are, within economics, the subject of the analysis of organizations, a field pioneered by Ronald Coase's classic article (1937). Although not much will be said about them here, both the view of competition as a discovery procedure and some remarks made by Hayek regarding organizations, to be quoted below, provide hints of possible contributions of a market-process approach to this analysis.

The task to be attempted by this section, however, is more limited. First, it will point out some important differences between non-market and market (entrepreneurial) hierarchies—the pairs included under (5) —from a market-process perspective. Then it will try to show why decentralized hierarchies could not be perfect substitutes for the price system—the pair included in (2) —as solutions to the problems of centralized planning.¹⁶

Non-market- and market-designed organizations

It is not always apparent in the writings of the bounded-rationality authors whether they see any significant theoretical differences between non-market- and market-designed organizations. In fact, the absence of a clear-cut distinction between them suggests that the differences are not believed either to exist or to be very important. However, some differences can be pointed out that are important.

The differences appear most clearly when comparing how, for example, government, on the one hand, and private individuals, on the other, decide between carrying out an activity through 'markets' (i.e. competition via a price mechanism) or through 'hierarchies' (or organizations, or planning); and, if deciding to create a 'hierarchy', how they choose the type of organization to adopt, the specific personnel to be included, and so on.

In government hierarchies, market competition is replaced by central direction, or planning. This, to a certain extent, is also true of firms. As Coase (1937:389) put it, 'the distinguishing mark of the firm is the supersession of the price mechanism'.¹⁷ Planning, from an entrepreneurial perspective, has the disadvantage of not providing a

framework stimulating a discovery process the way a market system does. On the other hand, there are a number of difficulties that can apparently be dealt with better by such a form of organization. The avoidance of the duplication of effort and investment that may be present in competitive situations (Kirzner 1984a:417), problems of co-ordination of decisions and activities, the reduction of transaction costs, and other problems studied by the economic theory of organizations, seem to sometimes make some forms of planned organization desirable. However, the choice of what arrangement to adopt is not made in the same way by profit-motivated private individuals and government planners. Knowledge regarding what the trade-offs are between markets, with their scope for entrepreneurship, and organizations, with their own advantages, cannot be assumed to be 'given' to these decision-makers.

As individuals are not assumed always to 'know what it is they do not know', the choice between carrying out an activity within an organization or leaving it in the market cannot be made in terms of costs and benefits: knowing the latter would require individuals to know and evaluate what could or could not be discovered if scope were left for entrepreneurship, a logical impossibility. In the market system, different combinations of planning and markets can be tried out and are allowed to compete against each other. Although in a world in which equilibrium is never achieved this will not lead to the revelation of the optimal extent of planning, it will at least lead to the abandonment of the less successful forms of organization. Although more research is necessary to make this last statement more precise,¹⁸ nothing similar can be said of government planning. As Kirzner (1984a:417) has argued,

if central planning is *imposed* on an otherwise free market, whether in comprehensive terms or not, such planning will almost always involve the knowledge problem, and to an extent *not* likely to be justified by any advantages that centralization might otherwise afford [emphasis in original].

The knowledge requirements of creating an organization should not be underestimated. Suppose there is an individual (a 'planner') who believes a certain task should be undertaken. This 'planner' has also somehow decided that it would be best to do so through a hierarchical organization. For this he must know which of the numerous

conceivable organizational forms is most appropriate for his purposes. The particular form he chooses will determine which functions it will include, what tasks and responsibilities the individual filling each one will have to carry out, and so on. The 'planner' will need to know which employable individuals in the market are the most competent to fill each position. All these facts must be discovered.

Profit-motivated entrepreneurs will be stimulated to discover as much as possible of the knowledge necessary for setting up the most convenient hierarchical organization. No similar incentive system seems available to government planners. Thus an entrepreneurial perspective shows significant differences between organizations chosen by agents in the market and organizations imposed by government. The latter are much less likely to represent a wise decision in terms of grappling with 'the economic problem'.

The following statement of Coase, although it claims too much for the market because it seems couched in equilibrium terms, already hinted in 1937 at this conclusion:

The important difference between [economic planning and planning in firms] is that economic planning is imposed on industry while firms arise voluntarily because they represent a more efficient method of organising production. In a competitive system, there is an 'optimum' amount of planning!

(1937:389, n.)

Of course, there is not an 'optimum' amount of planning in a competitive system—at least, not in the usual sense economists attribute to the word 'optimum'. The argument is instead that there are profits to be made in such a system as long as the most appropriate degrees and forms of planning have not been discovered and implemented. These profits keep entrepreneurs alert and flexible regarding organizational forms. Non-market-created hierarchies, on the other hand, would most likely be established with better alternatives remaining undiscovered, and could continue to exist regardless of whether they were appropriate or not.

Markets and decentralized bureaucracies

Nelson pointed out once that even though comprehensive central planning might be impossible to carry out successfully because of bounded-rationality considerations, this is not a sufficient argument for turning over to markets for a solution. There is still another alternative to consider: decentralized bureaucracies. In Nelson's words,

[Hayek] argued that...simulated socialist markets were likely to operate much more sluggishly than real capitalist markets. While he did not spell out in any detailed way why this should be the case, obviously he did not believe that bureaucratic decentralization would be effective. It is apparent that many contemporary economists agree with him.

(1977:134)

In spite of this apparent agreement among economists, Nelson states, it is not 'clear that it is impossible to devise responsive decentralized systems without going to the unregulated market, and there may be good reasons for trying' (ibid.: 141).

This is a reasonable argument from a perspective emphasizing bounded rationality as the main constraint on a central planner. After all, Simon (1979:501) mentions the division of 'the decision-making task among many specialists, coordinating their work by means of a structure of communications and authority relations' as a procedure already widely used by organizations to 'transform intractable decision problems into tractable ones'.¹⁹ In fact, Simon criticizes the economics literature for treating organizations 'as highly centralized structures in which all the important decisions are made at the center', without noticing that 'real-world organizations behave quite differently' (1981:48). Although the context of this specific argument suggests that Nelson has in mind decentralized *government* bureaucracies and that he is using the term 'markets' to include private organizations, this is just one instance of the absence, in the bounded-rationality approach, of differentiation between price-mediated transacting and decentralized organizations.

This absence is also noticeable in Simon's writings. Business organizations, Simon believes, '*like market economies*, are vast, distributed computers whose final choice processes are substantially decentralized' (ibid.: 49; emphasis added). This, as pointed out in

chapter 4, leaves unanswered the question of what differences, if any, there are between a market and a decentralized bureaucracy.

The distinction this section will make is between price-mediated transacting and appropriately decentralized organizations in general, although the differences pointed out in the previous section between market and non-market hierarchies are also important here.

Even though Simon may be right in criticizing some economists for holding excessively simplified notions of organizations, there is a sense in which economists are ultimately right in regarding any organization as being centralized. This can be better explained with the help of an example. Suppose that an individual believes that the organization he is creating should be designed as a decentralized hierarchy (perhaps because of the complexity he thinks characterizes the task to be performed). He will then specify which decisions will require his intervention and which will better be left to the different individuals in his organization.²⁰ He will perhaps even reward the latter, through some incentive system, for making what *he* regards as successful decisions (thus stimulating the use of the entrepreneurial abilities he may have perceived them to possess).

This description should start to suggest that markets and decentralized hierarchies, even though they might be considered equally good simplifying devices by a perspective that emphasizes the need for economizing the individual's limited computational capacities, are quite different when viewed from a perspective concerned with discovery. Even if this organization may be said to be decentralized, the perceived need for it and its specific design has come from a single entrepreneurial mind—that of the 'planner'—and the members of the organization operate within the guidelines, and for the purposes, set by this mind.

Hayek (1973: esp. 46–54) deals with a similar issue when he contrasts the 'rules' (as opposed to detailed *commands*) of an organization with those of a 'spontaneous order'—a category in which he includes the market. According to Hayek,

what distinguishes the rules which will govern action within an organization is that they must be rules for the performance of *assigned* tasks. They presuppose that the place of each individual in a fixed structure is determined by command and that the rules each individual must obey depend on the place which he has been

assigned and on the particular ends which have been indicated for him *by the commanding authority*.

(ibid.: 49; emphasis added)

the general rules of law that a spontaneous order rests on aim at an abstract order, the particular or concrete content of which is not known or foreseen by anyone; while the commands as well as the rules which govern an organization serve particular results aimed at by those who are in command of the organization.²¹

(ibid.: 50)

This makes ‘decentralized hierarchies’ different from a market. Even if decentralized organizations may allow room for some entrepreneurial initiative on the part of their different members, these initiatives will be constrained to occur within the limits determined by the overall design of the ‘planner’.²² Only those individuals approved by the ‘planner’ will be allowed to exercise this limited entrepreneurial initiative. And, in many cases, the creative individuals will have to convince the ‘planner’ of the advantages of their new ideas, so that the application of these ideas will depend on the planner’s entrepreneurial vision. Organizations may therefore be said to be *entrepreneurially centralized*.

Markets, on the other hand, leave much more room for entrepreneurial imagination and creativity. The results that entrepreneurs may achieve in the market are not limited to the possibilities discovered by any single limited mind. The profit incentives markets provide are rewards not for the successful achievement of tasks assigned by any overall planner, but rather for the task of better co-ordinating a diversity of individual plans. Any individual in the economy is free to exercise his entrepreneurial imagination and to try out ideas he believes to be promising. This is frequently referred to as ‘free entry’.²³ In this way the market taps everybody’s entrepreneurial abilities.

To the stimulus to innovation and discovery provided by pecuniary profits free entry also adds the powerful spur of rivalrous competition and potential pecuniary losses. Every individual in the market knows that if he does not strive to discover and exploit profit opportunities, others will, and will deprive him of them. In all these respects organizations, even decentralized ones, are at a disadvantage with regard to markets. Still, as mentioned before, there are sometimes other economic

reasons that make the adoption of such forms of organization desirable for market participants.

To summarize: from an entrepreneurial perspective, organizations can never be fully decentralized, in the sense that they will always, by definition, require a designing mind (or a few minds in collaboration). Markets, on the other hand, are a fully decentralized alternative, in the sense that they consist of a set of rules with no specific tasks or functions assigned to anyone. The important implication of this is that markets leave much larger scope for entrepreneurship to be exercised. Decentralized organizations may make almost no use of the entrepreneurial abilities of people other than their designer, or, if they attempt to do so, will stimulate alertness only in forms, and towards goals, already broadly perceived as desirable by the entrepreneurial mind of the designer. Thus, while from the bounded-rationality perspective there may be no significant differences between these alternative forms, from an entrepreneurial view there are differences and they are important.

CONCLUDING REMARKS ON BOUNDED RATIONALITY

The last two chapters have analysed critically the views of markets and planning that emphasize the specific cognitive limitation of human beings described as bounded rationality. Although the authors in this tradition express criticisms of the optimizing, equilibrium approach similar to those presented by market-process economists, there are significant differences between these perspectives.

The entrepreneurial perspective is concerned with a knowledge problem that has more to do with 'sheer' ignorance of the facts, whether many or few, than with an ignorance due only to the quantity of things that need to be known by minds with finite storage and computing capacities. This leads it to be concerned with ways of stimulating discovery and 'alertness', and to judge economic systems in terms of how well they do this.

The emphasis on computational and storage limitations by bounded-rationality theorists leads them to see the performance of alternative economic systems in terms of how well they enable individuals to cope with the facts produced by an 'information-rich' world. This apparently makes comprehensive central planning seem an impracticable alternative. But it also makes these theorists see the

main virtue of markets in the fact that they keep situations informationally simple for individuals by generating prices that serve as information summaries (as emphasized by Simon) and by operating with decentralized units that face smaller informational problems (as emphasized by Nelson). As prices are not 'perfect' summaries, as decentralized private firms may face co-ordination problems, and as planning systems can also be decentralized, this approach does not seem to show the market system as an alternative with net informational advantages. This is in contrast to an entrepreneurial approach. The understanding it provides of the knowledge problem, and of the consequent need for a discovery procedure, shows the market system as having an advantage in this regard and highlights difficulties with any type of planning that remain to be solved satisfactorily by its proponents.

Conclusions

THE ARGUMENT RESUMED

Hayek's 1945 essay gave rise to a much broader understanding of prices by economists. Prices are generally seen nowadays not only as incentives for individuals to make decisions consistent with the scarcity of resources, but also as performing some informational role. Both the economics of information and the bounded-rationality approaches have been interested in this role of prices and have acknowledged Hayek's influence. Economists working in these areas have analysed critically, in more or less detail, his arguments (and sometimes Mises's).

This book argues that some implications to be derived from Hayek's approach to prices have gone unperceived. While the economics of information and bounded-rationality theory have yielded interesting insights into some possible informational roles of prices, they have missed an important one. The previous chapters tried to remedy this failing by examining representative work done from these perspectives and contrasting it with a third approach to problems of knowledge in an economy. This third approach, a development of the work of Mises, Hayek, Kirzner, and other modern Austrians, highlights the disequilibrium role of prices. Economists have mistakenly tended to think that whatever insights the Austrians have to offer in this respect have already been absorbed into their theories. This book tries to show that such is not the case.

The three approaches to the informational role of prices can be described in summary fashion:

1. One approach taken by a large part of the literature is to analyse the informational role of prices within an equilibrium framework. This approach has been concerned mostly with what were shown in

chapter 3 to be two different informational roles that prices may perform. On the one hand, prices may serve as knowledge surrogates in the sense that individuals optimizing with respect to them act as if they knew more than they actually do. Prices thus serve to reduce the amount of information people need to have for them to make decisions that are correct from the standpoint of the economy as a whole. On the other hand, prices may serve as sources from which individuals may try to infer knowledge about the traded good.

Much work has tried to determine whether prices perform these roles effectively, that is, whether they are informationally efficient. Grossman and Stiglitz's conclusion, like that of other economists, is that, in most realistic situations, prices are not informationally efficient, because of different types of externalities. (This, of course, raised the issue of what a relevant standard of efficiency ought to be, an issue previously brought up by Harold Demsetz.)

2. Also interested in the informational role of prices, and of markets in general, are theorists adopting a bounded-rationality framework. Their main dissatisfaction with the equilibrium approach, as chapter 4 showed, is in its use of optimizing individuals facing costly information as a framework for the analysis of informational problems. Bounded-rationality theory emphasizes that the cognitive limitations of individuals are more serious, and that these limitations make optimizing behaviour impossible. However, at this stage of its development, the bounded-rationality analysis of the informational role of prices, at least as found in Simon's writings, does not differ significantly from that of equilibrium theory. Prices appear in Simon's work as useful summaries (knowledge surrogates) that allow individuals to make reasonably good decisions in spite of their bounded rationality.

Chapter 4 also tried to show that bounded-rationality theory, though it has pointed correctly at weaknesses of the omniscient, optimizing agent of equilibrium theorizing, still tends to underestimate the knowledge problem by emphasizing only certain limitations of human beings as information processors. This underestimation is related to Simon's adoption of a special notion of complexity and a theory of cognition that, as was shown, are highly questionable and have received serious criticism. The limitations of the bounded-rationality approach as it now stands should be noticed, particularly because it is often perceived as the only alternative to the economics of information for the analysis of informational issues.

3. The argument this book presents is that the informational role of prices is understood more fully if analysed in disequilibrium terms. Although contemporary economic theory has not been very interested in the analysis of disequilibrium, widely respected economists, cited in chapter 2, argue that many important phenomena may never be understood if economics is confined exclusively to the analysis of equilibrium states. The crucial role of prices as incentives to entrepreneurial discovery, this book argues, is such a phenomenon. The main case against analysing only equilibrium states is that in equilibrium many of the serious informational problems actually faced by individuals are already solved. Only when disequilibrium market processes are analysed can prices be seen as stimulating genuine entrepreneurial discovery. This very important informational role is not apparent in equilibrium. In fact, it is therefore not so much that this informational role has been neglected by most economists as that it has been missed because the equilibrium framework currently used by the economics of information does not encompass it.

For the disequilibrium analysis of prices, this book drew from the work of economists who are interested in the market process rather than in equilibrium states. In this approach, radical ignorance and entrepreneurial discovery play a central part. Rivalrous competition is the market's way of attempting to overcome, even if only partially, the knowledge problem. Prices perform a crucial informational role in this competitive process: disequilibrium prices provide profit opportunities that spark entrepreneurial discovery of new knowledge. They are not merely summaries or sources for inferring information. At the same time, this approach does not require denying the importance of these other informational roles of prices.

Disequilibrium prices might not perform those roles at all successfully if 'being in disequilibrium' were to mean that they are completely 'wrong' and changing almost randomly. The only reason why market prices are usually fairly dependable, and may be able to perform the equilibrium informational roles reasonably well, is that the profits they provide continuously stimulate a competitive discovery process that incorporates valuable knowledge into them. Without this capacity, prices would most likely perform no informational role at all. It is in this sense that the disequilibrium informational role of prices is literally fundamental. (Even someone who prefers to view market prices as equilibrium prices, it should be

noticed, would have to rely implicitly on this role to ensure that equilibrium is achieved and always maintained.)

Aside from emphasizing prices as summaries of information, the bounded-rationality approach, as exemplified by Nelson's work, attributes advantages to the market system that are to a large extent due to its being highly decentralized. To show why the same, or even better, results would most likely not be achievable by merely decentralizing a centrally planned economy, chapter 5 pointed out some important differences between market and non-market organizations and between decentralized hierarchies and markets. Market competition and profits stimulate entrepreneurs to discover when organizations may be worth while and as much knowledge as possible for setting up the most appropriate type of organization. Non-market-created hierarchies, on the other hand, would most likely be established with better alternatives remaining undiscovered, and could continue to exist regardless of whether they were appropriate or not. Regarding the differences between decentralized organizations and markets, it was argued that, from an entrepreneurial perspective, organizations can never be fully decentralized, because they require a designing mind that will also limit the range of entrepreneurship that can be exercised within them. Markets, on the other hand, are fully decentralized and leave a much larger scope for entrepreneurship. The conclusion then is that price-mediated transactions have some informational advantages even over decentralized organizations, and that organizations set up by the market will most likely be more economically worth while than those created otherwise.

Viewing prices as incentives to entrepreneurial discovery opens up potentially fruitful areas of research. As an example of the many issues that may be worthy of investigation, some points only tangentially touched on here can be briefly mentioned.

SOME AREAS FOR FUTURE RESEARCH

1. The ways in which the different informational roles of prices may be interrelated in the market process are one topic that should be examined. Chapter 3 took a small step in this direction when examining how market prices, that is, disequilibrium prices, could still possibly perform a role as knowledge surrogates quite effectively. The

way in which disequilibrium prices may serve as signals or sources of information remains to be analysed. Also, the fact that disequilibrium prices will sometimes perform these functions somewhat unreliably suggests reasons for the existence of many other institutions in a market economy that also perform informational roles and that complement or replace prices. Although many informational institutions and devices (such as rules, contracts, brand names, warranties, reputation, and so on) are already being studied, mostly from an economics-of-information approach, their possible usefulness in the face of sheer ignorance and uncertainty should also be considered.

2. Given the contrast drawn here between the theories of cognition implied in bounded-rationality and in market-process theory, it would be of interest to learn more about the ability of individuals to discover new opportunities and to act reasonably well in complex situations. Experimental economics of the type mentioned in chapter 3 may yield some interesting results, although it is likely that part of this research will exceed the competence of economists. Perhaps psychologists,¹ and researchers in Artificial Intelligence, will at some point be able to provide economists with useful knowledge about the entrepreneurial capacities of individuals.

3. The role of money in the discovery process. For historical reasons, market-process theory has traditionally been concerned with analysing the informational advantages of markets over different types of centrally planned systems. However, the incentive role attributed to pecuniary profit opportunities suggests that a monetary market economy also has advantages as a discovery procedure over barter economies. The market-process approach presented here appears to attribute more effectiveness as entrepreneurial incentives to pecuniary profits than to the 'psychic' profits obtainable in a moneyless economy. But this argument should not be interpreted as a resuscitation of *homo economicus*: the point instead may be that, in a disequilibrium world, money may not only facilitate exchange, as generally explained, but may also offer more scope for entrepreneurship by making more alternatives profitable to larger numbers of entrepreneurial individuals.

The absence of money may increase the complexity of the knowledge problem, thus making it much more likely that beneficial alternatives will go unnoticed.² Relatedly, because of the liquidity of money, pecuniary profits may perhaps attract more entrepreneurs than do profits in kind.

As a by-product, a better understanding of the role of money in the discovery process may also shed light on the effects of inflation on entrepreneurial discovery.

4. Much work remains to be done regarding the role and the effects of entrepreneurship in the market process. One issue that needs additional study is the determination of what entrepreneurship can be said precisely to achieve, given that 'equilibration' may not be a wholly satisfactory description. (This should not be interpreted as the adoption of a Schumpeterian position that finds 'disequilibration' to be a better description.)

A few considerations may serve to illustrate the point. The *data* faced by participants in a market economy consist of their perceptions of the availability of resources and of technology, and of other individuals' planned supplies and demands for goods. These perceptions lead to plans that, even if they turn out to be correct in the sense that they lead to profits, would not necessarily coincide during the market process with the 'correct' plans a hypothetical omniscient observer would recommend. This is because the plans an agent finds profitable are largely dependent on the behaviour, both correct and mistaken, of others: for example, a brick manufacturer could find it profitable to sell bricks to a builder who later realizes that his project was a mistake.

The implication of this is that profit opportunities may not always stimulate the discovery of courses of action that are 'correct' from an omniscient perspective. Instead, they may be incentives for co-ordinating individual plans, even when some of these plans may be mistaken. (Of course, profits and losses will also act as incentives for the correction of the mistaken plans.) This feature may explain why markets, without ever reaching something like equilibrium, still seem reasonably orderly. And it suggests that 'equilibrium' may perhaps be profitably substituted by some other term for the description of the degree of plan co-ordination achieved by a market system.

Notes

1 Introduction

- 1 For the sake of avoiding confusion, this book follows Machlup's (1980: 8) suggestion not to distinguish between 'knowledge' and 'information' when referring to '*what* people know or are informed about.' Although, as Machlup shows, many important distinctions can be drawn between these terms, and although 'all information, in the sense of the contents conveyed, is knowledge' but 'not all knowledge may properly be called information' (58), these differences are not of much significance here (see esp. 8–9, 56–8).
- 2 Some contemporary economists in this approach, wishing to rid economics of its dependence on any notion of equilibrium whatsoever, would probably object to the description of their work as 'disequilibrium' theory.

2 A theory of the market process

- 1 See also Hey (1981:3–4).
- 2 And thus, for example, there have been, and continue to be, discussions regarding whether macroeconomic models that allow for unemployment are or are not disequilibrium models. According to Fisher (1983:1), these 'are not truly disequilibrium investigations, although they are sometimes misnamed as such'.
- 3 Similarly, Hahn also says, 'an economy is in equilibrium when it generates messages which do not cause agents to change the theories which they hold or the policies which they pursue' (1984a:59). For a comparison of Hayek's and Hahn's notions of equilibrium, see Littlechild (1982) and White (1982).
- 4 According to Hey (1981:238), 'it is...important not to equate equilibrium in a market with market-clearing, and disequilibrium with non-market clearing'.
- 5 According to Fisher (1983:2), 'by building a full model of disequilibrium behavior we obtain considerable insight into a number of areas. These include the nature of fixed-price, quantity-constrained equilibria, the role of

money, the behavior of arbitraging agents, and the function of the stock market.'

- 6 See also the remarks in Arrow (1986: s. 387).
- 7 See also the criticisms in Littlechild (1977).
- 8 The founder of the tradition, Carl Menger (1871), had already shown a somewhat unusual concern with problems of knowledge. Even so, some recent interpretations, such as, for example, Streissler (1973), seem to exaggerate Menger's sophistication in this area.
- 9 In contrast to this view of competition, most economists can be described as seeing it only as a 'disciplinary procedure', to use Boudreaux's (1987) term.
- 10 Different types of ignorance are distinguished below.
- 11 For some differences between this view of entrepreneurship and Schumpeter's see Kirzner (1973:72–4, 79–81, and esp. 125–31).
- 12 However, this dichotomy should not be taken too literally: the Misesian approach 'treats both tasks—that of identifying the relevant ends-means framework and that of seeking efficiency with respect to it—as a single, integrated human activity' (Kirzner 1973:34).
- 13 For an attempt, see Littlechild and Owen (1980).
- 14 This is not to deny the usefulness of sometimes abstracting from the temporal element and dealing with current prices.
- 15 On this distinction, see Kirzner (1985a:34–6, 94–8), Kirzner (1973: 228–9), and Lavoie (1985a:143–4).
- 16 For support of Hayek's statement that his constitutes the 'older' view of competition, see McNulty (1967, 1968).
- 17 For example, Hayek (1940:191) has referred to 'the failure to understand the true function of the price mechanism, caused by the modern preoccupation with stationary equilibrium...'.¹
- 18 See Garrison (1982:131–8). For examples of different positions see O'Driscoll and Rizzo (1985), and the review of their book by Kirzner (1985b); Cowen and Fink (1985:866–9); Thomsen (1986); Fehl (1986: 72–86); and High (1986).
- 19 This makes Mises's entrepreneur different from Schumpeter's: the latter 'destroys' existing equilibria.
- 20 Although the argument may merit lengthier treatment, the same comments are applicable to 'search theory'. Kirzner (1973:34).
- 21 See, e.g., Kirzner (1983:137–53).
- 22 Buchanan (1985:19) makes a similar distinction when he mentions as motives for lacking some information that it may be 'because it is too costly to obtain, or because *it is simply overlooked*' (emphasis added).
- 23 See, e.g., Polanyi (1969).
- 24 See, e.g., Lavoie (1986).
- 25 See also Reder (1947:147–51).
- 26 See Buchanan (1985:26). Buchanan's book shows different difficulties with the usual efficiency analysis, but also points out unsolved problems in the comparative institutions approach. Regarding the latter, Buchanan (1985:44) says, 'there is not even a theory of intersystemic efficiency comparisons in the same sense in which there is a theory of intrasystemic comparisons'.

- 27 Unlike in Demsetz's framework, these mistakes and inefficiencies are due to 'sheer' ignorance.
- 28 Buchanan (1985:15–16) may be referring to this (although he may really have a more Schumpeterian framework in mind) when he describes as the 'diachronic efficiencies of the market' that 'competition in nonideal markets generates incentives for behavior that tends toward the more perfect satisfaction of the conditions of the ideal market...'.

3 Equilibrium prices and information

- 1 Also: 'the merit of perfect competition is that it would cause prices to transmit information reliably...' (Scitovsky 1954:150); 'a fundamental result of competitive analysis is that market prices contain all of the information required for consistent and efficient decision-making by firms and individuals' (Kihlstrom and Mirman 1975:357).
- 2 Grossman (1976, 1978); Grossman and Stiglitz (1976, 1980); Grossman (1981). Many of these essays have been reprinted in Grossman (1989). A very similar approach to Grossman and Stiglitz's appears in Green (1973, 1977).
- 3 See Fama (1970).
- 4 For a recent summary of Grossman's work in this area, see Kreps (1988).
- 5 The two quoted passages reflect frequently encountered interpretations of Hayek's argument. See, for example, Sowell (1980:38, 75, 79, 80); Arrow (1974:158); Frydman (1982); Hahn (1984b: esp. 128); Hirshleifer and Riley (1979:1412); McAfee and McMillan (1987:699–700, 732–3); Schotter (1985:39–40, 41–2); Kohler (1982:28 ff.), and Dolan (1983:62).
- 6 See, for example, Hurwicz (1960, 1972, 1973, 1977, 1984). Also see Almon (1963); Calsamiglia (1977); Davis and Whinston (1966); Marschak (1959, 1972); Mount and Reiter (1974); Osana (1978); Reiter (1977). For a relatively recent survey and more bibliography, see Groves and Ledyard (1987).
- 7 Most of this literature uses the term 'information' in the measurable sense meant in 'information theory'. It is in this context that Arrow (1974:159) complains about the lack of 'a more definitive measure of information and its costs, in terms of which it would be possible to assert the superiority of the price system over a centralized alternative'.
- 8 For critical comments on this point, see Lavoie (1986).
- 9 In fact, Grossman and Stiglitz (1976:248) seem to believe that Hayek is also saying so.
- 10 See Streit (1984) for some criticisms of this argument.
- 11 The classic article on 'breakdown' and 'thinness' of markets is Akerlof (1970).
- 12 Some problems with this example will be pointed out below.
- 13 Implicitly, Grossman and Stiglitz must assume this 'invisible hand' to be able to obtain the information costlessly. Otherwise, it would only become optimally informed.

- 14 It is not clear why the shoe seller does not count the complementary information among the benefits of his activity. If he did, the information externality problem would lead him to find a lower level of sales profitable, he would acquire less knowledge, and there would still be an efficiency problem from a standard perspective.

In another article, Grossman and Stiglitz mention briefly the case of costless information and claim that, in the model they are considering, an equilibrium set of prices exists, but at 'zero trade'. Because of the absence of trade, Grossman and Stiglitz (1976:251, n.) point out, 'there is no obvious mechanism for sustaining this particular set of prices, and this is a serious limitation'.
- 15 This type of problem is highlighted by Hirshleifer (1971).
- 16 For a refusal to regard entrepreneurship, a concept that includes the activity of arbitrage, as a costly resource, see Kirzner (1985a:24–5).
- 17 Contrary to this, arbitrage has generally been considered, at least implicitly, to be a disequilibrium activity invoked to ensure the establishment of equilibrium conditions. As Varian (1987:56) puts it: 'It is generally felt that part of the definition of equilibrium in a perfect market is that no opportunities for pure arbitrage exist.' Similarly, Grossman and Stiglitz (1980:393) point out that the 'conventional' position is that 'when markets are not arbitrated, there are profits to be made, and so equilibrium must entail perfect arbitrage: the profits accrue in the process of responding to some unspecified disequilibrium'. In other words, equilibrium is in the traditional view the result of arbitrage activity, and not an environment in which it takes place.
- 18 For a similar idea, see Schultz (1975). Grossman and Stiglitz have an interesting justification for their procedure: they argue that 'the limitation of our analysis to stationary stochastic processes is not a serious limitation; economic theory is concerned with identifying, describing, and explaining regularities in economic processes. Economic theory attempts to identify within a particular event those characteristics which it has in common with other events which have occurred. It is these regularities that are described by the stationary stochastic process' (1976:247, n.).
- 19 Stiglitz (1987) is a survey of the literature dealing with such cases.
- 20 Veljanovski also quotes D.Laidler as saying, 'general equilibrium theory is concerned [with] a situation of highly imperfect knowledge. There is no other way to justify the key role played by prices in that analysis' (1982:64).
- 21 The competitive activity of other traders 'determines not only the informational quality of prices but also the speed at which changes in beliefs and underlying information are disseminated' (Streit 1984:394).
- 22 The large number of economists who have interpreted Hayek's argument in equilibrium terms may not be a mere product of coincidence: it appears Hayek himself was unaware in 1945 of the theoretical framework he was using (see Kirzner 1984b:203).
- 23 The attempt here is to spell out implications of Grossman and Stiglitz's approach which they themselves have not worked out. On the other hand, the work of Leonid Hurwicz already cited is an example of the line of argument described in the text.

- 24 Bartley says that his is a generalization of 'Hayek's approach somewhat in full harmony with his intentions', suggesting he also has some doubts about the theoretical framework (i.e. equilibrium or disequilibrium) used by Hayek in his 1945 article.
- 25 Of course, agents have to know a number of other things, such as their preferences, available technologies, etc., to adopt the best course of action for given prices.
- 26 For another interpretation of Hayek's view in terms of prices as sufficient statistics, see Williamson (1975:5, 25).
- 27 The added emphasis is the same as in Garrison's article, quoted below. The same argument appears in Koopmans (1957:22, 53).
- 28 See, e.g., Grossman's (1981:545) statement: 'Of course, the theory does not explain how equilibrium comes about.'
- 29 In their example, described above, it matters whether the price of the asset increases because of a larger 'informed' demand or because of a smaller supply. This is not the case for Hayek's tin example, in which any of these causes requires the same adjustment.
- 30 'Hayek was not contending that prices as numbers are the only pieces of information that the market requires' (Lavoie 1985b:80–1). This more moderate claim is consistent with Frydman's (1982:664) remark that 'in addition to information contained in market prices, social norms (in particular business practices) imposing some restrictions and coherence on the individual decisions and information generated by institutions external to the market may play important roles in understanding decentralized market processes'. The informational role of institutions is analysed in Schotter (1981).
- 31 For a survey of his experimental work, see Smith (1982b). For an example of other experimental work on prices and information, see Plott and Sunder (1982). About the field in general, see Roth (1986).
- 32 Smith does not claim that his results constitute such a refutation.
- 33 However, these authors seem to be close to Demsetz's 'comparative institution approach' when they conclude, on the same page, that they 'cannot provide an answer to whether a centralized or decentralized organization is more efficient, without more knowledge of the costs of operating a centralized informational mechanism...'.
 34 Also see Streit (1983:9–10).
- 35 See also Hirshleifer and Riley (1979:1414).
- 36 'It is only in equilibrium that it can be claimed that a market participant guided by market prices is automatically steered toward those actions that will coordinate smoothly with the actions of all the other (similarly guided) market participants' (Kirzner 1984a:415). See also Kirzner (1984b:204–5).
- 37 Cf. 'Economic calculation, despite its imperfect configuration of disequilibrium relative prices, still...enables entrepreneurs to eliminate from consideration the innumerable possibilities of technologically feasible but uneconomic production processes' (Lavoie 1985a:57).
- 38 Of course, strictly speaking, profitable production of a good using the resource is also a form of arbitrage, in a broader than usual sense of the term.

- 39 See also Hahn (1984a:11) and Fisher (1983:4). According to Lavoie (1985a:35), 'there are elements of disco-ordination diffused throughout any market economy...But there are also the well-known general regularities, such as those between prices and costs of production, that are reflected in both the classical labor theory of value as well as the modern subjectivist theories of marginal utility and imputation.'
- 40 This view clearly underlies Mises's writings: in a world with no unpredictability no 'Misesian' action would be necessary (an equilibrium would, presumably, sooner or later, be achieved), and in a world with completely volatile and unpredictable change no such action could be successful or, therefore, worth while.
- 41 As in other instances, the imperfection here referred to uses the questionable standard of an omniscient agent's point of view.
- 42 Schumpeter (1934:62) describes equilibrium as 'never attained, continually "striven after" (of course not consciously) ...'.
- 43 In the world of price-taking agents of the standard version of this story, it is not clear who, aside from the makeshift device of a fictitious auctioneer, is in charge of increasing the price of tin. This deficiency of standard ('competitive') price theory leads Fisher (1970:195) to remark that 'the Invisible Hand is a little too invisible in this, the center of its activities'.
- 44 This role includes the possibility of transmitting information ('signalling') through prices.

4 'Bounded rationality' and the price system

- 1 For example, Simon (1981:41), Nelson and Winter (1982:361), and Williamson (1975:4–5) have interpreted Hayek as arguing against central planning from a bounded-rationality perspective. See also Langlois (1985:225–35). For an attempt to find common elements in these lines of thought, see Langlois (1986: esp. 225–41).
- 2 See, e.g., Radner (1968); Nelson and Winter (1982); Williamson (1975); Loasby (1976).
- 3 Cf. Kirzner (1973:32–7). Simon, like the Austrians, also distinguishes between subjective and objective rationality, and mentions the need for an understanding of how the means-ends framework is perceived by agents. An early distinction between objective and subjective rationality appears in Hutchison (1938:109–14).
- 4 Also see Ando (1979:86).
- 5 The Austrian flavour resides in the content of the passage and not in its reference to Schumpeter.
- 6 For a strong criticism of Simon's, and other researchers', treatment of human beings as information processors, see Dreyfus (1979) and Dreyfus and Dreyfus (1986).
- 7 Williamson also includes within bounded rationality what he calls 'language limits', which 'refer to the inability of individuals to articulate their knowledge or feelings by the use of words, numbers or graphics in ways

- which permit them to be understood by others'. This, however, does not appear in Simon's writings.
- 8 The 'boundedness' refers, then, to objective rationality, meaning by this the perfectly informed decisions that would be made by the theoretical fiction of an omniscient agent.
 - 9 On his idea of complexity, see also Simon (1962b). There is some similarity between Simon's and Hayek's (1964) notions of complexity. On complexity, see Grunberg (1978:546–9).
 - 10 Also see Simon (1976:430–1). Simon is fond of this example because, for him, chess is 'a microcosm that mirrors interesting properties of decision-making situations in the real world' (1972:412).
 - 11 This saturation could, presumably, occur at different stages such as that of reception, of storage, or of the processing of information.
 - 12 'In a world where information is relatively scarce, and where problems for decision are few and simple, information is almost always a positive good. In a world where attention is a major scarce resource, information may be an expensive luxury, for it may turn our attention from what is important to what is unimportant. We cannot afford to attend to information simply because it is there' (Simon 1978b:456).
 - 13 Simon (1970:167) defines an 'information-rich' world as one in which (1) the average search time required to discover a new opportunity is low, or (2) there are many highly valued opportunities. This idea involves a quite particular view of perception. For example, an entrepreneur could, apparently, become aware of too many opportunities, rather than find it difficult to discover any at all. This view of perception will be considered below.
 - 14 Baumol (1979:75–6) realizes that this is not what Simon is saying. On the attempt to make apparently non-optimizing behaviour compatible with maximization, see Winter (1971:242–3); Boland (1981: 1033); Langlois (1986:226).
 - 15 This is not so obviously true. It could be argued that the widespread use of computers has made the world more complex.
 - 16 However, this behaviour may approach objective rationality in some exceptional circumstances 'if the number of very important variables is small at any given time, *and* if this list of important variables does not change from time to time without the change being noticed' (Simon 1984:48; emphasis in original).
 - 17 The neoclassical agent would 'allocate just as much effort to scanning the horizon for newly important variables as is justified by the marginal value (in terms of improved decision making) of the scanning. This...alternative is vacuous, providing us with no clue as to how these marginal costs and benefits are to be estimated' (Simon 1957: 48).
 - 18 See, for example, Simon (1957:204–5) and Simon (1978a:455).
 - 19 'We need far more empirical knowledge than we now possess as to the bases on which aspirations are formed and the conditions under which they are modified' (Simon 1958:399). There is some similarity between this idea and Mises's (1949:13–14) notion of 'felt uneasiness' as a prompter of human action. The differences between them seem to be due to the different

- theories of cognition held by Simon and market-process economists, as argued below.
- 20 The market *system* allows individuals a choice between organizing activity through firms or through market trading according to what they perceive to be most advantageous.
 - 21 Simon's use of these concepts implies that, when he finds limitations in what he calls 'markets', his alternative is not necessarily government intervention: perhaps some other market alternative, such as firms, is the solution. In fact, the next chapter argues that the market system stimulates the discovery of solutions to possible limitations of price-mediated transactions.
 - 22 Simon credits this idea to Dahl and Lindblom (1953).
 - 23 The price system is 'an institution that reduces the amount of nonlocal information the actors must possess to make reasonable (i.e. satisficing) decisions' (Simon 1981:57).
 - 24 The lack of an explicit distinction between market and non-market hierarchies is evident here.
 - 25 'We must keep in mind that...externalities...are rarely absent. Markets can only be used in conjunction with other methods of social control and decision making...' (Simon 1983:89).
 - 26 'Bounds on rationality are interesting...only to the extent that the limits of rationality are reached—which is to say, under conditions of...complexity' (Williamson 1975:22).
 - 27 The intention of the example is to build a simple situation so that this is true. It would be easy to turn this same example into a complex situation by arguing that the small number of traders introduces strategic considerations and other bargaining difficulties.
 - 28 'One reason why computers have been so important to Man is that they enlarge a little bit the realm within which his computational powers can match the complexity of the problems' (Simon 1976:430).
 - 29 See Dreyfus (1979: esp. 91–129).
 - 30 If human beings have this ability, Simon's notion of satisficing may turn out to be of doubtful validity. (Individuals would not need to scan; their attention could be drawn directly to the most worth while alternative.)
 - 31 According to Dreyfus (1979:33), the 'knowledge representation problem' refers to 'the problem of how to structure and retrieve *data* in situations when anything might be relevant'.
 - 32 Again, in this view the planner neglects knowledge of specific circumstances because he is incapable of 'absorbing' it, not because he does not notice it or because it is, in some way, tacit and inarticulable.
 - 33 As pointed out, Hayek has also sometimes argued in this way.
 - 34 This may also explain his puzzling statement about the advantage in terms of optimality properties of markets over hierarchies.
 - 35 As will be pointed out, this does not mean that market participants will not, in spite of these disadvantages, find it useful sometimes to adopt hierarchical forms of organization for carrying out their activities.
 - 36 For example, market-process economists have clearly improved their understanding of the nature of ignorance and discovery by facing the challenges posed by the economics of information and search theory.

- 37 No example is drawn from Kirzner's writings because the issue of complexity hardly appears explicitly in them. It is largely due to his concentration on entrepreneurial discovery, even in simple contexts, that the need for the distinctions made here becomes evident.
- 38 For example, Hayek's concern has recently been interpreted to be over whether the central planner will be able to cope with the enormous amount of information, 'whether the king can handle the job of collecting and using the relevant information...', given that he 'has a limited ability to process [it]' (Farrell 1987:117, 121). See also Buchanan (1985:16), although he seems to be aware there is more to the problem.
- 39 See, for example, Hayek (1935b:155). Also Lavoie (1985a:161–2).
- 40 An idea that also appears in Schultz (1975:831–2).
- 41 See, for example, O'Driscoll (1977)
- 42 This, for example, is what seems to be happening in those anecdotes about socialist economies that describe plant managers as unable to comply with their output quotas because the production of the required inputs was not contemplated.
- 43 Although co-ordination of actions can coexist with disco-ordination of knowledge, it is not clear whether there could be co-ordination of knowledge and disco-ordination of actions—the latter due to 'complexity'. This would depend on whether co-ordination of knowledge includes knowledge of the appropriate arrangements to overcome complexity successfully. (Co-ordination of knowledge would perhaps lead to optimal disco-ordination of actions!)
- 44 See, for example, High (1986:114).
- 45 The full elucidation of this point and of the whole issue would require much more space than can be devoted to it here.
- 46 Another implication of the distinction made here is that, although in the habitual use disco-ordination of actions refers to observable problems, those of Kirzner's type that take the form of missed opportunities may never become visible. This is important because, from this entrepreneurial perspective, the disco-ordination that occurs in a planned economy, for example, is not limited to the observed shortages or surpluses of goods. It also includes the unknown number of alternative 'facts' (types and qualities of products that could be made, what quantities of them to produce, with what production technologies, what resources are available and in what quantities, etc.) that were never discovered and exploited. The ignorance of these facts constitutes probably the most serious source of inefficiency in those economies.

5 Change, responsiveness and co-ordination

- 1 Nelson (1977:132–43; 1981). The argument also appears, unmodified, in Nelson and Winter (1982), but it seems reasonable to attribute it to Nelson.
- 2 That is, (1), that under certain assumptions a competitive equilibrium is Pareto-optimal, and (2), that any Pareto-optimal outcome can be achieved as a competitive equilibrium by a suitable distribution of initial endowments.

- 3 Economists, Nelson and Winter (1982:358) say, are aware of the fact that not all firms act as price-takers, that there are public goods and 'externalities are pervasive', that economic agents may lack perfect information, and so on.
- 4 Nelson (1981:95) credits Simon for the 'most forceful articulation of the meaning and significance of bounded rationality...'
- 5 Nelson (1977:136) points out the failure of economists—whom he does not identify— 'to recognize the contradiction between those arguments for market structure which are based on the premise that behavior of a competitive regime will be Pareto optimal, and those arguments for real markets which are focused on the alleged flexibility and responsiveness of such a regime...'
- 6 Nelson (1981:95) notices that both Mises and Hayek, 'the two advocates of private enterprise in this debate, pose their arguments in a form much more distant from that of contemporary welfare economics than does Lange, the advocate of Socialism'.
- 7 The other exception to the 'static perspective,' in his view, is 'the early Schumpeter'. Because of his questionable differentiation of Mises's and Hayek's argument, Nelson seems to believe that Lange's was an appropriate response to Mises's challenge.
- 8 In the light of the economic analysis of regulation and the theory of public choice, Nelson may have been too hasty in attributing such altruistic motives to these government interventions. However, it should be noticed, he ends by not advocating any particular regime because he does not 'have any strong notions about the answers' to these organizational questions (Nelson 1981:110).
- 9 See the quotation of Gordon and Hynes (1970) in chapter 2.
- 10 For a survey of this literature, see Fisher (1983).
- 11 The tone of this statement is cautious because a market economy need not actually achieve equilibrium even under imaginary unchanging conditions. However, as long as this equilibrium state was not reached, a market system would still stimulate entrepreneurial activity, whereas no equivalent discovery process would be at work in a planned regime.
- 12 Mises and Hayek used a similar assumption, although Mises (1936: 142) correctly pointed out that to do so 'we need only avoid asking how this stationary condition is achieved'. See, e.g., Mises (1920: 109–10), and Hayek (1940:188).
- 13 In Nelson's book with Winter, consideration of these possibilities appears to have been deliberately suppressed (Nelson and Winter 1982:38).
- 14 The terms 'hierarchy' and 'organization' will be used interchangeably.
- 15 The term 'non-market hierarchies' is used because although it will most often refer to government-generated organizations, what is meant, more generally, is organizations created by individuals whose decisions are not motivated by profits in the market and who are significantly immune from its competitive pressures (in the sense that they do not have to compete for the acquisition of resources and may not be overly concerned about pecuniary losses caused by their decisions). In the case of government, this independence from the market is due to the possibility of obtaining resources through non-voluntary means.

- 16 An implication of this will be that decentralized bureaucracies are also targets of the entrepreneurial criticisms of planning. For an argument in favour of central planning that is largely based on an unawareness of both distinctions made here, see Wanless (1987:52–68).
- 17 Coase (1937:388) also quotes D.H.Robertson's description of firms in the market as 'islands of conscious power in this ocean of unconscious co-operation...'.¹
- 18 See Hurwicz's (1984:423–4) brief criticism.
- 19 For a more recent paper adopting a similar approach, see Camacho and Persky (1988:367–80).
- 20 Hayek (1973:49) describes this as 'guiding the actions of individuals by rule rather than specific commands'.²
- 21 This distinction is also present when Coase (1937:387) says that 'an economist thinks of the economic system as being co-ordinated by the price mechanism and society becomes not an organisation but an organism'. For a criticism of the use of the term 'organism' for this purpose rather than 'spontaneous order', see Hayek (1973:52–4).
- 22 For Mises (1949:304–5) a 'manager' will have 'entrepreneurial functions...assigned to him within a limited and precisely determined sphere of action'. For several comments of relevance to this section, see, in general, Mises (1949:303–8).
- 23 The term 'free' here does not mean 'costless' or 'equally costly to all'. It means 'the freedom of potential competitors to discover and to move to exploit' perceived profit opportunities (Kirzner 1985a:142). See also Armentano (1982).

6 Conclusions

- 1 See, for example, Gilad (1981, 1982).
- 2 This appears to be hinted at in Mises (1949:353).

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